
ZTR Trouble Shooting And FAQ List

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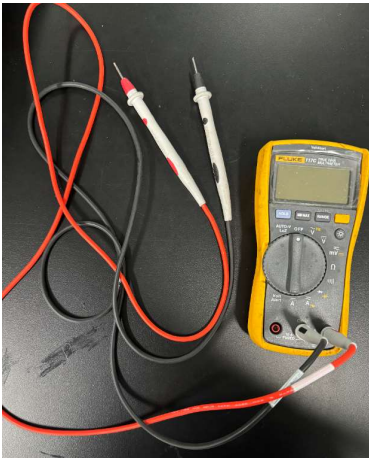
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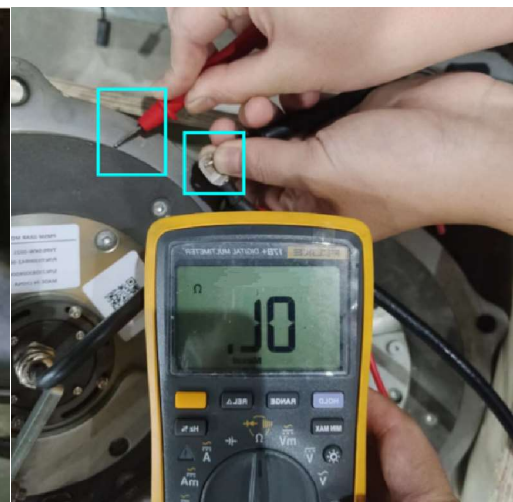
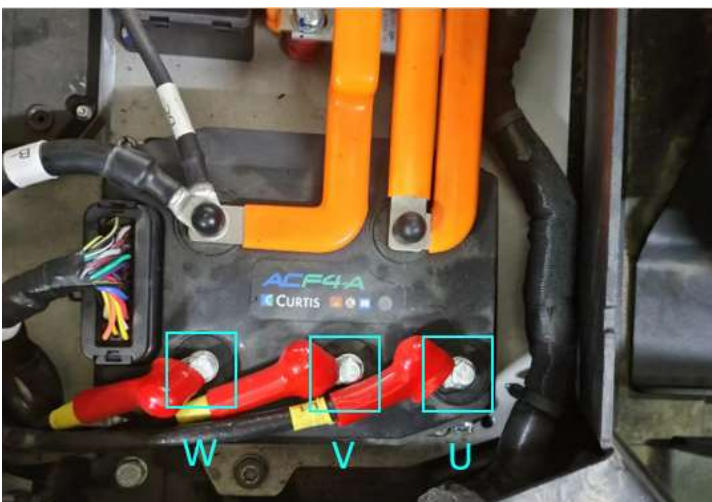
TR 12/TL 12 Right/Left Wheel Motor Controller Overcurrent

Tool: Multimeter、Computer、PCAN、Debugging wire



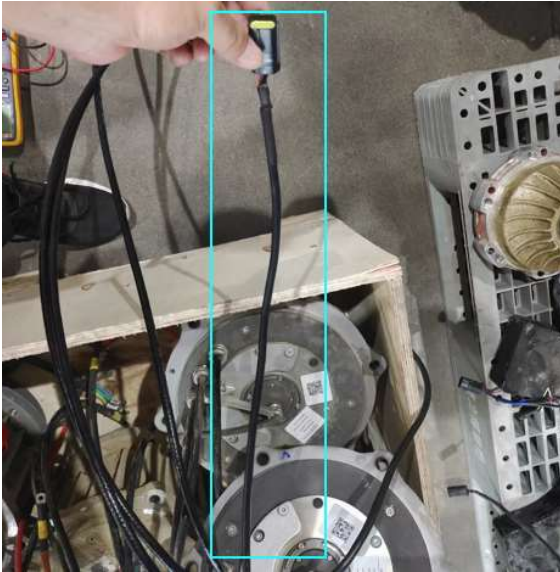
Check:

1. Display pop up TR 12/TL 12 fault code.
2. Remove all the load of the vehicle, restart and observe whether the instrument is still reported this fault. If the Fault disappears the cause of the fault is vehicle overload.
3. Check whether the insulation skin of motor U \ V \ W phase wire is broken and copper wire is together at the damage, If the wiring harness is damaged, dispose of the damaged area.
4. Remove the U/V/W phase line of the motor controller, Measure the resistance value between U \ V \ W phases and the motor housing using a multimeter resistance range,



5. If the resistance value is less than $20M \Omega$, there is a short circuit between the internal winding and the casing of the motor, and the motor needs to be replaced.
6. If the resistance measured by the motor phase line and the housing is infinite, observe whether there is any damage to the outer packaging of the motor encoder's wiring harness. If the wiring harness is damaged, replace the

motor encode.



7.If the wiring harness of the motor encoder is intact, measure the resistance value between the controller U \ V \ W phases. If the value displayed on the multimeter is below 10 Ω , it indicates that the diode has broken down and caused a short circuit, and a new motor controller needs to be replaced.

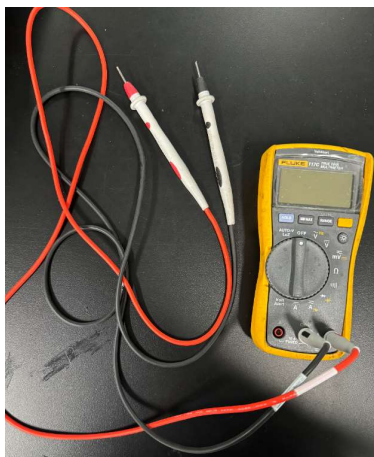


8.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor

auto-matching test.

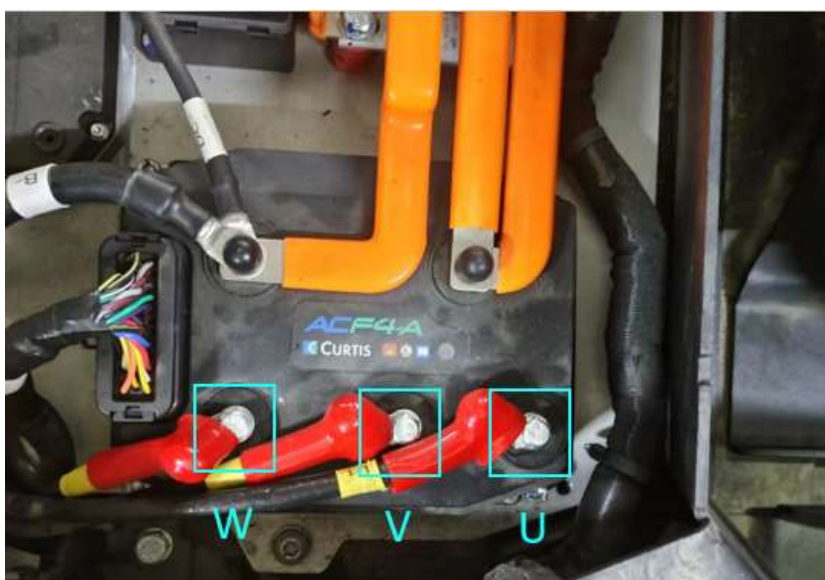
TR 13/TL 13 Right/Left Wheel Motor Controller Current Sensor Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 13/TL 13 fault code
2. Check whether the insulation skin of motor U \ V \ W phase wire is broken and copper wire is together at the damage, If the wiring harness is damaged, dispose of the damaged area.
3. Remove the U/V/W phase line of the motor controller



4. restart the vehicle. If the fault continues to occur, it indicates that the internal current sensor of the motor is damaged and a new motor controller needs to be replaced,
5. If the fault disappears, Measure the resistance value between U \ V \ W phases and the motor housing using a

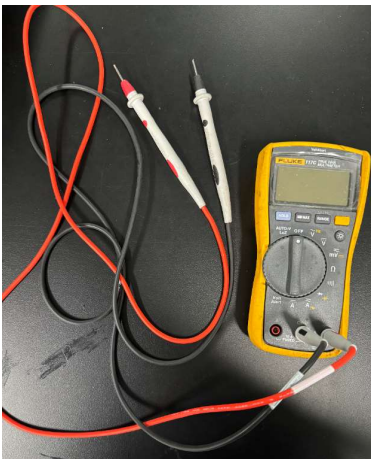
multimeter resistance range, If the resistance value is less than $20\text{M } \Omega$, there is a short circuit between the internal winding and the casing of the motor, and the motor needs to be replaced



4. If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor auto-matching test.

TR 14/TL 14 Right/Left Wheel Motor Controller Precharge Failed

Tool: Multimeter、Computer、PCAN、Debugging wire



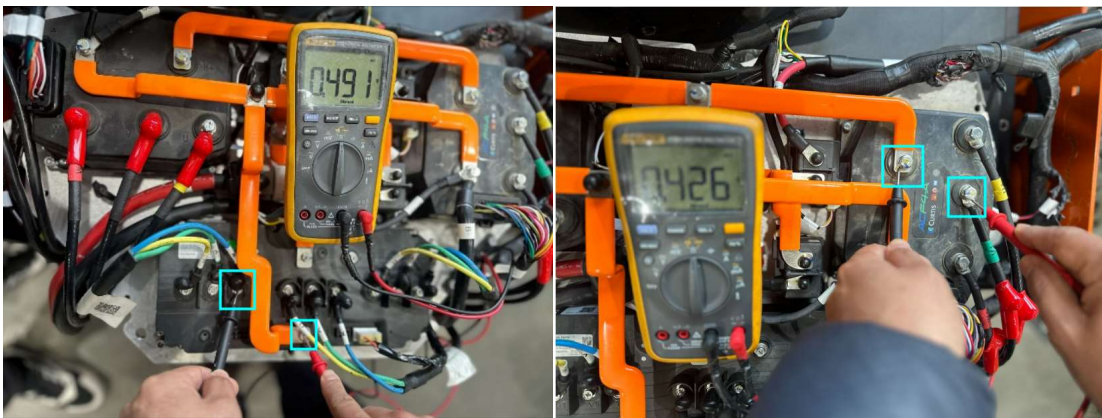
Check:

1. Display pop up TR 14/TL 14 fault code
2. keep the vehicle power off state.
3. Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the positive

electrode.



4. Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the negative electrode.



5. If the range is 0.45-0.52V, the MOS is normal. Otherwise, it is judged that the MOS is damaged and the controller needs to be replaced.

6. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 15/TL 15 Right/Left Wheel Motor Controller Severe Undertemp

Tool: Infrared Thermometers、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 15/TL 15 fault code
2. Check the current ambient temperature. If the ambient temperature is below -40°C , it exceeds the minimum temperature limit. Please wait until the ambient temperature is above -40°C before using the vehicle.
3. If the ambient temperature is higher than -40°C , the motor controller temperature 0x502 in the vehicle CAN message can be read through the upper computer, and the message can be compared with the ambient temperature. If the difference is too large, it indicates that the internal temperature sensor of the motor controller is faulty and the controller needs to be replaced.

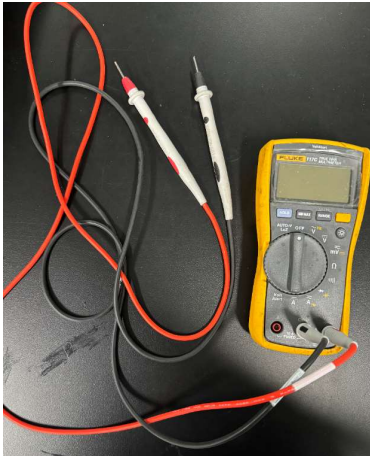


45. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

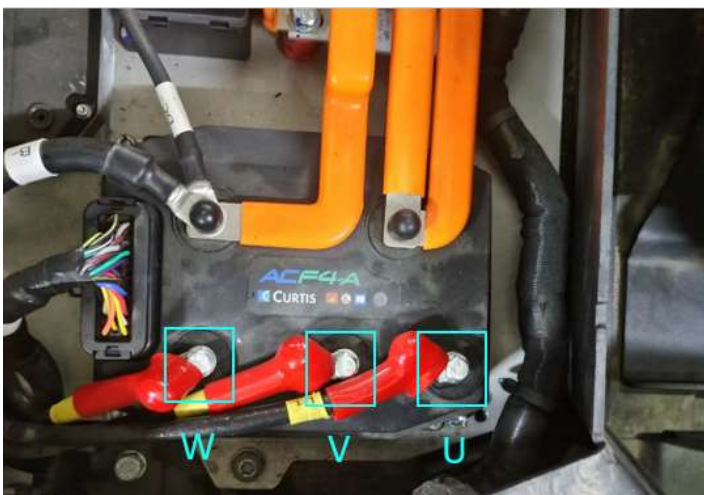
TR 16/TL 16 Right/Left Wheel Motor Controller Severe Overtemp

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 16/TL 16 fault code
2. Remove all the load of the vehicle, restart and observe whether the instrument is still reported this fault. If the Fault disappears, the cause of the fault is vehicle overload.
3. restart the vehicle. If the fault continues to occur, Check the connection of the motor U/V/W phase line on the controller and tighten the loose bolts



4. restart the vehicle. If the fault continues to occur the motor controller is faulty and the controller needs to be replaced
5. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

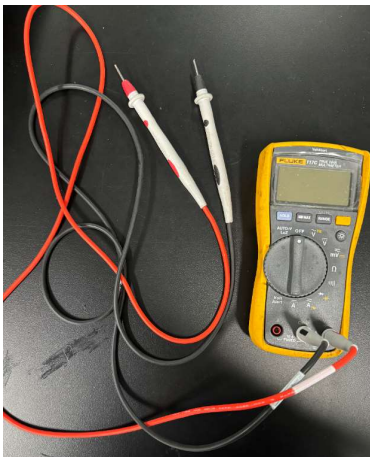
TR 19/TL 19 Right/Left Wheel Motor Controller Speed Limit Supervision

Check:

- 1.Display pop up TR 19/TL 19 fault code
- 2.Please restart the vehicle.
- 3.Please make sure that the weight of the vehicle and the driving slope are within the specified range.

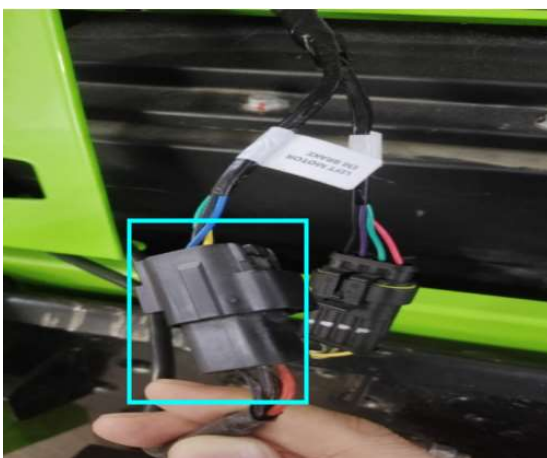
TR 1A/TL 1A Right/Left Wheel Motor Not Stopped

Tool: Multimeter、Computer、PCAN、Debugging wire



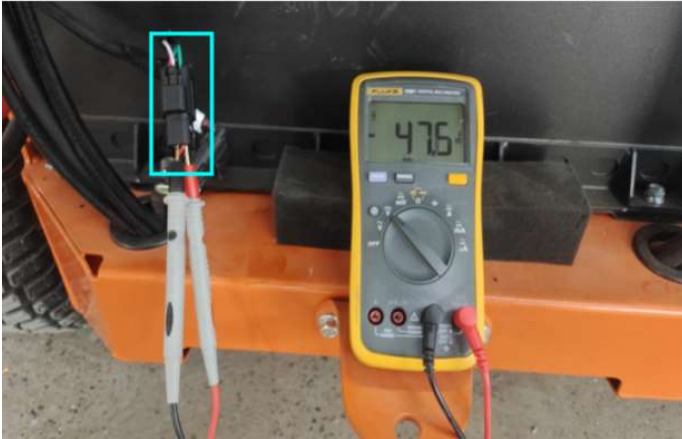
Check:

- 1.Display pop up TR 1A/TL 1A fault code
- 2.Please check whether the connector of electromagnetic brake is well connected.



- 3.Measure the DC voltage of the electromagnetic brake using a multimeter,If the value is proximity 48V But the

vehicle is in a stopped state the motor controller is faulty and the controller needs to be replaced



4.If the vehicle can drive normally, when the vehicle comes to a stop the electromagnetic brake does not work check if the electromagnetic brake release handle is open (The electromagnetic brake of the vehicle is closed in the factory state, If the release handle is tilted up, it indicates that the electromagnetic brake is in the manual release state) if the electromagnetic brake release handle is not opened, the electromagnetic brake is damaged.and the drive motor, needs to be replaced



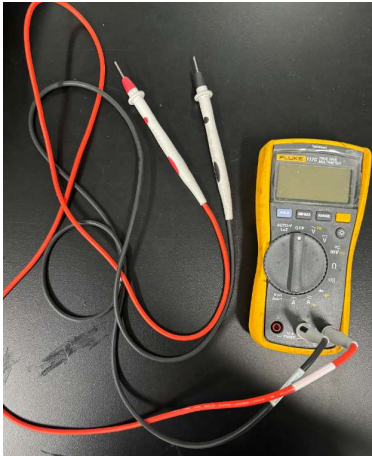
5The electromagnetic brake must be closed manually.

6.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor

auto-matching test.

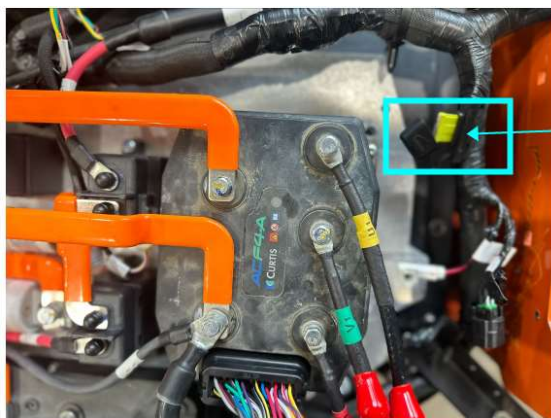
TR 1D/TL 1D Right/Left Wheel Motor Controller Reset Rejected

Tool: Multimeter、Computer、PCAN、Debugging wire



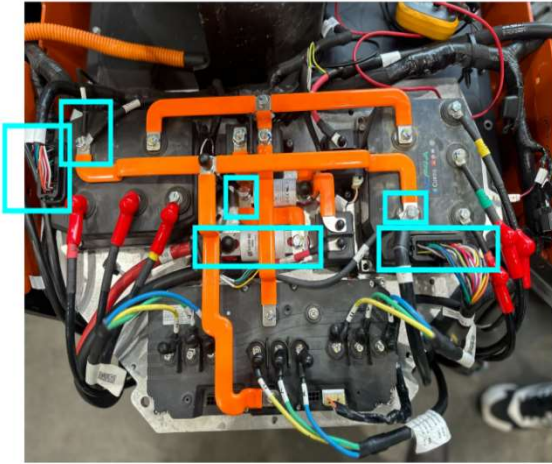
Check:

1. Display pop up TR 1D/TL 1D fault code
2. Please check whether the fuse on the wiring harness is well plugged as shown in the figure below.



Fuse

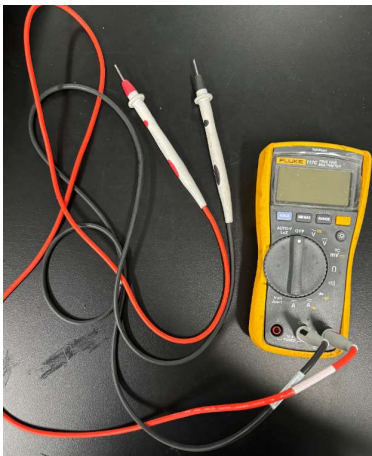
3. Please check whether the connectors are well connected as shown in the below figure.



4.Restart Vehicle.

TR 1E/TL 1E Right/Left Wheel Motor Short

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR 1E/TL 1E fault code
- 2.Check whether the insulation skin of motor U \ V \ W phase wire is broken and copper wire is together at the

damage, If the wiring harness is damaged, dispose of the damaged area.

3. Measure the resistance value between U \ V \ W phases and the motor housing using a multimeter

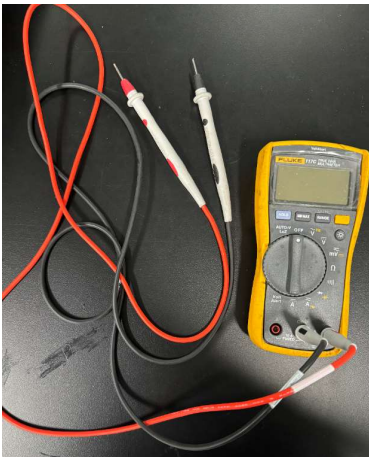


4. If there is a resistance value during the test, there is a short circuit between the internal winding and the casing of the motor, and the motor needs to be replaced.. (under normal circumstances, the resistance value of the phase line and the motor housing is infinite.)

5. If you have replaced a new motor, remember to follow the self-learning steps for the motor auto-matching test.

TR 22 Right Wheel Motor Controller Overtemp Cutback

Tool: Multimeter、Computer、PCAN、Debugging wire



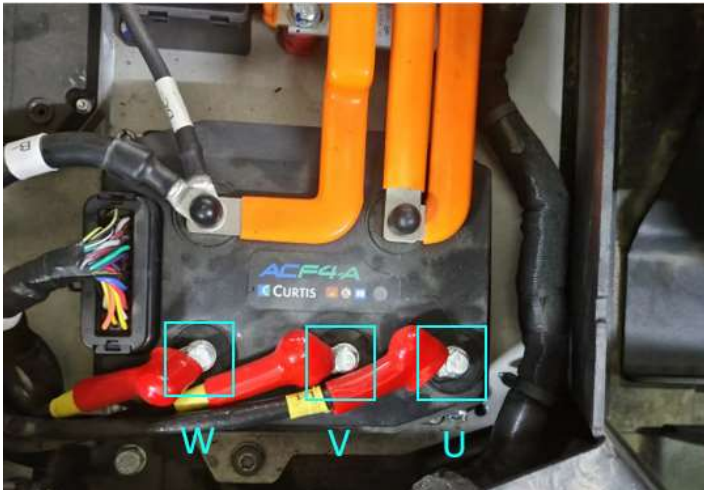
Check:

1. Display pop up TR 16/TL 16 fault code

2. Remove all the load of the vehicle, restart and observe whether the instrument is still reported this fault. If the Fault disappears, the cause of the fault is vehicle overload.

3. restart the vehicle. If the fault continues to occur, Check the connection of the motor U/V/W phase line on the

controller and tighten the loose bolts



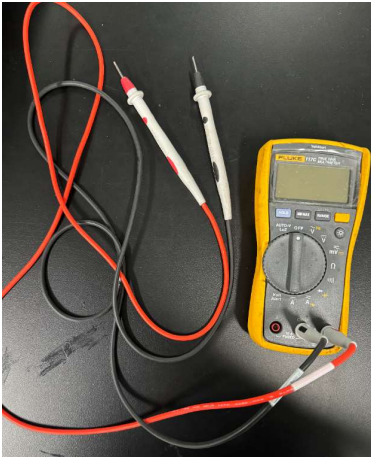
4.restart the vehicle. If the fault continues to occur the motor controller is faulty and the controller needs to be replaced

5.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 25 Right Wheel Motor Controller Ext 5V Supply Failure

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

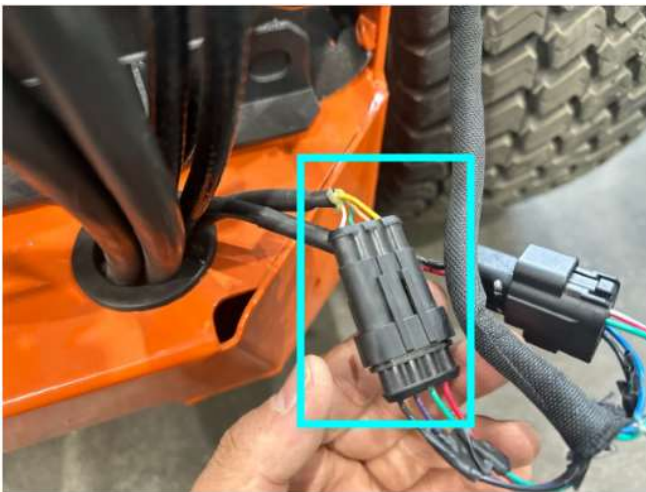
1. Display pop up TR 25 fault code
2. Please check whether the encoder is well assembled.



3. Disconnect the connector of the controller and measure the load between of pin 7 and pin 26.



4.If the load is less than 50 ohms,Please check whether the connector of encoder is well connected.

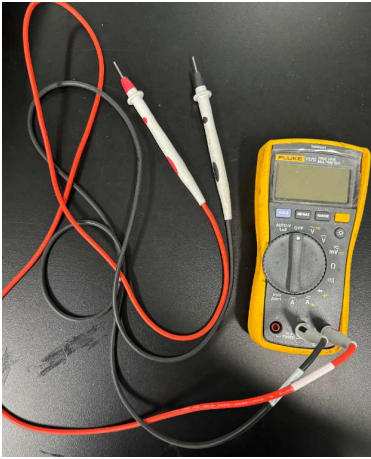


5.If you have reassembled the encoder, remember to follow the self-learning steps for the motor auto-matching

test.

TR 26 Right Wheel Motor Controller Ext 12V Supply Failure

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

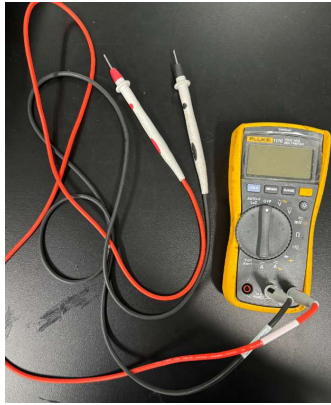
- 1.Display pop up TR 26 fault code
- 2.Disconnect the connector of the controller and measure the load between of pin 7 and pin 27.



- 3.If the load is less than 60 ohms,Please check if the connector harness of the controller is properly connected

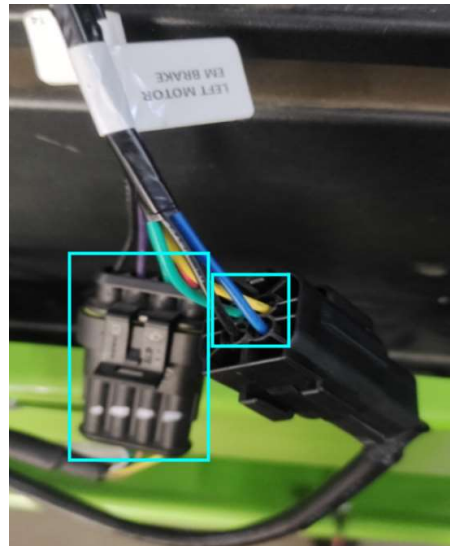
TR 28 Right Wheel Motor Slight Overtemp

Tool: Infrared Thermometers、Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 28 fault code
2. Remove all the load of the vehicle, Wait for a period of time to restart the vehicle and observe whether the instrument is still reported this fault. If the Fault disappears, the cause of the fault is vehicle overload.
3. restart the vehicle. If the fault continues to occur, Check the current ambient temperature. measure the resistance value of the motor temperature sensor,



4. Check the table against the actual temperature to see if the resistance value is correct (If at room temperature 25°C environment, it returns to about 560Ω, it means that the temperature sensor is normal. If the resistance value is

above 1282Ω, it indicates that the temperature sensor has been damaged, so the motor must be replaced.)

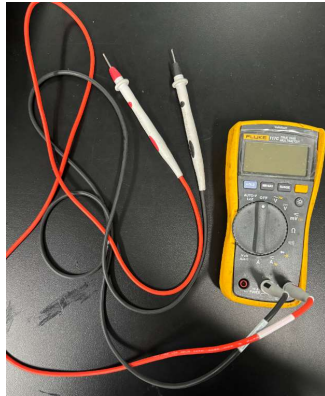
℃	℉	%/ (K)	(Ω)			(K)
			MIN	TYP	MAX	
-40	-40	0.97	294	322	350	±8.85
-30	-22	0.94	327	356	385	±8.76
-20	-4	0.91	361	392	423	±8.7
-10	14	0.88	397	430	463	±8.65
0	32	0.87	434	469	504	±8.61
10	50	0.85	475	512	549	±8.58
20	68	0.82	517	556	595	±8.55
25	77	0.80	540	580	620	±8.54
30	86	0.79	562	603	644	±8.53
40	104	0.78	610	653	696	±8.5
50	122	0.75	659	704	749	±8.46
60	140	0.73	711	758	805	±8.42
70	158	0.67	768	814	860	±8.37
80	176	0.63	827	873	919	±8.31
90	194	0.62	887	935	983	±8.25
100	212	0.62	950	1000	1050	±8.17
110	230	0.62	1011	1068	1125	±8.66
120	248	0.58	1077	1138	1199	±9.17
130	266	0.52	1148	1209	1270	±9.69
140	284	0.51	1215	1282	1349	±10.24
150	302	0.50	1279	1352	1425	±10.8

5.If you have replaced a new motor, remember to follow the self-learning steps for the motor auto-matching

test.

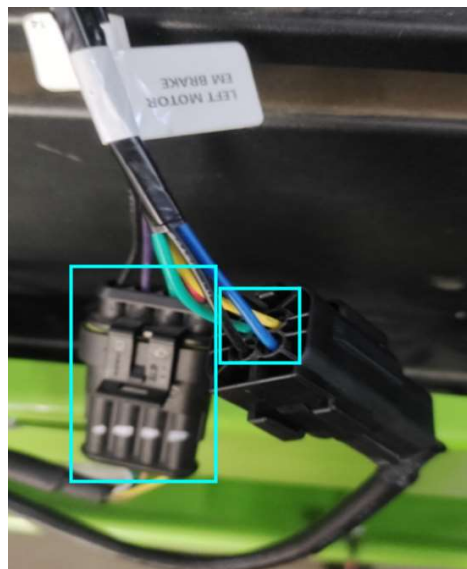
TR 29 Right Wheel Motor Temperature Sensor Abnormal

Tool: Infrared Thermometers、Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 29 fault code
2. Remove all the load of the vehicle, Wait for a period of time to restart the vehicle and observe whether the instrument is still reported this fault. If the Fault disappears, the cause of the fault is vehicle overload.
3. restart the vehicle. If the fault continues to occur, Check the current ambient temperature. measure the resistance value of the motor temperature sensor.



4. Check the table against the actual temperature to see if the resistance value is correct (If at room temperature 25°C environment, it returns to about 560Ω, it means that the temperature sensor is normal. If the resistance value is

above 1282Ω, it indicates that the temperature sensor has been damaged, so the motor must be replaced.)

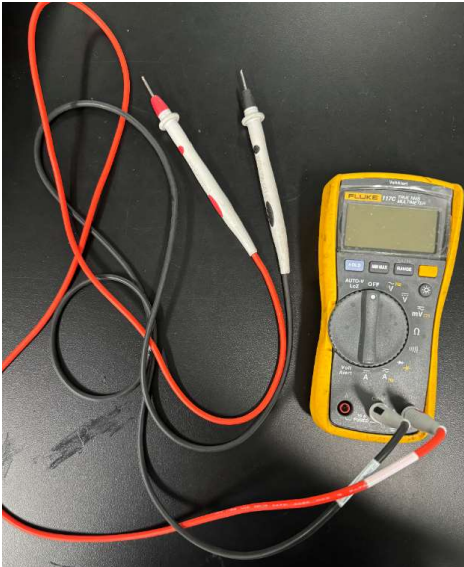
℃	℉	%/ (K)	(Ω)			(K)
			MIN	TYP	MAX	
-40	-40	0.97	294	322	350	±8.85
-30	-22	0.94	327	356	385	±8.76
-20	-4	0.91	361	392	423	±8.7
-10	14	0.88	397	430	463	±8.65
0	32	0.87	434	469	504	±8.61
10	50	0.85	475	512	549	±8.58
20	68	0.82	517	556	595	±8.55
25	77	0.80	540	580	620	±8.54
30	86	0.79	562	603	644	±8.53
40	104	0.78	610	653	696	±8.5
50	122	0.75	659	704	749	±8.46
60	140	0.73	711	758	805	±8.42
70	158	0.67	768	814	860	±8.37
80	176	0.63	827	873	919	±8.31
90	194	0.62	887	935	983	±8.25
100	212	0.62	950	1000	1050	±8.17
110	230	0.62	1011	1068	1125	±8.66
120	248	0.58	1077	1138	1199	±9.17
130	266	0.52	1148	1209	1270	±9.69
140	284	0.51	1215	1282	1349	±10.24
150	302	0.50	1279	1352	1425	±10.8

5.If you have replaced a new motor, remember to follow the self-learning steps for the motor auto-matching

test.

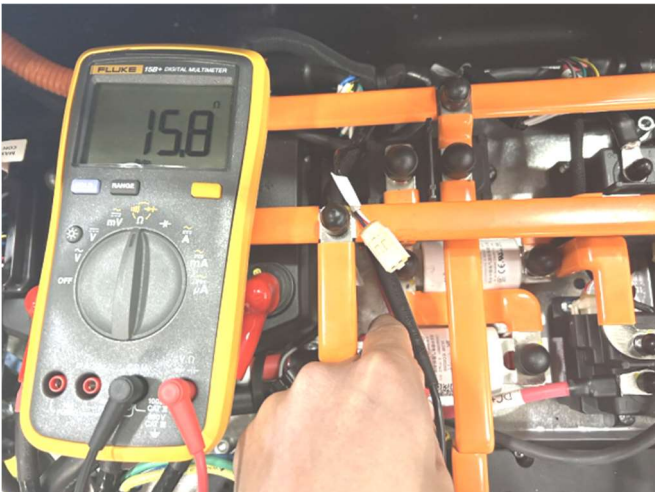
TR 31 Main Contactor Coil Driver Short /Open

Tool: Multimeter



Check:

1. Display pop up TR 31 fault code
2. Measure the resistance value between the main contactors coil using a multimeter's resistance range. when the resistance is 15.8Ω means The coil is ok, no open(just measure both side of the pin1 and pin 2)



3. Measure whether the cable from master controller connector to main contactor coil connector is open, 0.7Ω

means the cables are OK.



4.Pull out these two connectors, Measure whether the two cables(Pin13 and Pin6) are short, 0Ω means the cables are OK. No short happens.



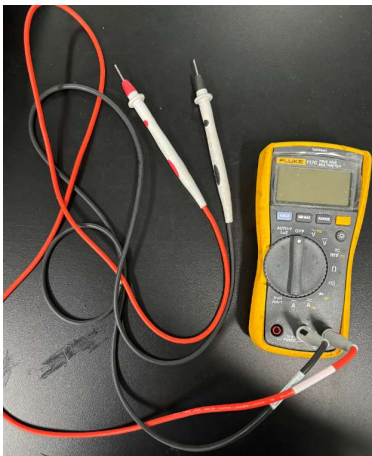
5.Measure whether the cable from master controller connector to main contactor coil connector are short to

Mover body, 0Ω means the cable are OK



TR 36/TL 36 Right/Left Wheel Motor Sin/Cos Encoder Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire

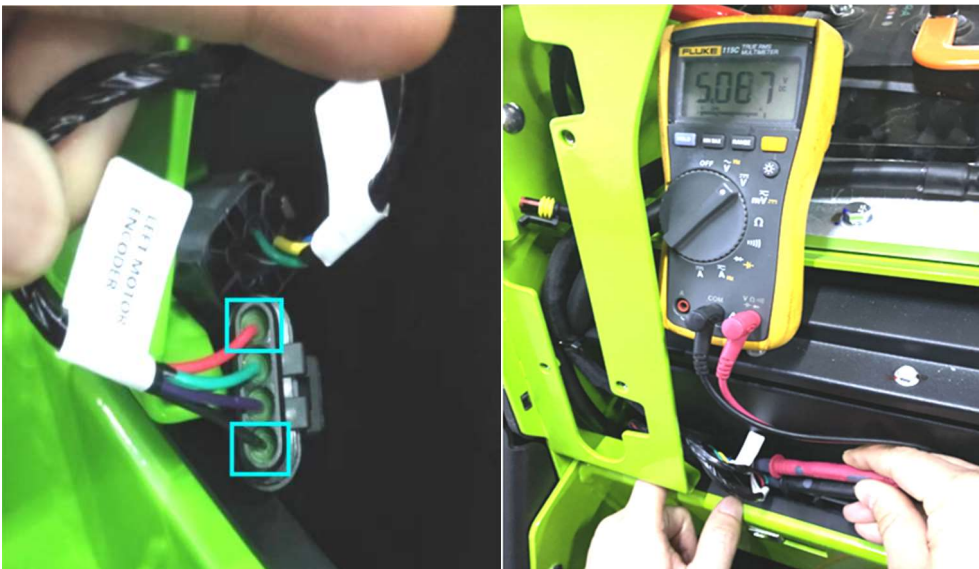


Check:

1. Display pop up TR 36 fault code
2. Check whether the connector joint is secure. Check whether the connector is securely connected.



3. Use a multimeter voltage range to measure the voltage between the red and black wires in Figure 1. If the voltage is below 5V or above 5V, it is determined that the 5V power supply of the controller is faulty and the controller needs to be replaced.



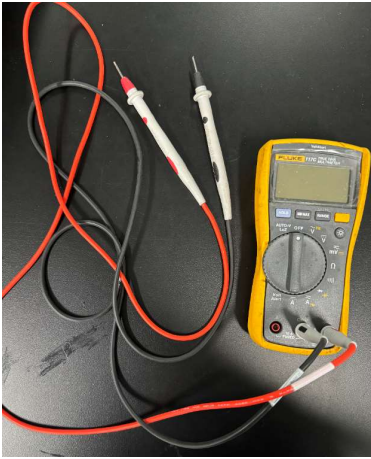
4. If the encoder 5V power supply is normal, but the fault still exists, it is considered that the encoder is faulty. replace the motor encoder.

5. If you have reassembled the encoder, remember to follow the self-learning steps for the motor auto-matching

test.

TR 37 Right Wheel Motor Phase Open

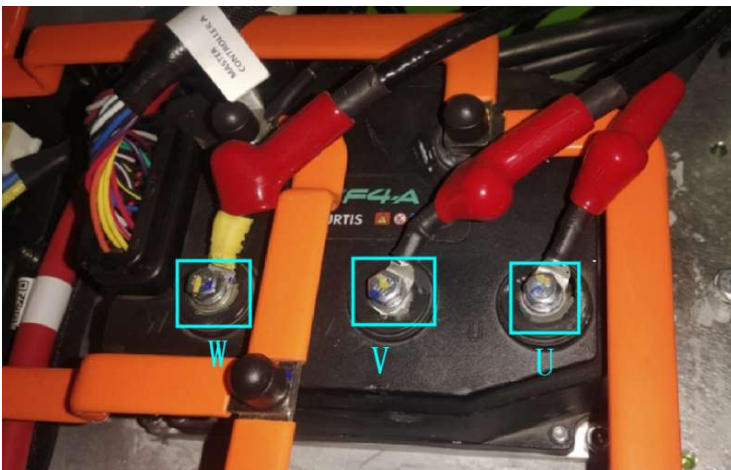
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

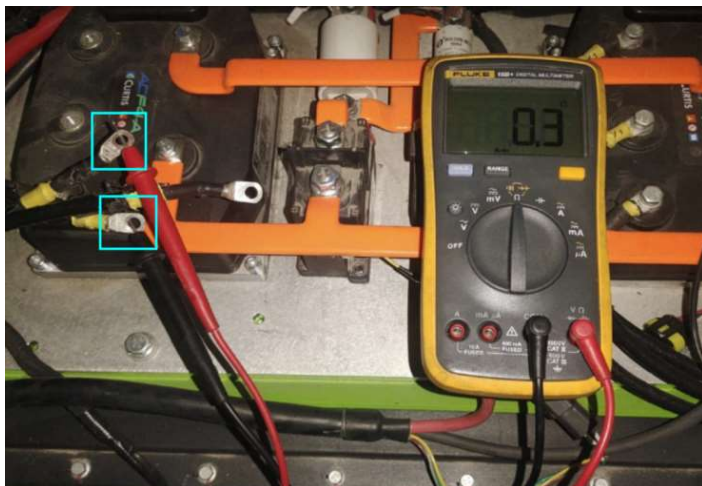
1.Display pop up TR 37 fault code

2.Check whether the motor phase line is reliably connected to the controller. Check the positions of the three screws. If they are loose, tighten the loose bolts and Check if screws are found to be damaged. Replace the new screws.if the controller screw hole is damaged and the controller needs to be replaced



3.restart the vehicle. If the fault continues to occur measure the resistance value between the motor U \ V \ W phases. If the value displayed on the multimeter is 0 or infinity it means that the motor has been damaged inside and

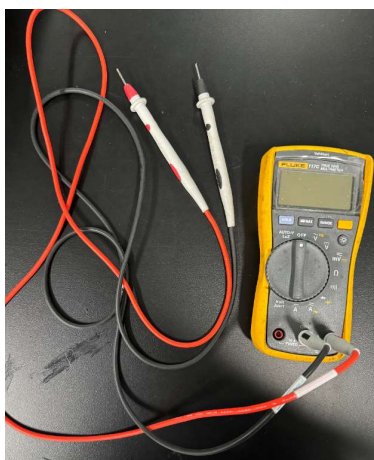
a new blade motor needs to be replaced.



4.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor auto-matching test.

TR 38 Main Contactor Welded

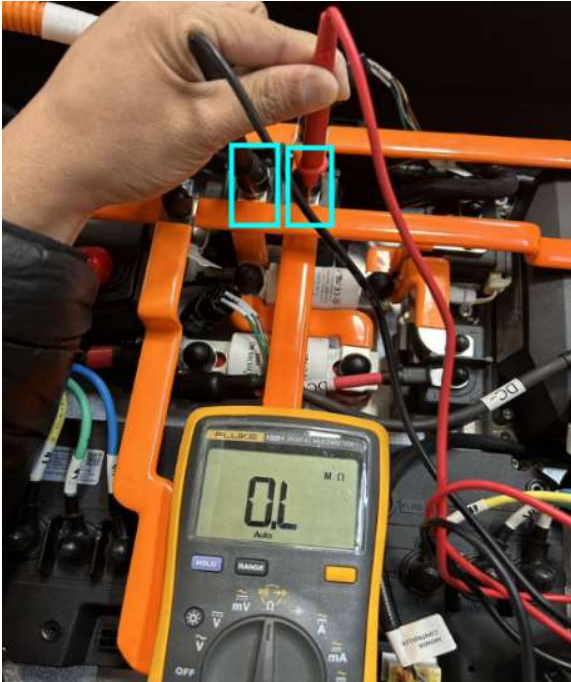
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR 38 fault code
- 2.Measure the resistance value between the main contactors using a multimeter's resistance range. when the resistance is 0 Ω , It means the contactor is ok, if the value is about 1.5 Ω , it means the contactor is exactly welded,

you just need to replace the contactor.



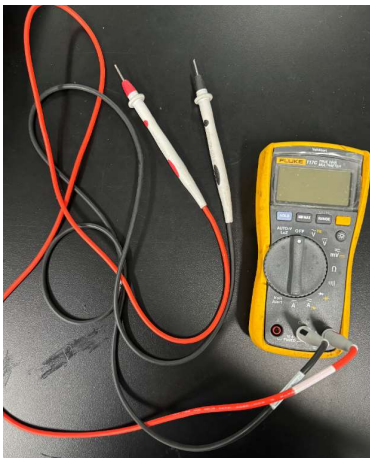
3. Tear down the contactor, measure the resistance between these two wiring terminals. when the resistance is $0\ \Omega$, It means the contactor is ok, if the value is about $1.5\ \Omega$, it means the contactor is exactly welded, you just need

to replace the contactor.



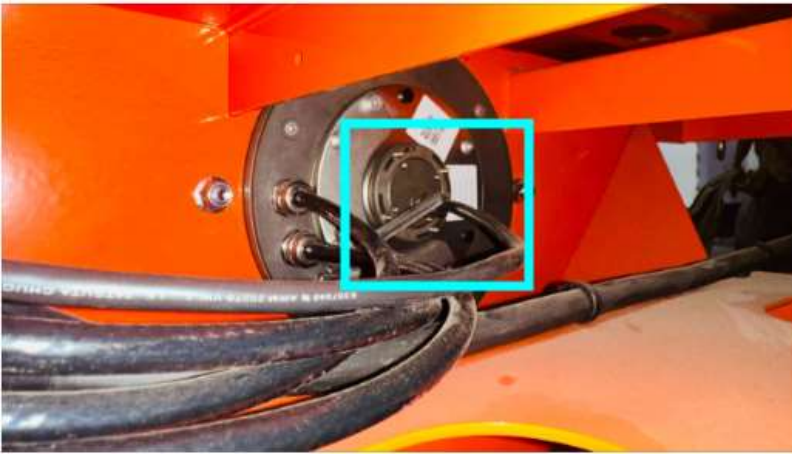
TR 3A Right Wheel Motor Controller Setup Needed

Tool: Computer、PCAN、Debugging wire

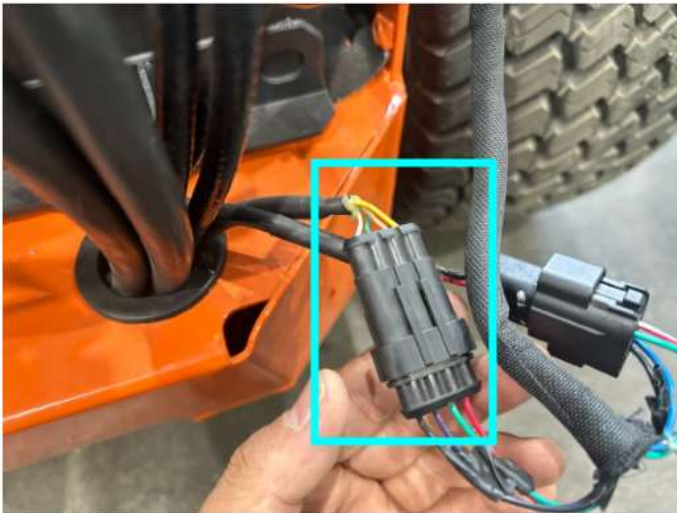


Check:

1. Display pop up TR 3A fault code
2. Please check whether the encoder is well assembled.



3. Please check whether the connector of encoder is well connected.



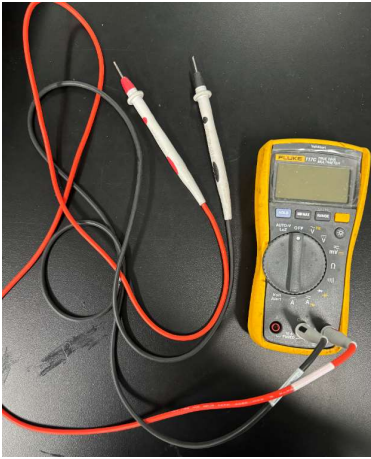
4. Run the appropriate motor commissioning sequences.

5. If you have reassembled the encoder, remember to follow the self-learning steps for the motor auto-matching

test.

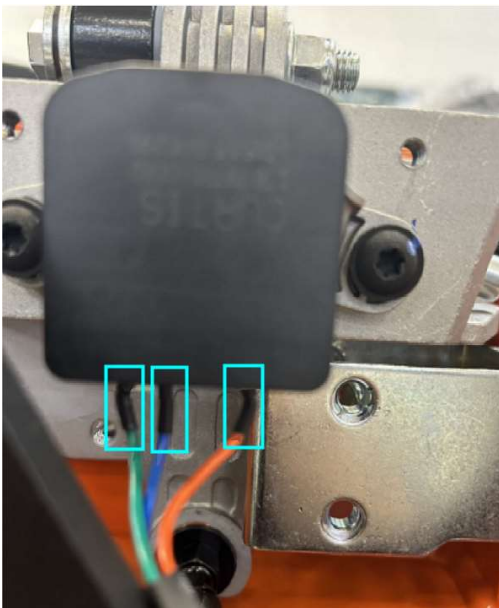
TR 42 Right Throttle Input Abnormal/ LEFT Throttle Input Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



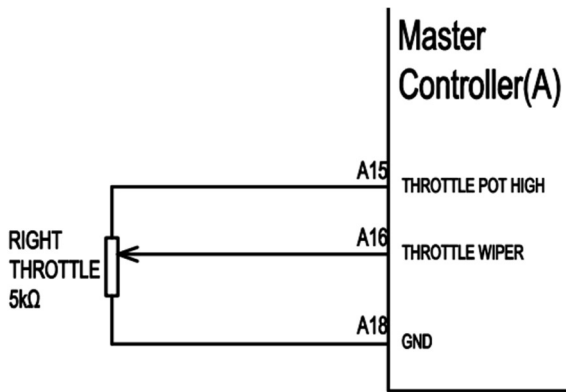
Check:

- 1.Display pop up TR 42 fault code
- 2.Check the soldering point of the sensor. Whether there is obverse damage on the root part.



- 3.measure the resistance of the throttle sensor,,the resistance between A15 and A18 is 5k Ω .In the neutral position

the resistance between A15 and A16, the resistance between A18 and A16 are around 2.5k Ω .



4.Measure any of these two pins between A15, A16 and A18, when the resistance is 0 Ω , It means there is

something wrong with the sensor,we need to replace the sensor.



5.If you replace a new potentiometer, please remember to do the auto-matching of throttle.

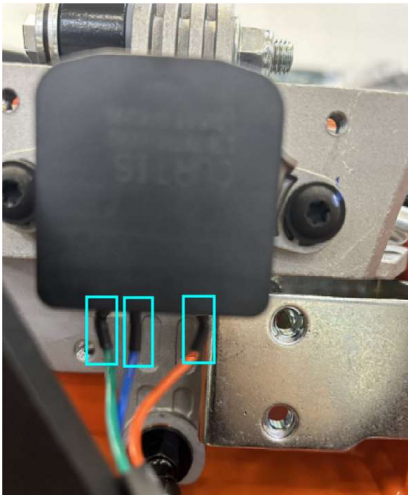
TR 43 Right Pot2 Wiper High

Tool: Computer、PCAN、 Debugging wire



Check:

- 1.Display pop up TR 43 fault code
- 2.Check the soldering point of the sensor. Whether there is obverse damage on the root part.



3.observe whether there is any damage to the outer packaging of the potentiometer wiring harness. If the wiring harness is damaged, replace the potentiometer.

4.If you replace a new potentiometer, please remember to do the auto-matching of throttle.

TR 46 Right Wheel Motor Controller NV Memory Failure

Tool: Computer、PCAN、Debugging wire



Check:

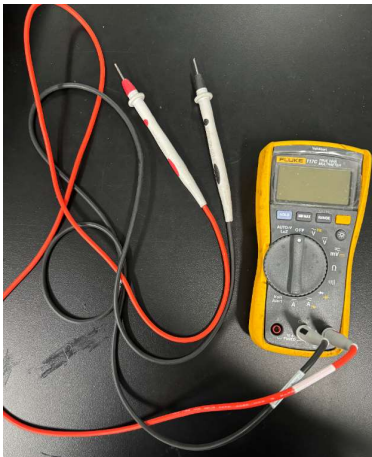
- 1.Display pop up TR 46 fault code
- 2.Restart Vehicle.
- 3.If this fault code persists, please replace a new controller.Refer to the maintenance manual for the specific disassembly and assembly steps
- 4.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 51 Display CAN Communication Abnormal

TR 52 Left Wheel Motor Controller CAN Communication Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. refer to the maintenance manual to remove the plastic parts, locate the left walking controller, observe the main control indicator light for 20 seconds, and check if the indicator light is flashing.

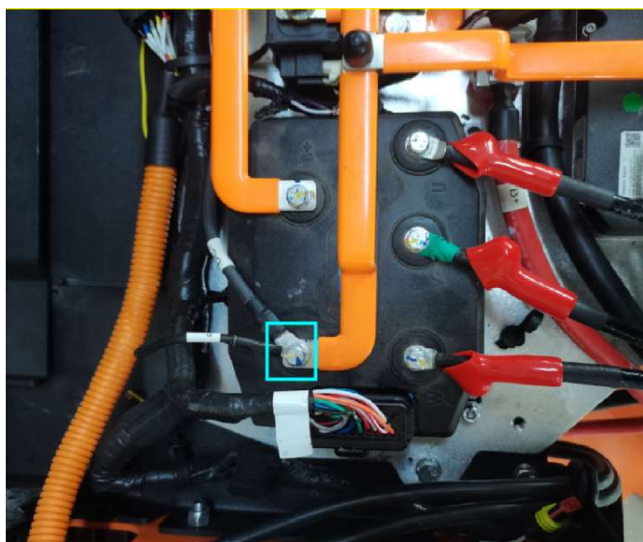


2. If the indicator light of the left walking controller flashes, it indicates that the power supply of the left walking controller is normal. Check if the left walking controller plug-in is loose, check if the left walking controller plug-in CAN-H and CAN-L have withdrawn the needle, and if there is garbage on the surface of the plug-in, which causes

poor contact of the plug-in



3.If the indicator light of the left walking controller does not flash, it indicates that the power supply of the left walking controller is abnormal. Use a wrench to confirm if the bolts at B - are tight and there are no signs of looseness.



4.If the bolts are tightened, use a multimeter to check if the voltage at point Pin 1 is greater than 70V. If the voltage at Pin 1 is less than 70V, observe whether the terminal at Pin 1 is loose or detached. If everything is normal at (1), replace the left controller



5.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

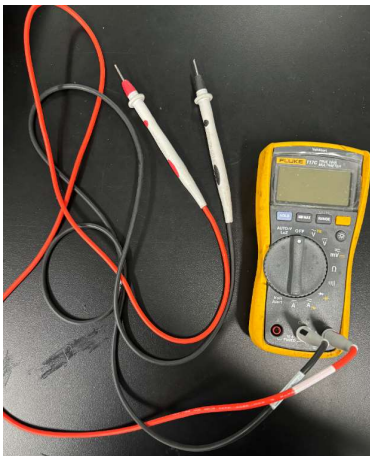
TR 53 Vehicle Initial State Detection Abnormal

Check:

1. Display pop up TR 56 fault code
2. The right parking switch is not in parked position .Please return the right lever to the park position to enable operation.
3. The left parking switch is not in parked position .Please return the left lever to the park position to enable operation.
4. The PTO Switch (Blade Switch) is not in OFF position .Please ensure PTO switch is pressed into the off position.
To enable operation.
5. Right side throttle lever is not in neutral position .Please ensure right throttle lever is in neutral position to enable operation.
6. Left side throttle lever is not in neutral position .Please ensure left throttle lever is in neutral position to enable operation.

TR 55 BMS CAN Communication Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Check if the BMS plugin is loose, check if the BMS plugin CAN-H and CAN-L have retracted needles, and if

there is any debris on the surface of the BMS plugin, causing poor contact of the plugin



2.If there is no problem with the battery pack plugin, use a multimeter to check if the CAN-H and CAN-L connections between the right walking controller plugin and the battery pack plugin are conductive. If there is no continuity, replace the main harness of the entire vehicle.



TR 56 Operator Presence Detection Abnormal

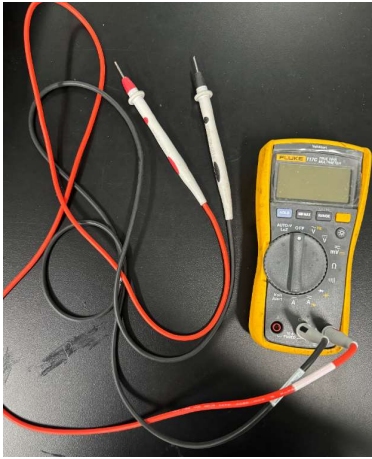
Check:

- 1.Display pop up TR 56 fault code
- 2..The operator is not in the correct operating position. .Please ensure you are in the correct operating position to resume operation.
- 3.Right side throttle lever is not in neutral position. .Please ensure right throttle lever is in returned to park position to resume operation.
- 4.Left side throttle lever is not in neutral position. .Please ensure left throttle lever is in returned to park position

to resume operation.

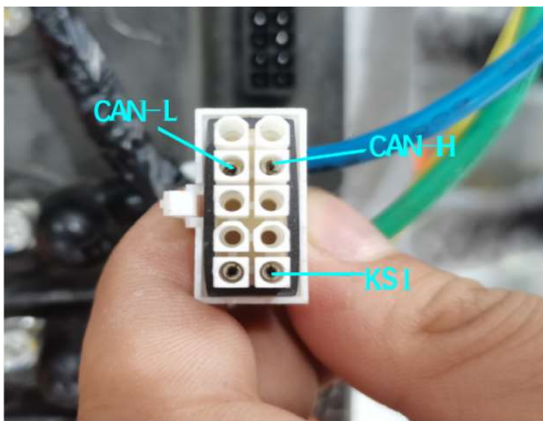
TR 58 Right Blade Motor Controller CAN Communication Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



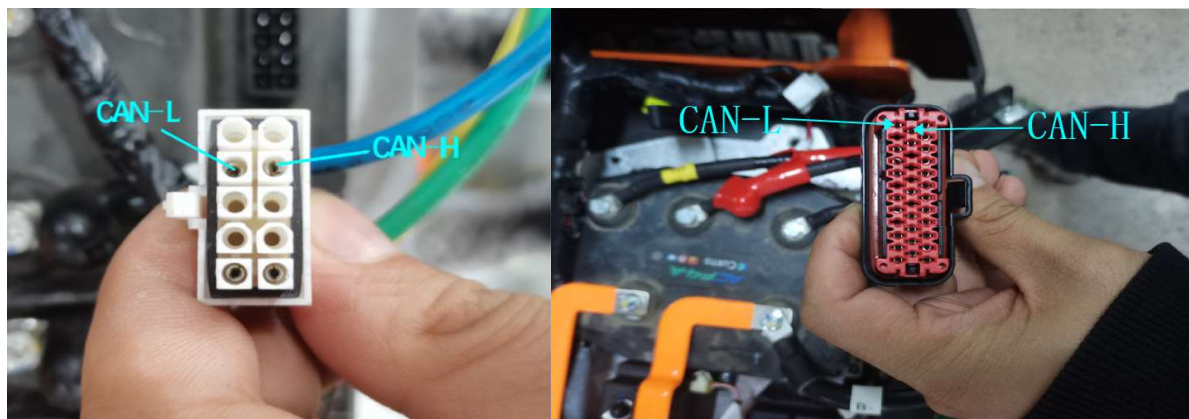
Check:

1. Display pop up TR 58/63/65 fault code
2. Check whether the Trinity cutter controller plug-in is loose, check whether the Trinity cutter controller plug-in CAN-H, KAN-L, KSI, and the three-in-one cutter controller plug-in surface has garbage, resulting in poor contact with the plug-in

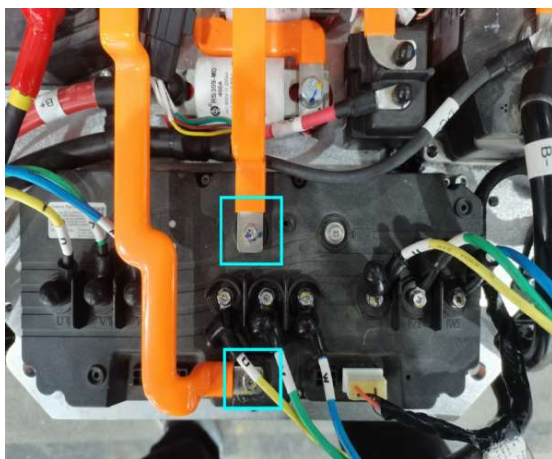


3. If this fault code persists, Use the multimeter to check whether the CAN-H, CAN-L conductivity. If not

conductivity, replace the vehicle main line harness.



4.restart the vehicle. If the fault continues to occur,Check the connection line on the controller and tighten the loose bolts



5.Use the multimeter voltage range to check whether the voltage at the KSI pin is greater than 70V. If less than 70V, replace the main line bundle. If it is greater than 70V, brush the Trinity cutter controller program .

6.restart the vehicle. If the fault continues to occur replace the Trinity cutter controller

7.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 5A Internal communication failure

Tool: Computer、PCAN、Debugging wire



Check:

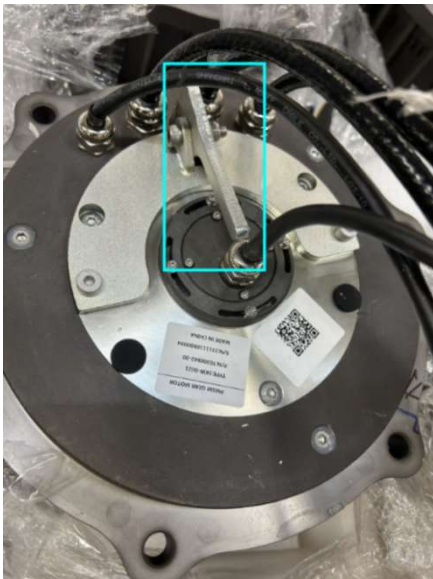
1. Display pop up TR **5A** fault code
2. keep the vehicle power off state, check whether wheels can be turned freely.
3. If the wheels cannot rotate, Please update the controller program
4. If this fault code persists, please replace a new controller. Refer to the maintenance manual for the specific disassembly and assembly steps
5. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 5B Right Parking Brake Manually Disengaged and Not Reset

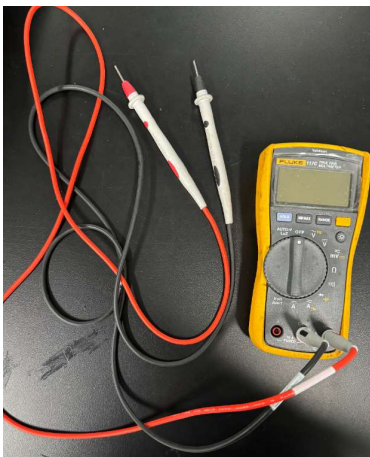
Check:

1. Display pop up TR 5B fault code
2. please to keep the brake lever at unreleased state.



TR 63 Left Blade Motor Controller CAN Communication Abnormal

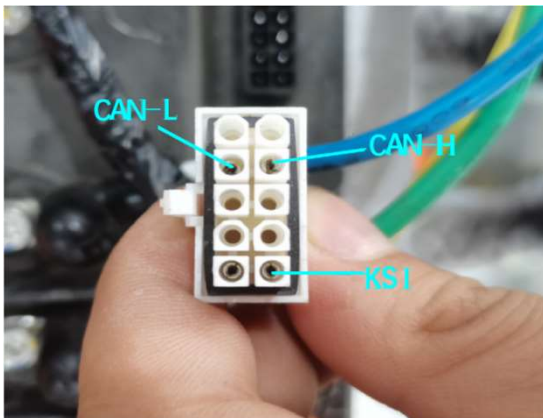
Tool: Multimeter、Computer、PCAN、Debugging wire



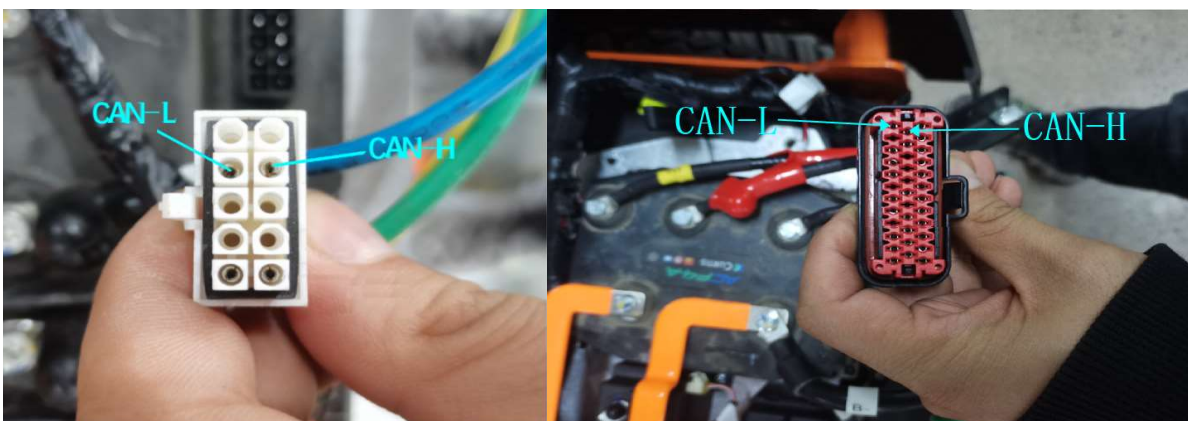
Check:

1. Display pop up TR 58/63/65 fault code
2. Check whether the Trinity cutter controller plug-in is loose, check whether the Trinity cutter controller plug-in CAN-H, KAN-L, KSI, and the three-in-one cutter controller plug-in surface has garbage, resulting in poor contact

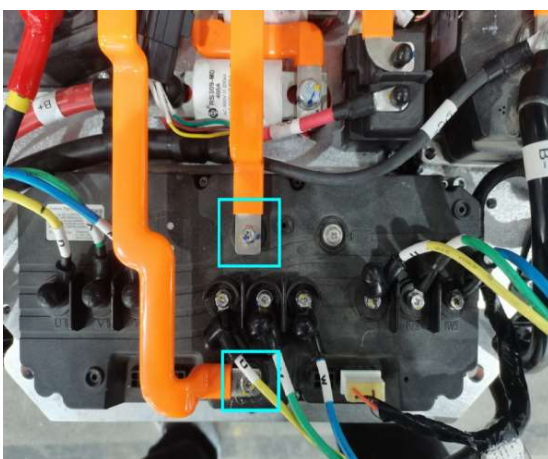
with the plug-in



3.If this fault code persists, Use the multimeter to check whether the CAN-H, CAN-L conductivity. If not conductivity, replace the vehicle main line harness.



4.restart the vehicle. If the fault continues to occur,Check the connection line on the controller and tighten the loose bolts



5.Use the multimeter voltage range to check whether the voltage at the KSI pin is greater than 70V. If less than 70V, replace the main line bundle. If it is greater than 70V, brush the Trinity cutter controller program .

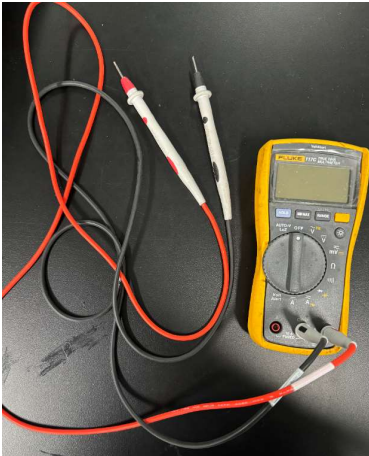
6.restart the vehicle. If the fault continues to occur replace the Trinity cutter controller

7.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

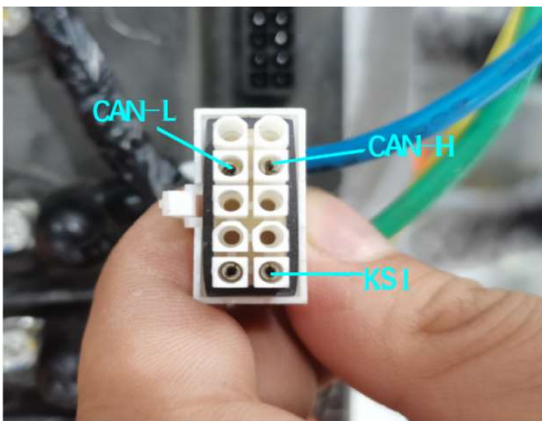
TR 65 Middle Blade Motor Controller CAN Communication Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



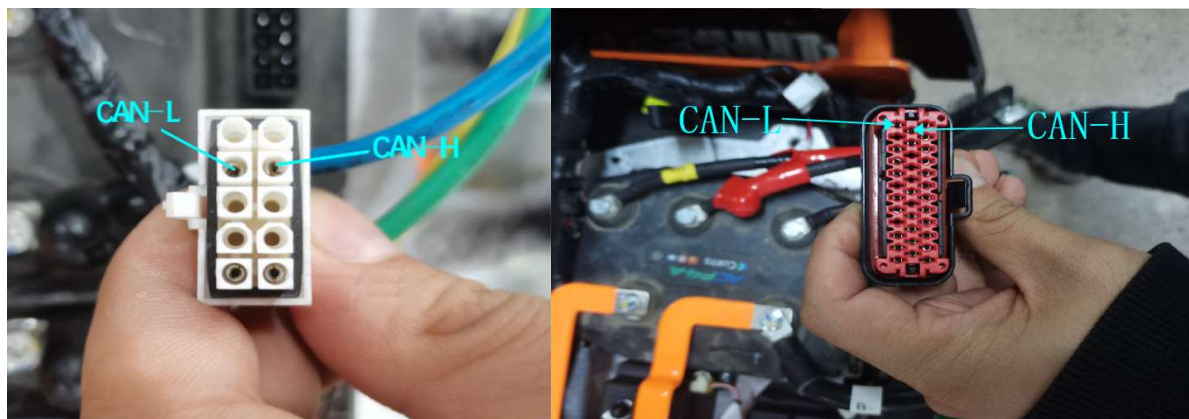
Check:

1. Display pop up TR 58/63/65 fault code
2. Check whether the Trinity cutter controller plug-in is loose, check whether the Trinity cutter controller plug-in CAN-H, KAN-L, KSI, and the three-in-one cutter controller plug-in surface has garbage, resulting in poor contact with the plug-in

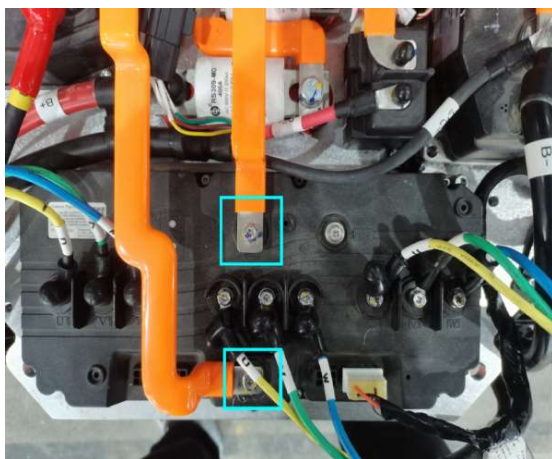


3. If this fault code persists, Use the multimeter to check whether the CAN-H, CAN-L conductivity. If not

conductivity, replace the vehicle main line harness.



4.restart the vehicle. If the fault continues to occur,Check the connection line on the controller and tighten the loose bolts



5.Use the multimeter voltage range to check whether the voltage at the KSI pin is greater than 70V. If less than 70V, replace the main line bundle. If it is greater than 70V, brush the Trinity cutter controller program .

6.restart the vehicle. If the fault continues to occur replace the Trinity cutter controller

7.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

TR 64/66 Second Left/Right Blade Motor Controller CAN Communication Abnormal

TR 67 Blade speed exceeds limit

Tool: Computer、PCAN、Debugging wire



Check:

1. Display pop up TR 67 fault code
2. Restart Vehicle.
3. If this fault code persists, please replace a new controller. Refer to the maintenance manual for the specific disassembly and assembly steps
4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

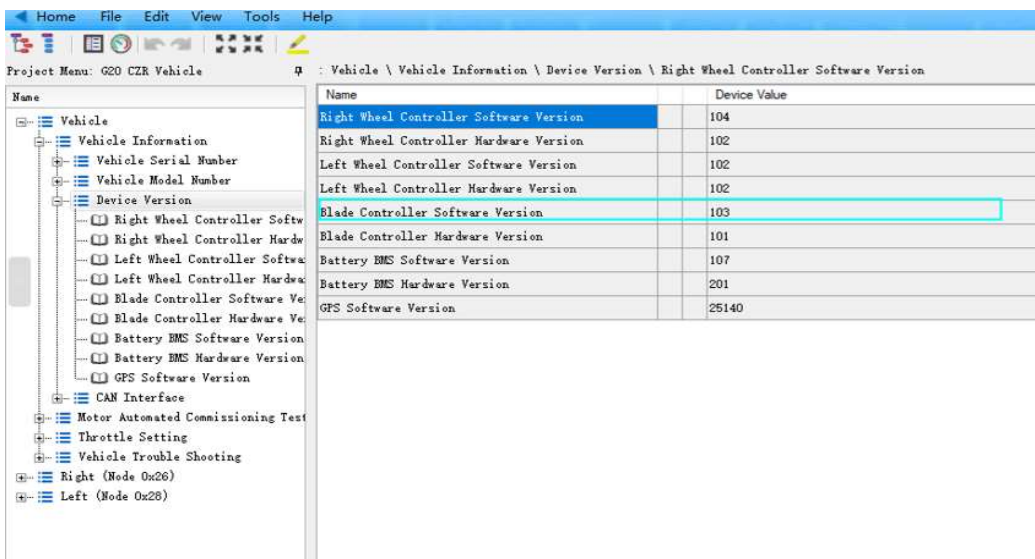
TR 73/TL 73 Right/Left Wheel Motor Stall Detected

Tool: Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR 73/TL 73 fault code
- 2.Use the CURTIS controller application Check whether the controller software version is correct, If the version is incorrect, rewrite the correct version.



Name	Device Value
Right Wheel Controller Software Version	104
Right Wheel Controller Hardware Version	102
Left Wheel Controller Software Version	102
Left Wheel Controller Hardware Version	102
Blade Controller Software Version	103
Blade Controller Hardware Version	101
Battery EMS Software Version	107
Battery EMS Hardware Version	201
GPS Software Version	25140

- 3.restart the vehicle. If the fault continues to occur,If this fault code persists, please replace a new blade controller.Refer to the maintenance manual for the specific disassembly and assembly steps
- 4.restart the vehicle. If the fault continues to occur.Check whether the motor power is insufficient.If the vehicle in the downhill, due to the motor power is not enough, resulting in motor stall. the Blade Motor needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps
- 5.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor

auto-matching test.

TR 83 Right Wheel Motor Controller Internal Hardware

Tool: Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR 83 fault code
- 2.Restart Vehicle.
- 3.If this fault code persists, please replace a new controller.Refer to the maintenance manual for the specific disassembly and assembly steps
- 4.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

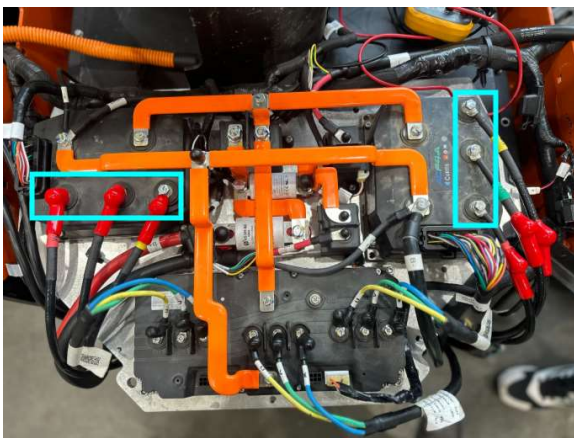
TR 87 Right Wheel Motor Controller Motor Characterization Fault

Tool: Computer、PCAN、 Debugging wire

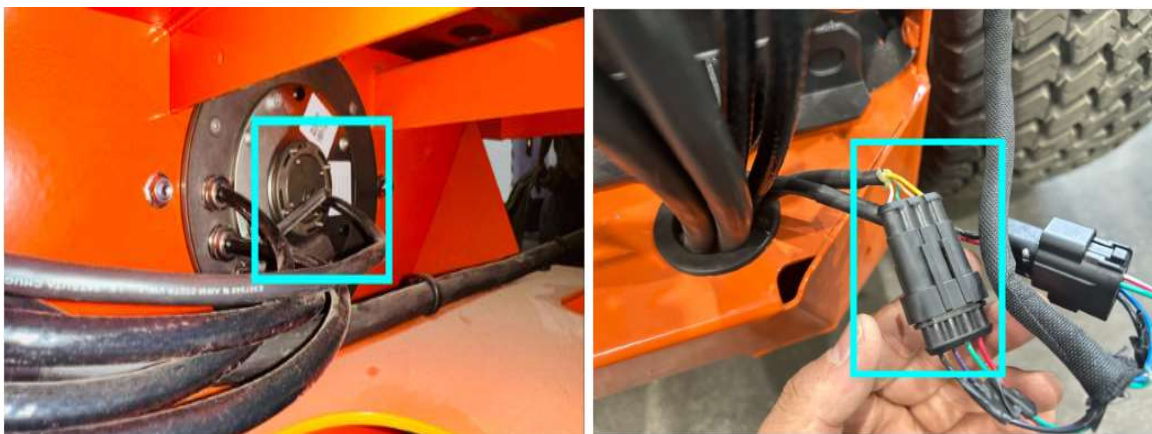


Check:

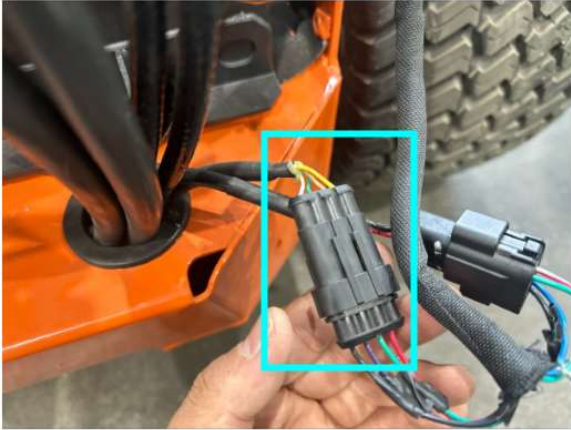
- 1.Display pop up TR 87 fault code
- 2.Please check that the phases of the motor are well connected to controller.



- 3.Please check that the encoder of the motor is well assembled, including installation and connectors..



4. Please check whether the connector of encoder is well connected.

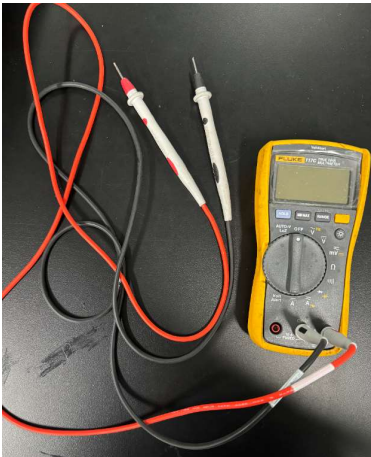


5. Run the appropriate motor commissioning sequences.

6. If you have reassembled the encoder, remember to follow the self-learning steps for the motor auto-matching test.

TR A1 Rear ETO Contactor Coil Driver Short/Open

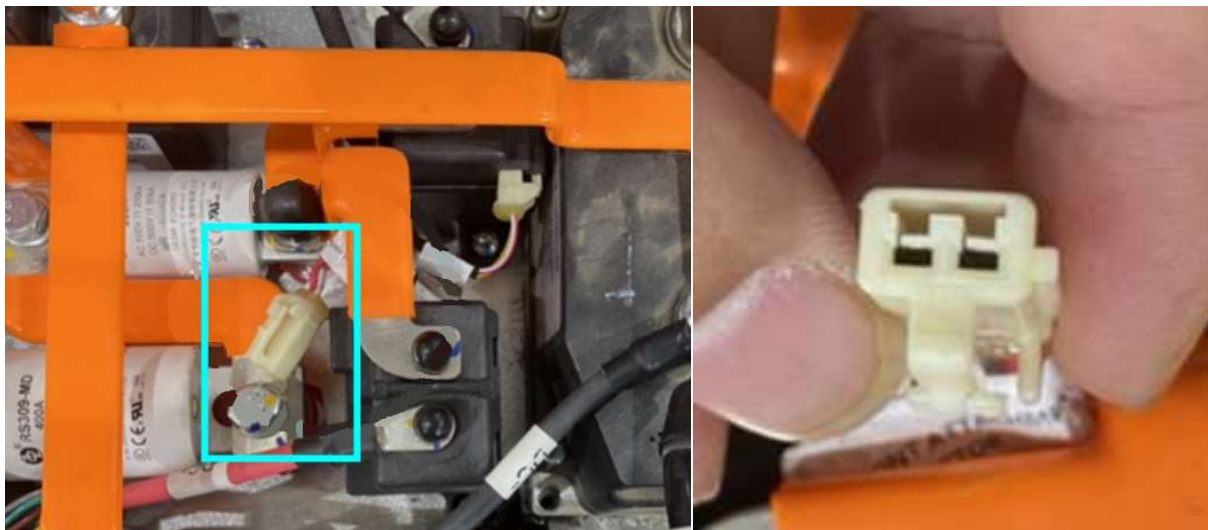
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR A1 fault code
2. First pull out the communication plug-in on the rear ETO relay, check whether the terminal in the connector

is out of the pin, if the pin is out, please deal with it, and insert the connector into the relay.



3. Power on the device again, if the fault does not disappear, pull out the main control communication plug-in and the relay, and measure whether the left PIN foot of the connector on the relay is connected to the 4 PIN foot on the main control plug-in.

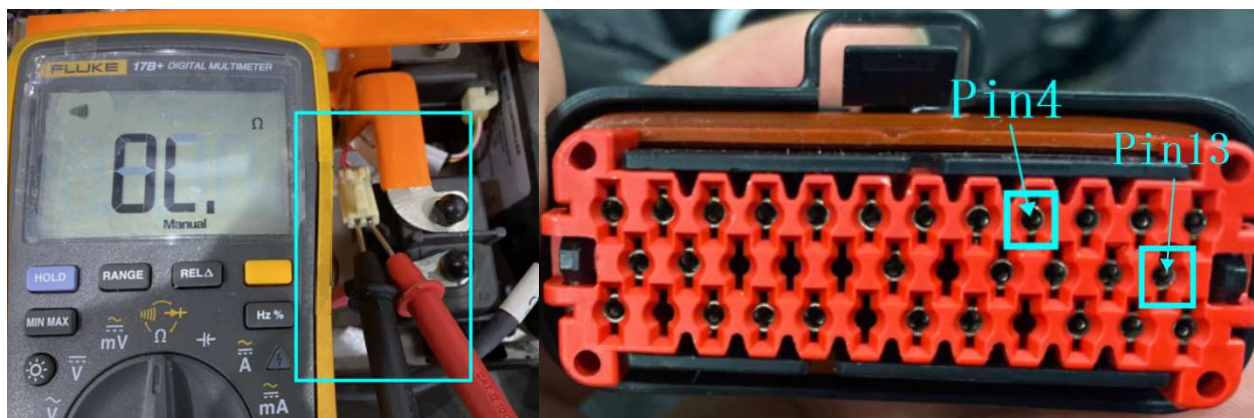


4. pull out the main control communication plug-in and the relay, and measure whether the right PIN foot of the connector on the relay is connected to the 13 PIN foot on the main control plug-in.



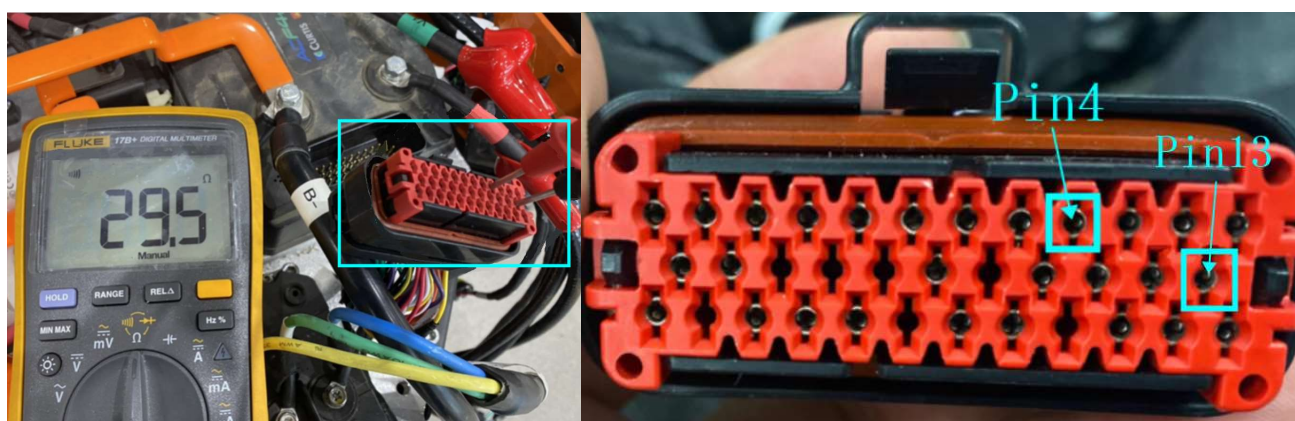
5. Pull out the main control communication plug and the plug on the relay. Use a multimeter to select the on/off

gear to measure whether the left and right PIN pins of the connector on the relay are connected, or measure the 4PIN and 13PIN pins on the main control communication plug. If there is a path on both sides,



6.If the left and right ends are paths, it means that the main line bundle is damaged and needs to be replaced.

7.The multimeter is set to on-off, plug in the plug-in on the relay, and measure whether the 4 PIN feet and 13 PIN feet on the master communication plug-in are disconnected. If the circuit is open, the front ETO relay is damaged and needs to be replaced.



8.Insert the main control communication plug, unplug the relay plug, use a multimeter diode gear, connect the red probe on the left side of the plug, and connect the black probe on the right side of the plug. If there is no value,

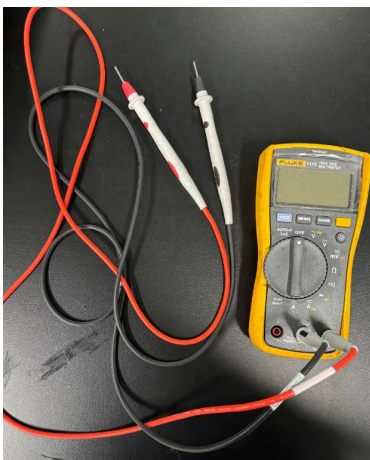
the controller is damaged and needs to be replaced.



4.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching test.

TR A2 Right Wheel Motor Controller EM Brake Driver Short/Open

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR A2 fault code
- 2.Pull out the electromagnetic brake communication plug and the main controller communication plug on the right drive motor, check if there is a pin on the connector that returns the PIN, and check if the PIN pin and the

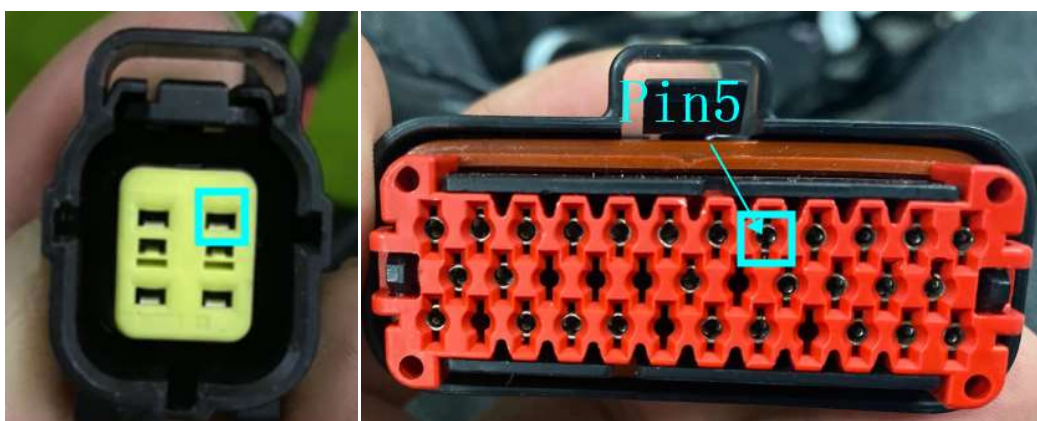
connector can be fixed.



3.Reinsert the plugin and restart the vehicle. if the fault still persists.Use a multimeter to turn on and off the gear, and measure whether the left pin in the first row is connected to the 13PIN pin on the main control communication plug.



4.Use a multimeter to turn on and off the gear, and measure whether the pin on the right side of the first row is connected to the 5PIN pin on the main control communication plug.



5.Measure the resistance between the left and right pins of the first row of the electromagnetic brake plug on the drive motor using a multimeter in the ohm range. If it does not exist or approaches zero infinitely, it indicates that

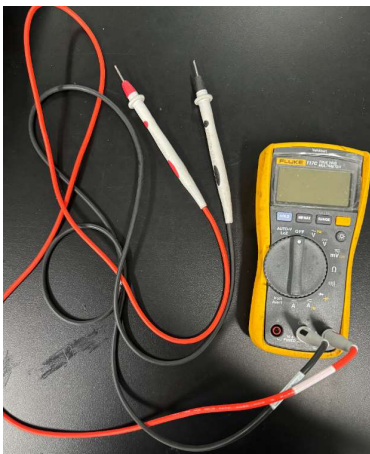
the electromagnetic brake on the drive motor is damaged and needs to be replaced.



6.If you have replaced a new motor, remember to follow the self-learning steps for the motor auto-matching test.

TR A3 Front ETO Contactor Coil Driver Short/Open

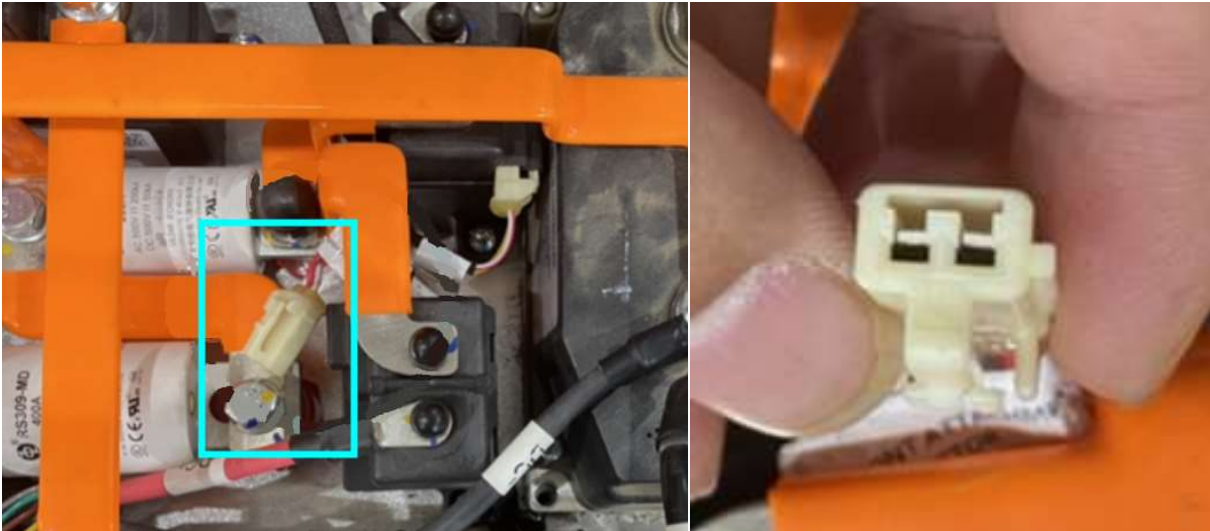
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR A3 fault code
- 2.First pull out the communication plug-in on the rear ETO relay, check whether the terminal in the connector

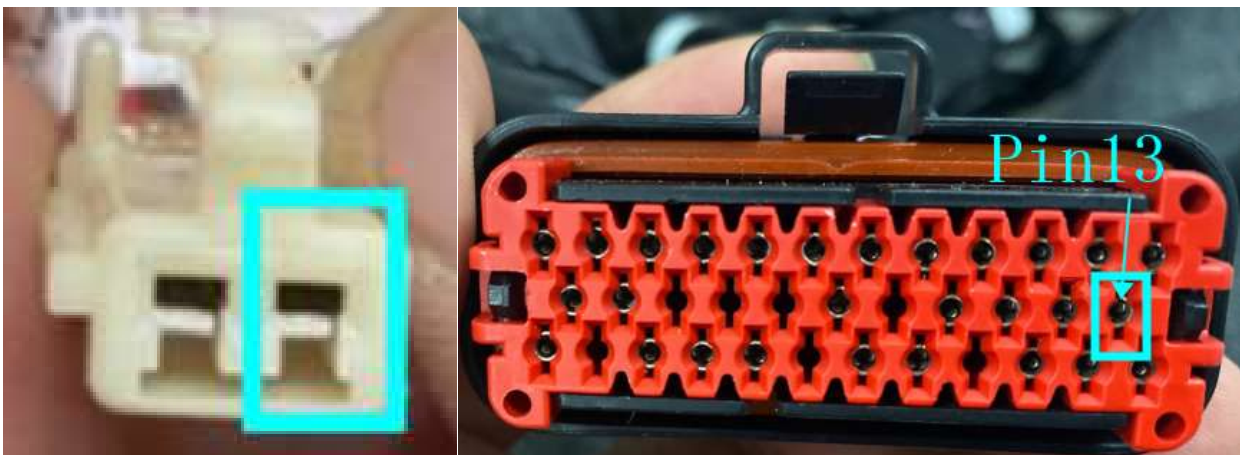
is out of the pin, if the pin is out, please deal with it, and insert the connector into the relay.



3.Power on the device again,if the fault does not disappear.pull out the main control communication plug-in and the relay, and measure whether the left PIN foot of the connector on the relay is connected to the 4 PIN foot on the main control plug-in.

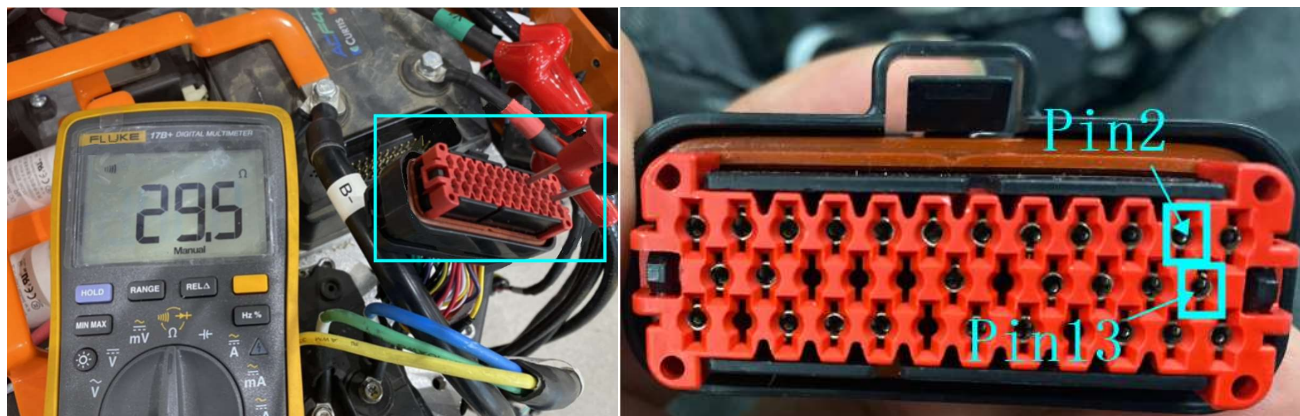


4.pull out the main control communication plug-in and the relay, and measure whether the right PIN foot of the connector on the relay is connected to the 13 PIN foot on the main control plug-in.



6.The multimeter is set to on-off, plug in the plug-in on the relay, and measure whether the 2 PIN feet and 13

PIN feet on the master communication plug-in are disconnected. If the circuit is open, the front ETO relay is damaged and needs to be replaced.



7. Insert the main control communication plug, unplug the relay plug, use a multimeter diode gear, connect the red probe on the left side of the plug, and connect the black probe on the right side of the plug. If there is no value, the controller is damaged and needs to be replaced.

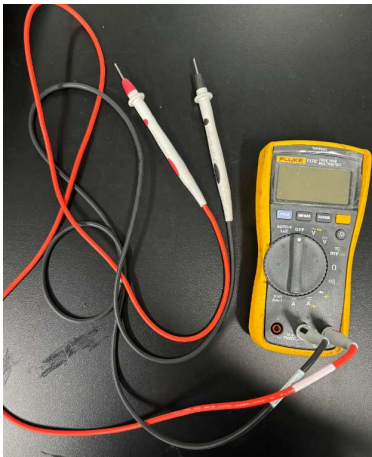


4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

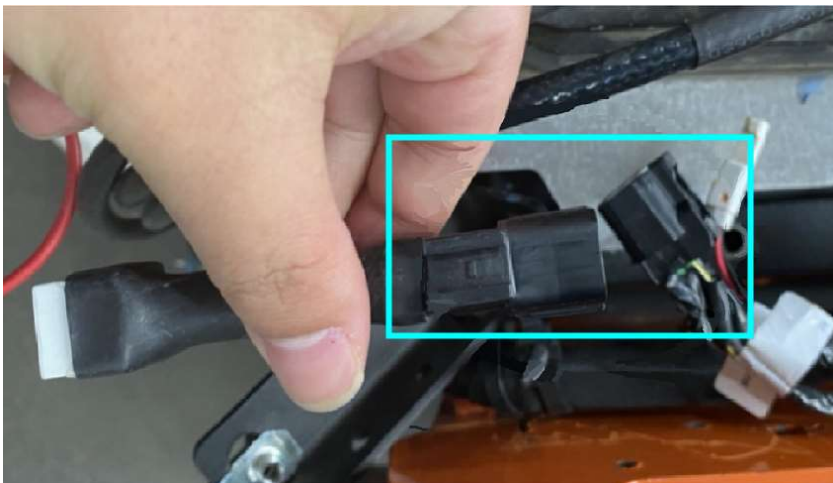
TR A4 Auto power off Contactor Coil Driver Short/Open

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up TR A4 fault code
2. Unplug the one click start relay from the vehicle wiring harness and check if there is a PIN retraction phenomenon at the terminals in the connector. If there is, reinsert the PIN pin into the connector and check if the terminals still fall off.



3. Reinsert the plugin and restart the vehicle if the fault still persists. Pull out the communication plug between the relay and the main control, and use a multimeter to turn on and off the gear. Measure whether the left PIN pin on the second row of the connector between the wire harness end and the relay is connected to the 3PIN pin on the

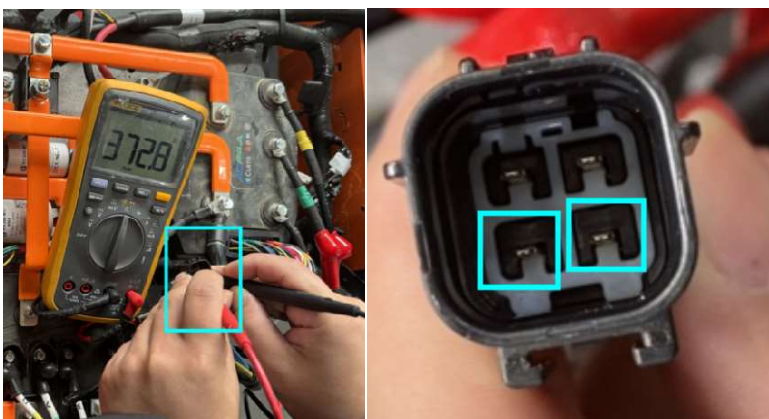
main control communication plug.



4.Pull out the communication plug between the relay and the main control. Use a multimeter to turn on and off the gear. Measure whether the second row right PIN pin on the connector between the wire harness end and the relay is connected to the 13PIN pin on the main control communication plug.

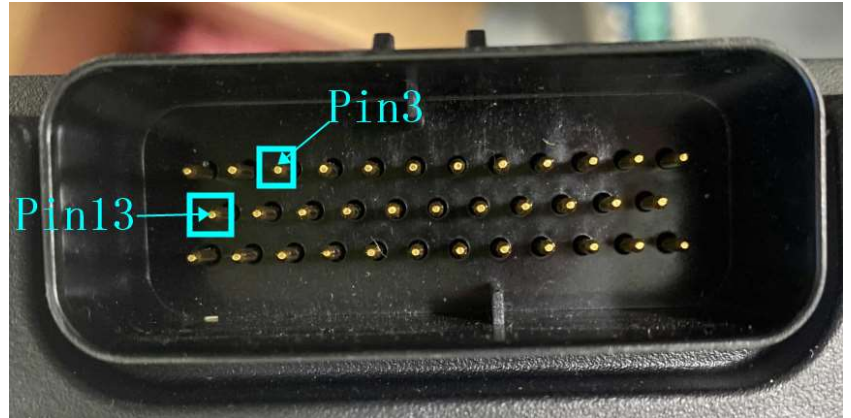
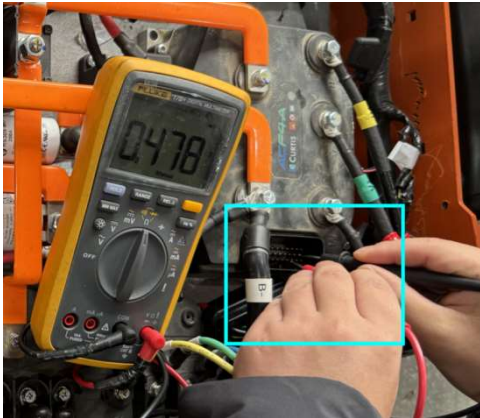


5.Using a multimeter ohmmeter, measure whether there is resistance value on the left and right sides of the second row of the connector on the relay. If not, it indicates that the one click start relay is damaged and needs to be replaced.



6.Unplug the main control communication plugin. Using a multimeter diode range, connect the red probe to pin 3 of the main controller and the black probe to pin 13 of the main controller. If the diode is not present, it indicates

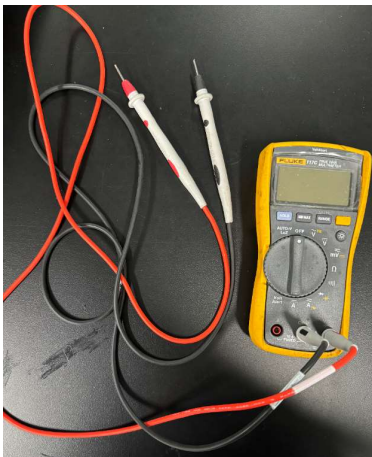
that the controller is damaged and needs to be replaced.



4.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching test.

TR A9 Right Wheel Motor Controller Coil Supply

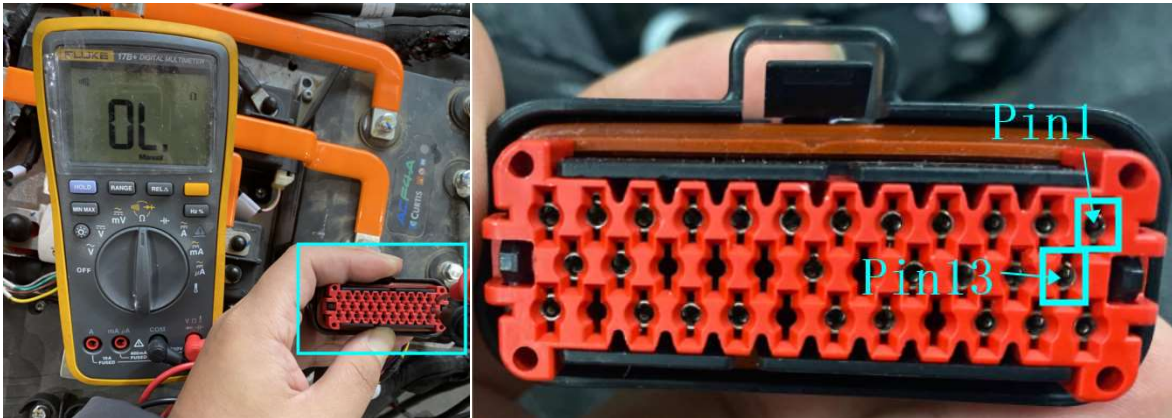
Tool: Multimeter、Computer、PCAN、Debugging wire



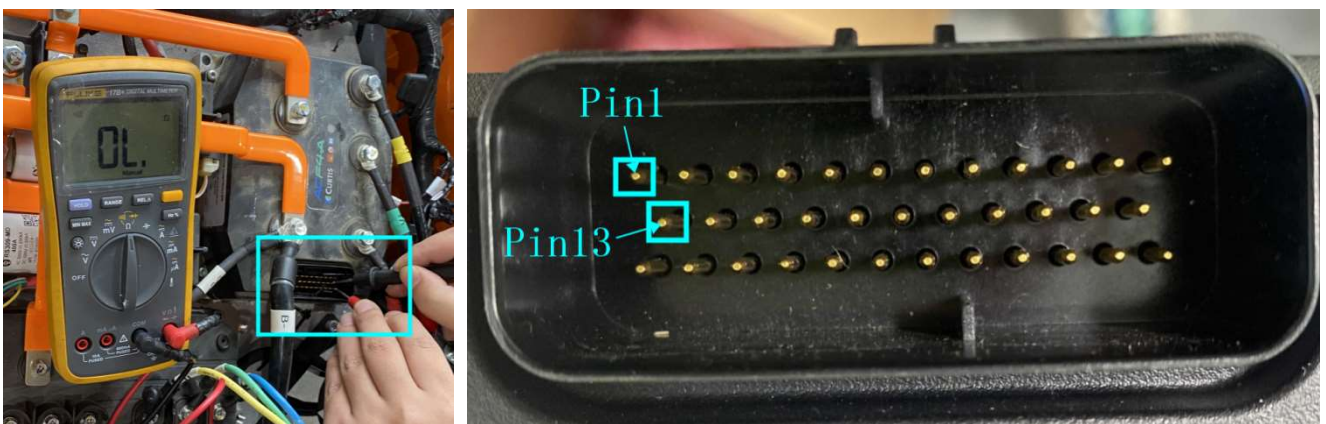
Check:

- 1.Display pop up TR A9 fault code.
- 2.Unplug the main controller communication plug, use a multimeter to turn on and off the gear, and measure whether pins 1 and 13 on the main controller communication plug are connected. If pins 1 and 13 are connected,

there is a problem with the main harness and it needs to be replaced.



3. Use a multimeter to turn on and off the gear. Measure whether pins 1 and 13 on the controller are connected. If pins 1 and 13 on the controller are connected, the main controller is damaged and needs to be replaced.



4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

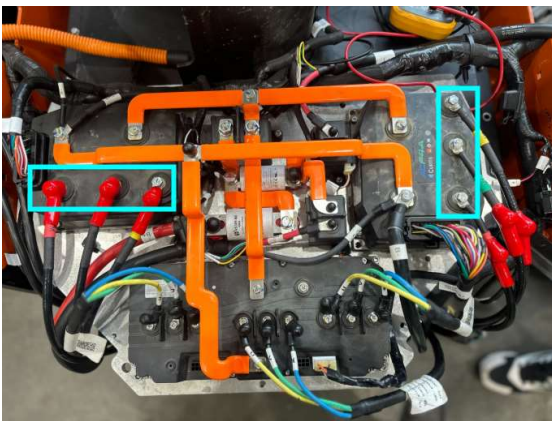
TR D2 Right Wheel Motor Controller Phase PWM Mismatch

Tool: Computer、PCAN、Debugging wire



Check:

- 1.Display pop up TR D2 fault code
- 2.Please check that the phases of the motor are well connected to controller.



- 3.Please check whether the encoder is well assembled.



- 4.Please check whether the connector of encoder is well connected.

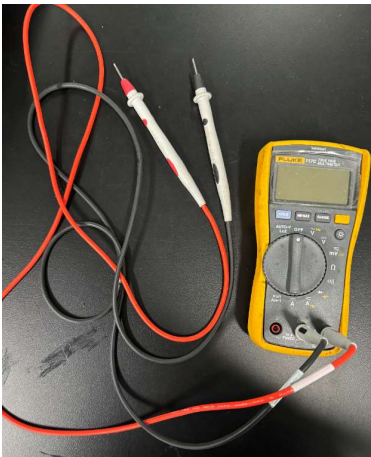


5.Run the appropriate motor commissioning sequences.

6.If you have reassembled the encoder, remember to follow the self-learning steps for the motor auto-matching test.

ML 11 /MM 11 /MR 11 Left/Middle/Right Blade Motor Controller Hardware Overvoltage or Overcurrent

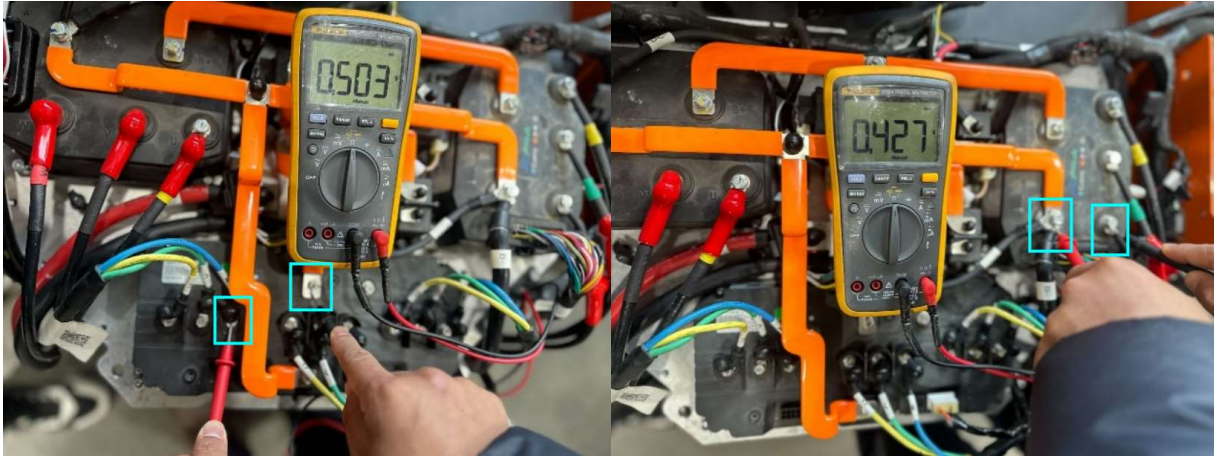
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up ML 11 /MM 11 /MR 11 fault code
- 2.If a malfunction occurs when the vehicle is powered on
- 3.Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the positive

electrode.



4.Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the negative electrode.



5. If the range is 4.7-5.0V, the MOS is normal. Otherwise, it is judged that the MOS is damaged and the controller needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps.

6. If Normal when the vehicle is powered on,Fault occurs after pulling up the PTO switch,keep the vehicle power off state.Check for foreign objects under the header

7.raise the header and manually move the blades to check if the motor can rotate smoothly. If the motor is

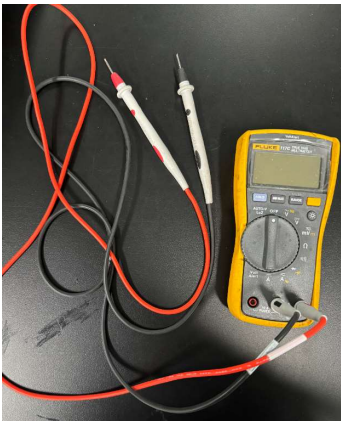
noticeably stuck, it is necessary to replace the cutter motor.



8.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor auto-matching test.

ML 12/MM 12/ML 12 Right/Middle/Left Blade Motor Controller Overcurrent

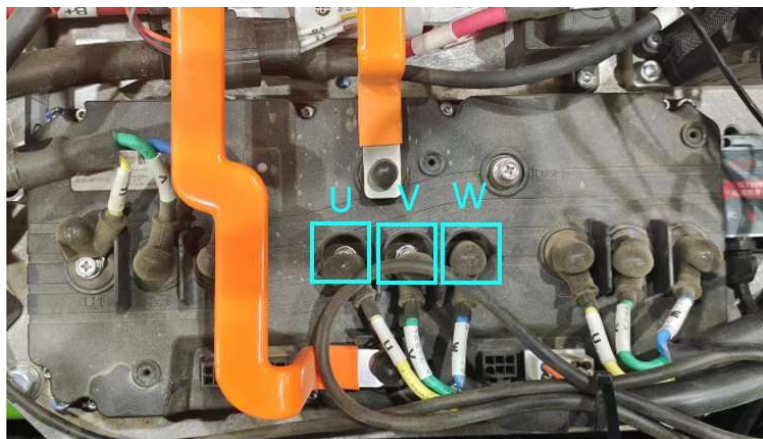
Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

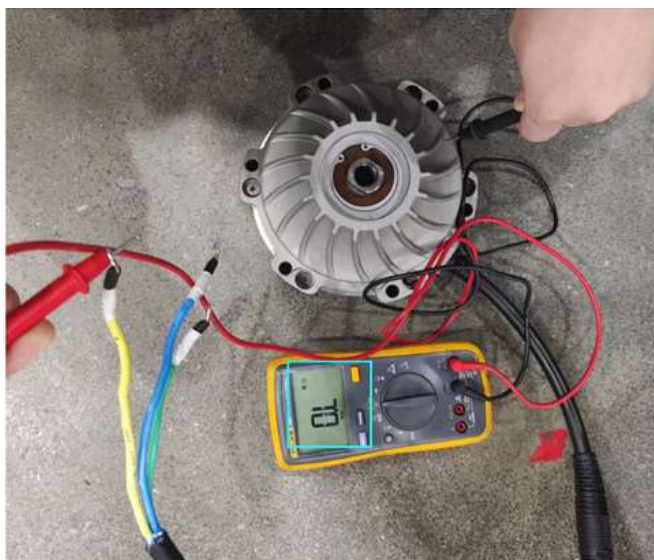
- 1.Display pop up ML 12/MM 12/ML 12 fault code
- 2.Move the vehicle to an open area without grass, restart the cutter motor.If the Fault disappears, the cause of the fault is the controller heating caused by the overload of the cutter motor, should be reduce mowing density.
- 3.Check whether the insulation skin of motor U \ V \ W phase wire is broken and copper wire is together at the damage,If the wiring harness is damaged, dispose of the damaged area.
- 4.Remove the U/V/W phase line of the motor controller,Measure the resistance value between U \ V \ W phases

and the motor housing using a multimeter resistance range.



5.If the resistance value is less than $20\text{M } \Omega$, there is a short circuit between the internal winding and the casing of the motor, and the motor needs to be replaced.

6.If the resistance measured by the motor phase line and the housing is infinite, observe whether there is any damage to the outer packaging of the motor encoder's wiring harness. If the wiring harness is damaged, replace the motor encode.



7.If the wiring harness of the motor encoder is intact, measure the resistance value between the controller U \ V \ W phases. If the value displayed on the multimeter is below $10\text{ } \Omega$, it indicates that the diode has broken down and

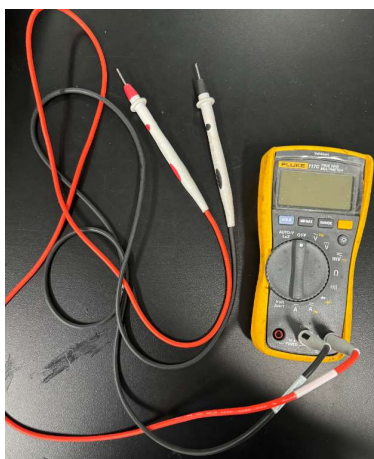
caused a short circuit, and a new motor controller needs to be replaced.



8.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor auto-matching test.

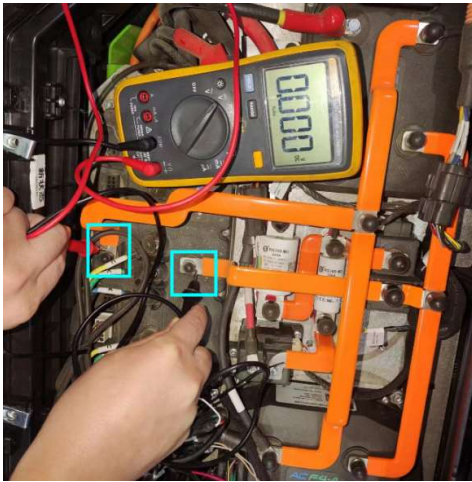
ML 13/MM 13/MR 13 Left/Middle/Right Blade Motor Controller Overvoltage

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

- 1.Display pop up ML 13/MM 13/MR 13 fault code
- 2.Restart Vehicle.
- 3.If this fault code persists, Vehicle power on Please use a multimeter to measure the bus DC voltage of the controller



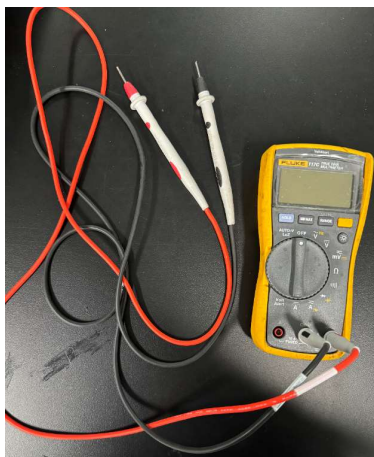
4.If the value displayed on the multimeter is less than 96V,it indicates that the internal voltage sensor of the motor controller is faulty and the controller needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps.

5.If the value displayed on the multimeter is greater than 96V,it indicates a battery failure and the battery system

needs to be checked,

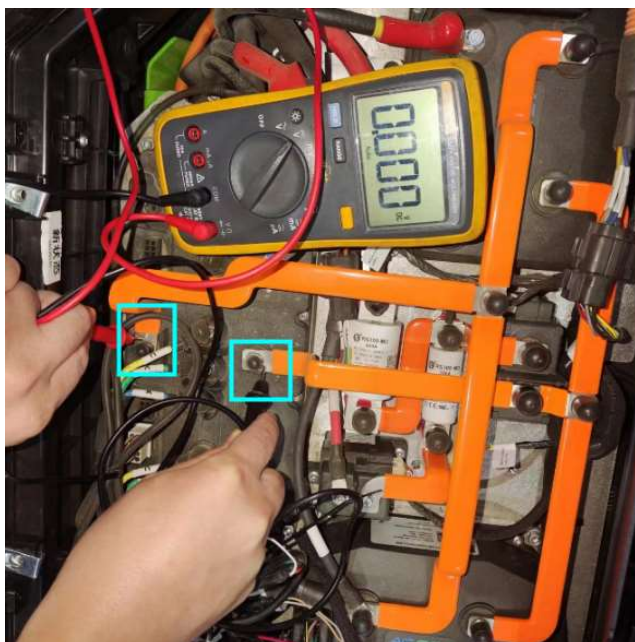
ML 14/MM 14/MR 14 Left/Middle/Right Blade Motor Controller Undervoltage

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up ML 14/MM 14/MR 14 fault code
2. Restart Vehicle.
3. If this fault code persists, Please use a multimeter to measure the bus DC voltage of the controller,



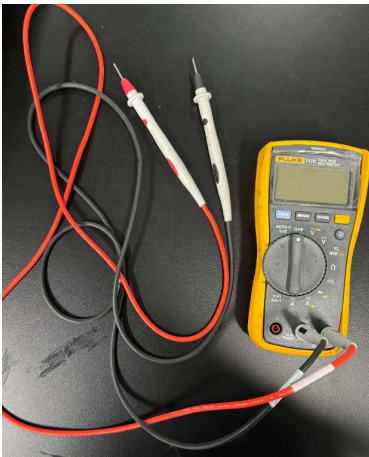
4. If the value displayed on the multimeter is greater than 72V, it indicates that the internal voltage sensor of the motor controller is faulty and the controller needs to be replaced..Refer to the maintenance manual for the specific

disassembly and assembly steps.

5.If the value displayed on the multimeter is less than 72V, it indicates a battery failure and the battery system needs to be checked,

ML 16/MM 16/MR 16 Left/Middle/Right Blade Motor Phase Open

Tool: Multimeter、Computer、PCAN、Debugging wire

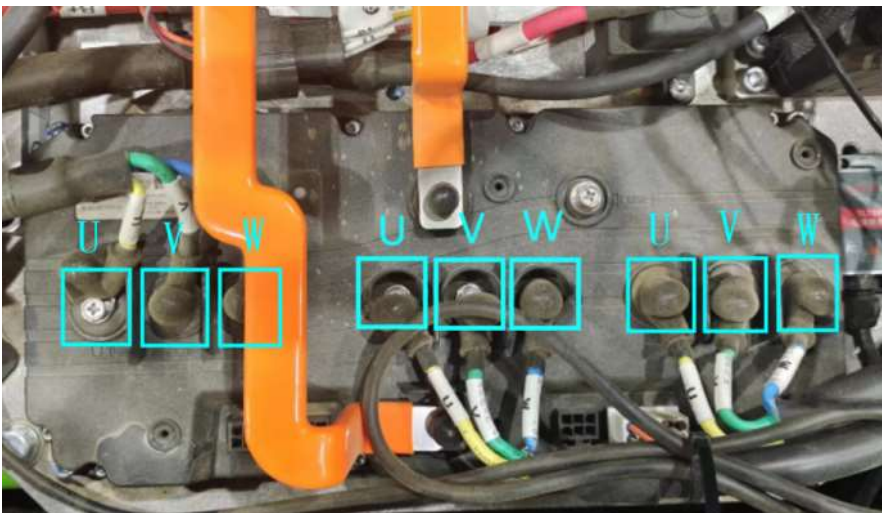


Check:

1.Display pop up ML 16/MM 16/MR 16 fault code

2.,Check whether the motor phase line is reliably connected to the controller. Check the positions of the six screws.

If they are loose. tighten the loose bolts and Check if screws are found to be damaged. Replace the new screws.if the controller screw hole is damaged and the controller needs to be replaced



3. restart the vehicle. If the fault continues to occur measure the resistance value between the motor U \ V \ W phases. If the value displayed on the multimeter is 0 or infinity it means that the motor has been damaged inside

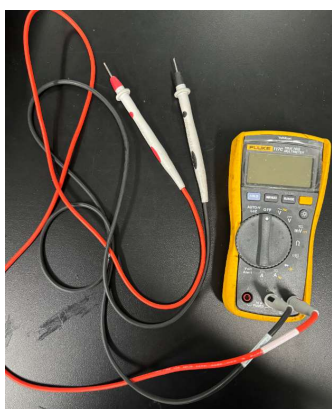
and a new blade motor needs to be replaced.



4.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor auto-matching test.

ML 17/MM 17/MR 17 Left/Middle/Right Blade Motor Controller Severe Overtemp

Tool: Infrared Thermometers、Multimeter、Computer、PCAN、Debugging wire

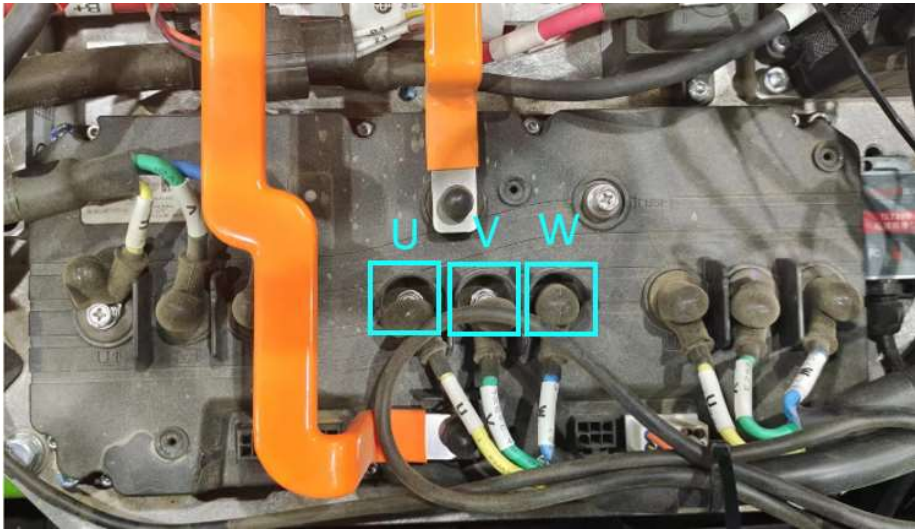


Check:

- 1.Display pop up ML 17/MM 17/MR 17 fault code
- 2.Move the vehicle to an open area without grass, restart the cutter motorIf the Fault disappears, the cause of

the fault is the controller heating caused by the overload of the cutter motor, should be reduce mowing density.

3. restart the vehicle. If the fault continues to occur,Check the connection of the motor U/V/W phase line on the controller and tighten the loose bolts.



4. restart the vehicle. If the fault continues to occur. Measure the temperature of the blade controller, and read the feedback temperature of the blade control through the upper computer ToolsForCAN-PlatformChecker for comparison, If the difference is significant it indicates that the internal temperature sensor of the controller is faulty and needs to be replaced with a new cutting blade controller.

ToolsForCAN-PlatformChecker - Made by Mason Xu Release at 20231127

左控制器硬件版本:	右控制器硬件版本:
左电机转速rpm:	右电机转速rpm:
左电机相电流A:	右电机相电流A:
左电机温度:	右电机温度:
左控制器温度:	右控制器温度:
故障代码:	故障代码:

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左控制器软件版本:	右控制器软件版本:
左控制器硬件版本:	右控制器硬件版本:
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左控制器温度:	右控制器温度:
故障代码:	故障代码:

车型:

GPS软件版本:

整车工作时h:

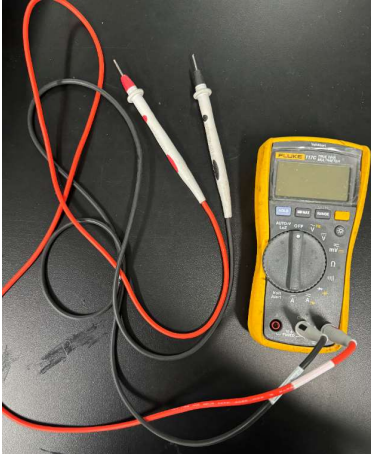
割刀工作时h:

5.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

ML 18 /MM 18 /MR 18 Left/Middle/Right Blade Motor Controller EEPROM Abnormal

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display pop up ML 18/MM 18/MR 18 fault code
2. Restart Vehicle.
3. If this fault code persists, please replace a new blade controller. Refer to the maintenance manual for the specific disassembly and assembly steps
4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

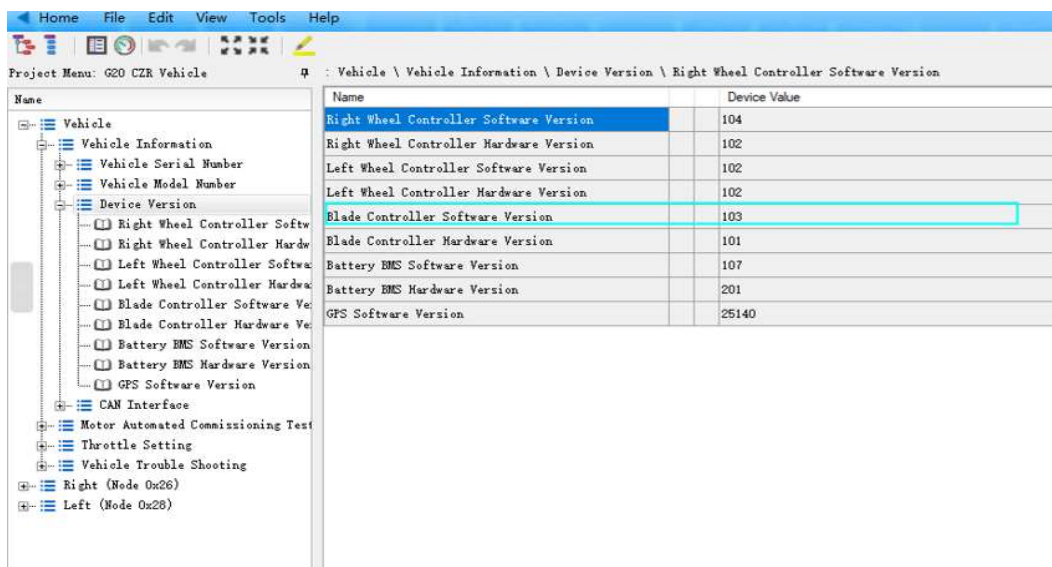
ML 21 /MM 21 /MR 21 Left/Middle/Right Blade Motor Stall Detected

Tool: Computer、PCAN、 Debugging wire



Check:

- 1.Display pop up ML 21 /MM 21 /MR 21 fault code
- 2.Use the CURTIS controller application Check whether the controller software version is correct, If the version is incorrect, rewrite the correct version.



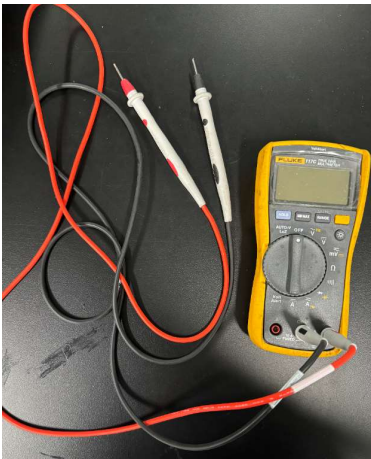
- 3.restart the vehicle. If the fault continues to occur,If this fault code persists, please replace a new blade controller.Refer to the maintenance manual for the specific disassembly and assembly steps
- 4.restart the vehicle. If the fault continues to occur.Check whether the motor power is insufficient.If the vehicle in the downhill, due to the motor power is not enough, resulting in motor stall. the Blade Motor needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps
- 5.If you have replaced a new controller or a new motor, remember to follow the self-learning steps for the motor

auto-matching test.

ML 24 CAN Communication with Right Wheel Motor Controller Abnormal

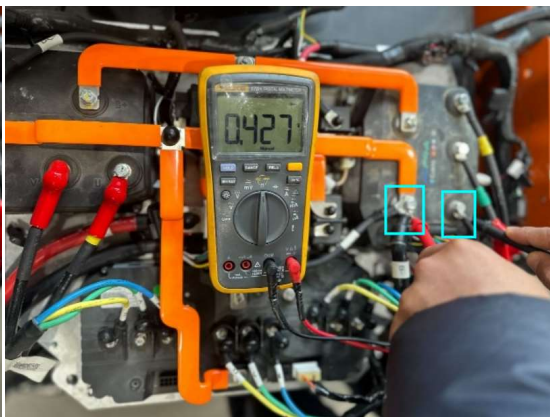
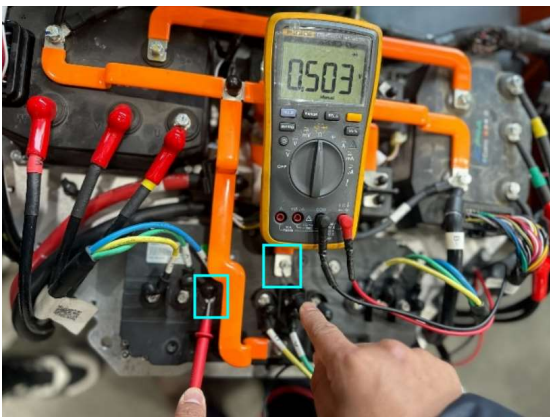
ML 23/MM 23/MR 23 Left/Middle/Right Blade Motor Controller Precharge Failed

Tool: Multimeter、Computer、PCAN、Debugging wire



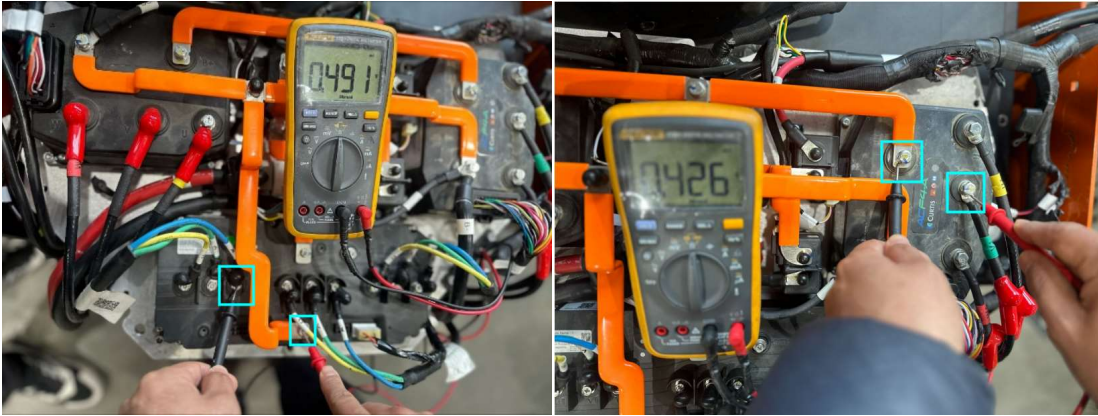
Check:

1. Display pop up ML 23/MM 23/MR 23 fault code
 2. keep the vehicle power off state.
 3. Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the positive electrode.
- electrode.



4. Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the negative

electrode.

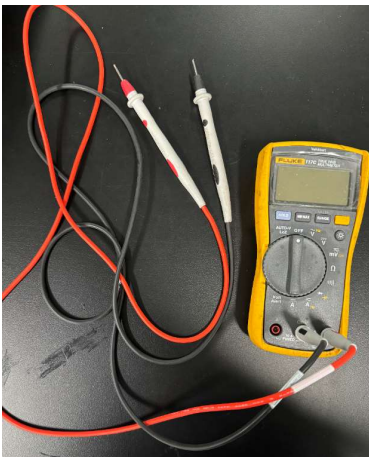


5.If the range is 0.45-0.52V, the MOS is normal. Otherwise, it is judged that the MOS is damaged and the controller needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps.

6.If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching test.

ML 26/MM 26/MR 26 Left/Middle/Right Blade Motor Controller MOSFET Abnormal

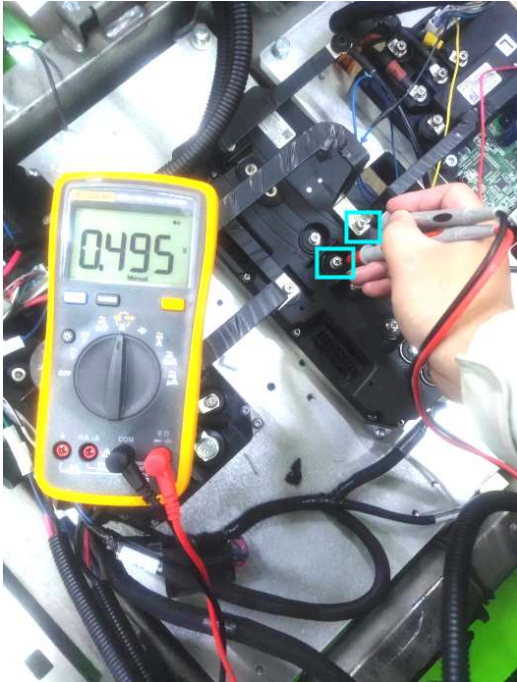
Tool: Multimeter、Computer、PCAN、Debugging wire



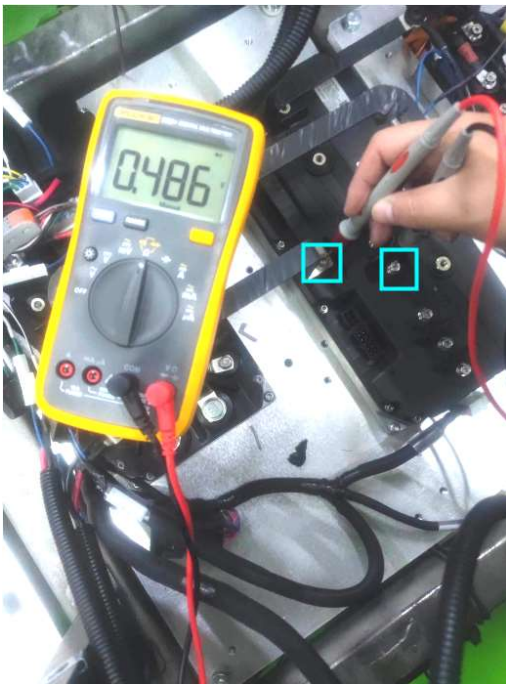
Check:

- 1.Display pop up ML 26/MM 26/MR 26 fault code
- 2.keep the vehicle power off state.
- 3.Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the positive

electrode.



4. Set the Multimeter to the diode position and measure the voltage between the U \ V \ W phase and the negative electrode.



5. If the range is 4.7-5.0V, the MOS is normal. Otherwise, it is judged that the MOS is damaged and the controller needs to be replaced..Refer to the maintenance manual for the specific disassembly and assembly steps

4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

ML 27/MM 27/MR 27 Left/Middle/Right Left Blade Motor Controller Temperature Sensor Abnormal

Tool: Computer、PCAN、Debugging wire



Check:

1. Display pop up ML 27/MM 27/MR 27 fault code
2. Restart Vehicle.
3. If this fault code persists, please replace a new blade controller. Refer to the maintenance manual for the specific disassembly and assembly steps
4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

ML 28/MM 28/MR 28 Left/Middle/Right Blade Motor Controller Self-Check Abnormal

Tool: Computer、PCAN、 Debugging wire



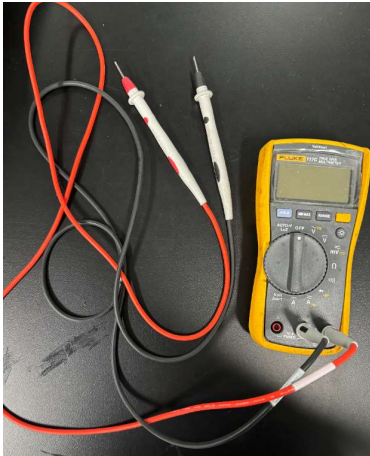
Check:

1. Display pop up ML 28/MM 28/MR 28 fault code
2. Restart Vehicle.
3. If this fault code persists, please replace a new blade controller. Refer to the maintenance manual for the specific disassembly and assembly steps
4. If you have replaced a new controller, remember to follow the self-learning steps for the motor auto-matching

test.

DS 11 Abnormal Display CAN Communication

Tool: Multimeter、Computer、PCAN、Debugging wire



Check:

1. Display only pop up a fault code DS 11
2. Try to control the car normally to see if the car can move forward and backward normally. If the vehicle can move forward normally, move backward.
3. Check whether the instrument plug-in is loose, check whether the CAN-H and CAN-L pins of the instrument plug-in are withdrawn, and whether there is any garbage on the surface of the instrument plug-in, causing poor contact of the plug-in.



4. Use a multimeter to check whether CAN-H and CAN-L of the instrument plug-in and CAN PORT plug-in are

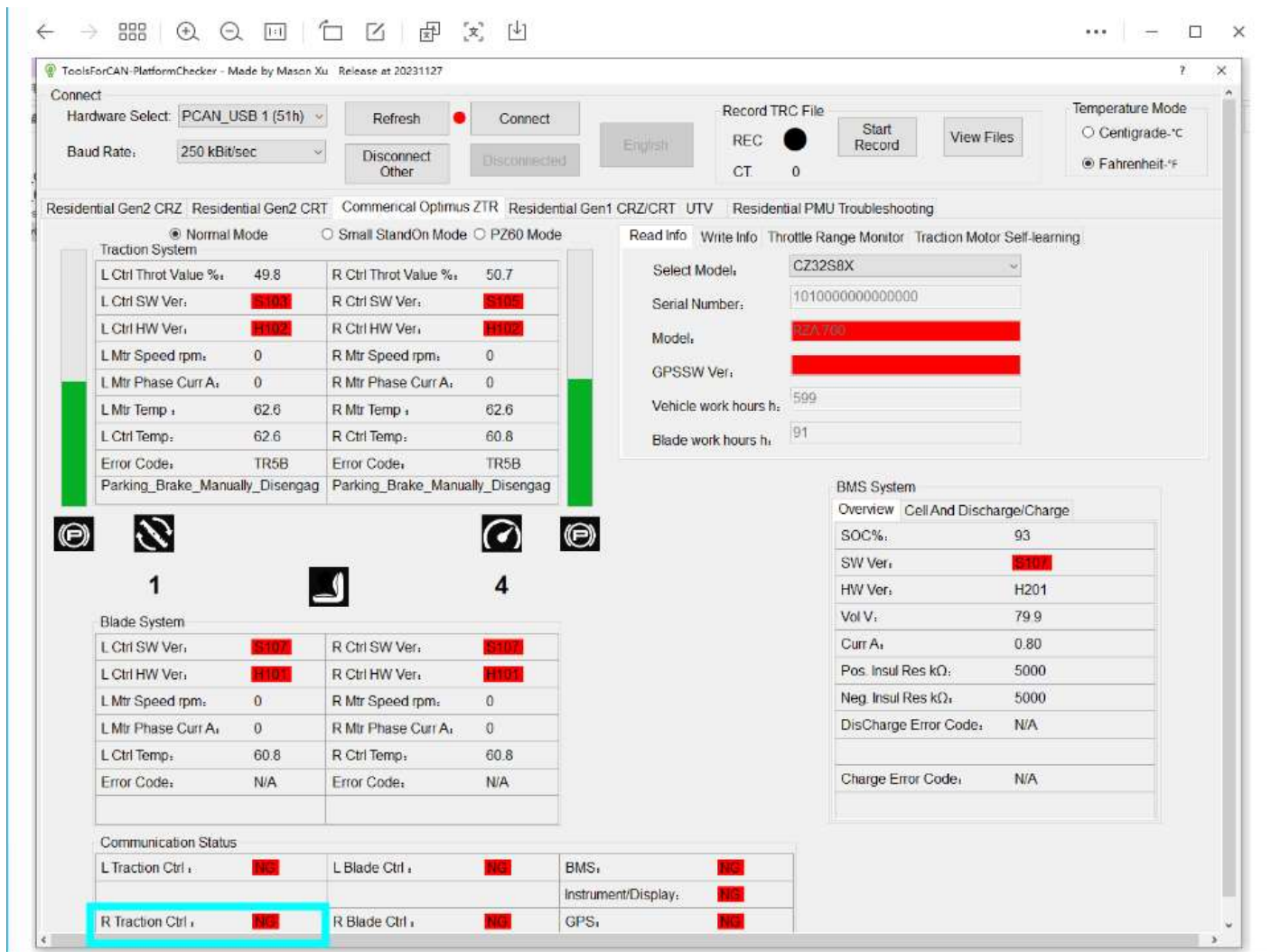
connected. If there is no continuity, replace the vehicle's main wiring harness.



5.If the instrument's CAN-H and CAN-L wiring harness are normal, replace the instrument.

6.If the vehicle cannot move forward or backward normally.  **ToolsForCAN-PlatformChecker** Use P-CAN, open the software, and check whether the main controller is online (displaying "OK" means online; displaying "NG"

means not online). If it is not online, it can be determined that the main controller is partially faulty.



7. Observe the main control indicator light for 20S seconds to see if the indicator light flashes. If the main controller indicator light flashes, it means that the main control power supply is normal. Check whether the main controller plug-in is loose, and check whether the CAN-H and CAN-L pins of the main control plug-in are withdrawn. Is there any garbage on the surface of the plug-in, causing poor plug-in contact.

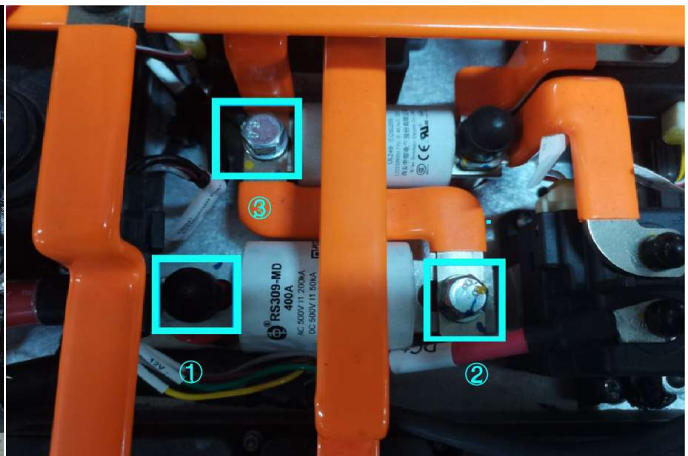
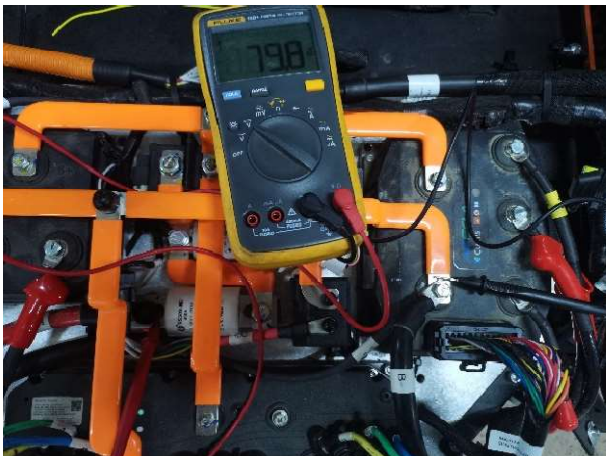


8. If the main control plug-in is normal, use a multimeter to check whether CAN-H and CAN-L of the instrument

plug-in and the main control plug-in are connected and whether there is an open circuit. If there is no continuity, replace the vehicle's main wiring harness.



9.If the indicator light of the main controller does not flash, it means that the power supply of the main controller is not normal. First use a wrench to confirm that the three bolts ①,② and ③ are tightened and there is no sign of looseness. If the bolts are tight, use the DC voltage range of the multimeter to check whether the voltage to ground at ①,② and ③ is greater than 70V.

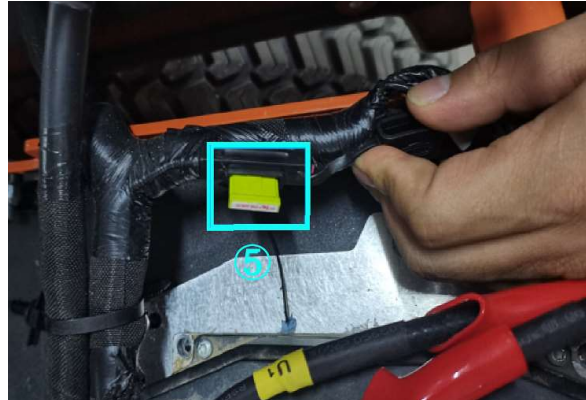


10.If the voltage at point ① is less than 70V, the power supply of the battery pack needs to be inspected.

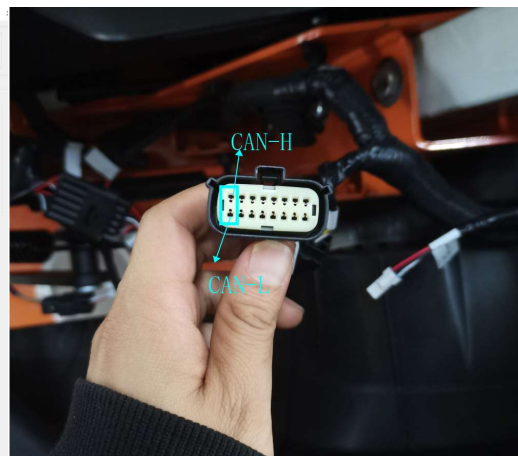
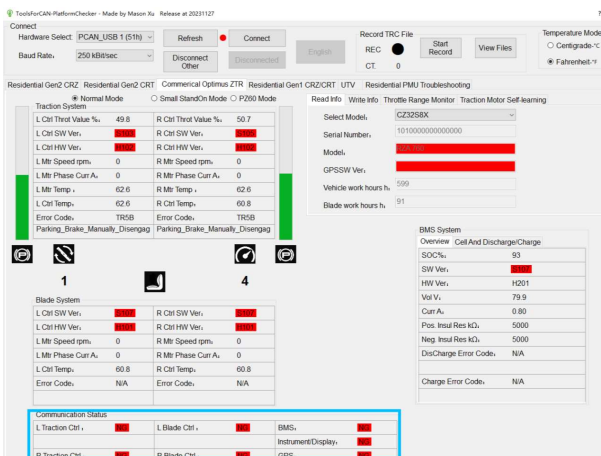
11.If the voltage at ① is greater than 70V and the voltage at ② is less than 70V, the 400A fuse between bolts ① and ② needs to be replaced.

12.If the voltage at ①,② and ③ is greater than 70V and the voltage at ④ is less than 70V,First check whether the ⑤-20A fuse is loose. If it is normal, use a multimeter to check whether the 20A fuse is damaged; if it is damaged,

replace the 20A fuse. If ⑤-20A fuse is normal, replace the main control harness.



13. Use P-CAN, open the software  **ToolsForCAN-PlatformChecker**, if all components are not online (all components display "NG" means not online), it means CANH and CANL are short-circuited, or you can use the on-off setting of a multimeter to measure whether the CAN-PORT plug-in CANH and CAN-L are short-circuited. The DS 11 and TR55 fault codes pop up on the display at the same time.



14. Use the on-off switch of a multimeter to measure CAN-H and CAN-L of the CAN-PORT plug-in, keep it there, and then unplug all CAN communication plug-ins in the vehicle in order to locate whether the vehicle harness CAN

-H and CAN-L is short-circuited or the component CAN -H and CAN-L are short-circuited.



①Light on the left



②Instrument plug-in



③GPS plug-in



④Three-in-one cutter controller plug-in



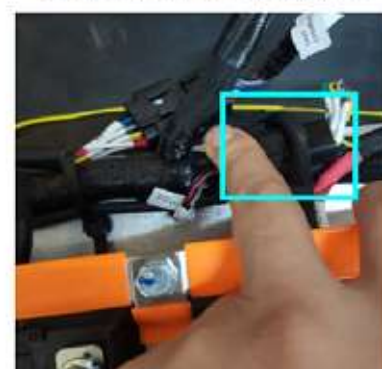
⑤Left travel controller plug-in



⑥Right walking controller plug-in



⑦Ex-ET0 plugin



⑧Post ET0 plug-in



⑨BMS plug-in

15.If plug-in No. ① is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of the left lamp component are short-circuited. If there is a short

circuit, replace the left lamp assembly.



16.If plug-in No. ② is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether the instrument components CAN-H and CAN-L are short-circuited. If there is a short circuit, replace the instrument components.



17.If plug-in No. ③ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether the GPS components CAN-H and CAN-L are short-circuited. If there is a short circuit, replace the GPS.



18.If plug-in No. ④ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of the three-in-one cutter controller components are short-

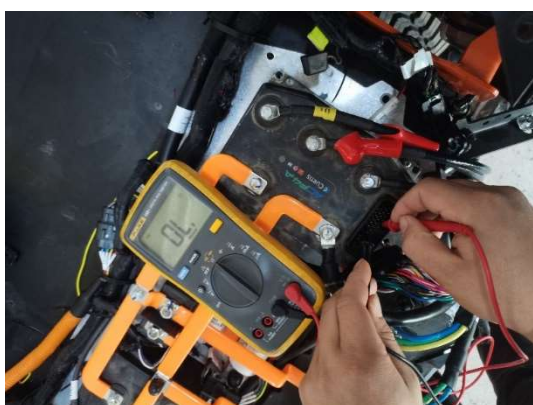
circuited. If there is a short circuit, replace the three-in-one cutter controller.



19.If plug-in No. ⑤ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of the left travel controller components are short-circuited. If there is a short circuit, replace the left travel controller.



20.If plug-in No. ⑥ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of the right travel controller components are short-circuited. If there is a short circuit, replace the right travel controller.



21.If plug-in No. ⑦ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether the CAN-H and CAN-L of the front ET0 are short-circuited. If there is a short circuit,

replace the front ETO harness.



22.If plug-in No. ⑧ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of ET0 are short-circuited. If there is a short circuit, replace the rear ETO harness.



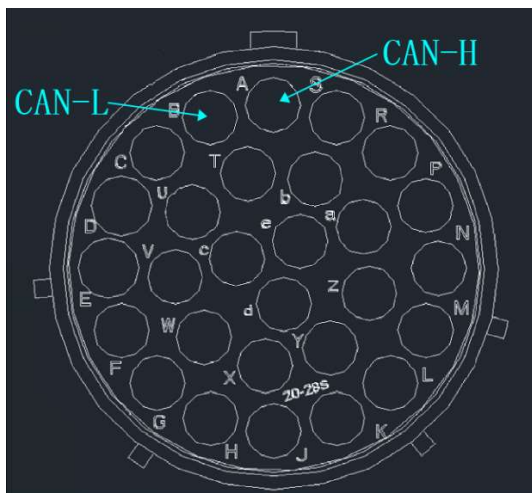
23.If plug-in No.⑨ is unplugged, the multimeter buzzer will not sound. Then use the on-off setting of the multimeter to measure whether CAN-H and CAN-L of the battery pack are short-circuited. If there is a short circuit, replace the battery pack.



24.If plug-ins No. ①、②、③、④、⑤、⑥、⑦、⑧, and ⑨ are unplugged at the same time, the multimeter

buzzer will not sound. Replace the main controller wiring harness.

25. Check whether the BMS plug-in is loose, check whether the BMS plug-in CAN-H and CAN-L pins are withdrawn, and whether there is garbage on the surface of the BMS plug-in, causing poor plug-in contact.



26. If there is no problem with the battery pack plug-in, use a multimeter to check whether the CAN-H and CAN-L of the instrument plug-in and the battery pack plug-in are connected. If there is no continuity, replace the vehicle's

main wiring harness.



BC 11 Input Power Undervoltage

Check:

1. Display pop up BC 11 fault code
2. Restart charge.
3. Check whether the grid voltage is abnormal.

BC 12 Input Power Overvoltage

Check:

1. Display pop up BC 12 fault code
2. Restart charge.
3. Check whether the grid voltage is abnormal.

BC 13 Output Undervoltage

Check:

1. Display pop up BC 13 fault code
2. Restart charge.
3. Check whether the load voltage is abnormal.

BC 14 Output Overvoltage

Check:

1. Display pop up BC 14 fault code
2. Restart charge.
3. Check whether the load voltage is abnormal.

BC 15 Output Overcurrent

Check:

1. Display pop up BC 15 fault code
2. Restart charge.
3. Check whether the charging loop current is abnormal and whether there is an output short circuit phenomenon.

BC 16 Charger Overtemp

Check:

1. Display pop up BC 16 fault code.
2. When the internal temperature dropped to 69°C (156 °F), the charger automatically resumed charging.
3. Charger performance reduced while temperature is high. Once temperature is within standard range fault will automatically clear and normal charging operation will resume.

BC 17 Output Short Circuit

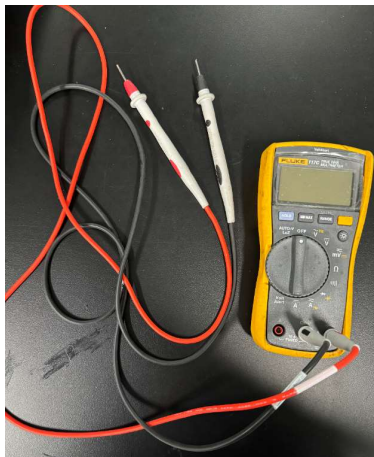
Check:

1. Display pop up BC 17 fault code.
2. Please discontinue charging. Ensure use of a known good charger. If this does not correct the fault, contact

after-3.sales service.

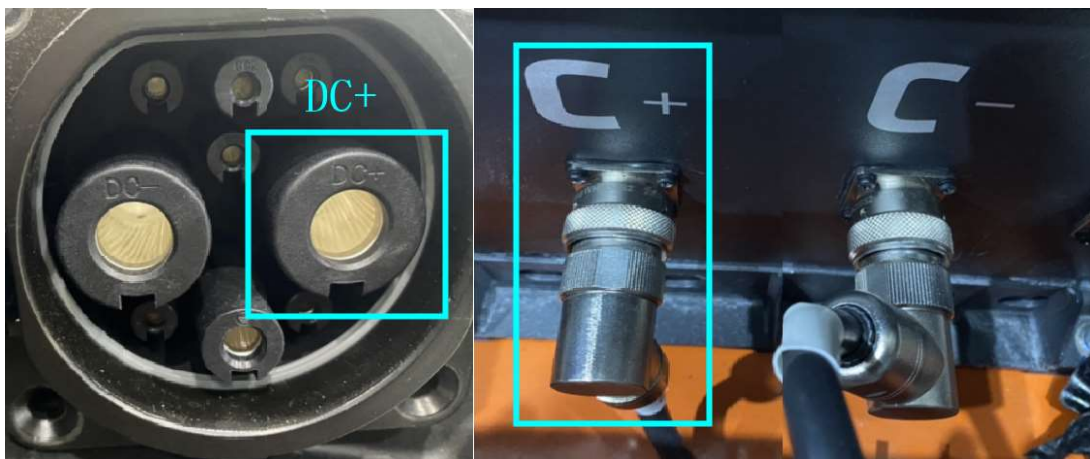
BC18 Battery Output Polarity Reversal

Tool: Multimeter



Check:

- 1.Display pop up BC 18 fault code.
- 2.Check whether the positive and negative electrodes of the charging socket correspond to the positive and negative electrodes connected to the battery pack.
- 3.Select the multimeter, remove the C + connector from the battery pack, and measure whether the C + connector and the positive electrode of the charging socket DC + are turned on



- 4.Select the multimeter, remove the C-connector from the battery pack, and measure whether the C-connector

and the charging socket negative electrode DC-go on



BC 19 No Battery Output Voltage

Check:

1. Display pop up BC 19 fault code.
2. Restart the charger, and reinsert charger plug.
3. Check for any debris that may be in charging port or plug and clear if present. If these actions for not correct the fault, discontinue use of the vehicle and contact after-sales service.

BC 21 Non-Load

Check:

1. Display pop up BC 21 fault code.
2. Restart the charger, and reinsert charger plug.
3. Check for any debris that may be in charging port or plug and clear if present. If these actions for not correct

the fault, discontinue use of the vehicle and contact after-sales service.

BC 22 CAN Communication Abnormal

BC 23 Internal Communication Abnormal

BC 24 Charger CC1 Abnormal

BMS 11 Battery Discharging Slight Overtemp

Check:

1. Display pop up BMS 11 fault code.
2. Vehicle has been put into a state of reduced performance while cooling. Once temperature is back within standard range, the fault will automatically clear and normal operation will resume.

BMS 12 Battery Discharging Severe Overtemp

Check:

1. Display pop up BMS 12 fault code.
2. Power off
3. Waiting 30 min
4. Restart the vehicle

BMS 13 Battery Discharging Slight Undertemp

Check:

1. Display pop up BMS 13 fault code.
2. Vehicle has been put into a state of reduced performance while cooling. Once temperature is back within

standard range, the fault will automatically clear and normal operation will resume

BMS 14 Battery Discharging Severe Undertemp

Check:

1. Display pop up BMS 14 fault code.
2. Power on
3. Waiting 30 min
4. Restart the vehicle

BMS 15 Battery Cells Slight Temperature Difference

1. Display pop up BMS 15 fault code.
2. Vehicle has been put into a state of reduced performance while cell temperatures return to normal. Once

temperature is back within standard range, the fault will automatically clear and normal operation will resume.

BMS 16 Battery Cells Severe Temperature Difference

Check:

- 1.Display pop up BMS 16 fault code.
- 2.Power off
- 3.Waiting 6 hours
- 4.Restart the vehicle

BMS 17 Battery Cells Slight Overvoltage

Check:

- 1.Display pop up BMS 17 fault code.
- 2.Stop charging
- 3.Normal driving

BMS 18 Battery Cells Severe Overvoltage

Check:

- 1.Display pop up BMS 18 fault code.
- 2.Stop charging
- 3.Restart the vehicle
- 4.Normal driving

BMS 19 Battery Cells Slight Undervoltage

Check:

- 1.Display pop up BMS 19 fault code.
- 2.Reduce load
- 3.Charging

BMS 21 Battery Cells Severe Undervoltage

Tool: Computer、PCAN、Debugging wire、Charger



Check:

- 1.Display pop up BMS 21 fault code.
- 2.Connect to host computer View computer interface

ToolsForCAN-PlatformChecker - Made by Mason Xu Release at 20231127

Connect
Hardware Select: Refresh English
Baud Rate: 250 kBit/sec

Record TRC File
REC ☐ CT: 0 Temperature Mode
☐ Centigrade-°C ☒ Fahrenheit-°F

Residential Gen2 CRZ Residential Gen2 CRT Commercial Optimus ZTR Residential Gen1 CRZ/CRT UTV Residential PMU Troubleshooting

☒ Normal Mode ☐ Small StandOn Mode ☐ PZ60 Mode

Traction System

L Ctrl Throt Value %:	0	R Ctrl Throt Value %:	0
L Ctrl SW Ver:		R Ctrl SW Ver:	
L Ctrl HW Ver:		R Ctrl HW Ver:	
L Mtr Speed rpm:	0	R Mtr Speed rpm:	0
L Mtr Phase Curr A:	0	R Mtr Phase Curr A:	0
L Mtr Temp :	0	R Mtr Temp :	0
L Ctrl Temp:	0	R Ctrl Temp:	0
Error Code:	N/A	Error Code:	N/A

0 0

Blade System

L Ctrl SW Ver:	R Ctrl SW Ver:
L Ctrl HW Ver:	R Ctrl HW Ver:
L Mtr Speed rpm:	0
L Mtr Phase Curr A:	0
L Ctrl Temp:	0
Error Code:	N/A

Communication Status

L Traction Ctrl :	L Blade Ctrl :	BMS:
		Instrument/Display:
R Traction Ctrl :	R Blade Ctrl :	GPS:

Read Info Write Info Throttle Range Monitor Traction Motor Self-learning

Select Model: CZ32S8X
Serial Number:
Model:
GPSSW Ver:
Vehicle work hours h:
Blade work hours h:

BMS System
Overview | Cell And Discharge/Charge

SOH%:	0
Cell Highest Vol V:	0 0
Cell Lowest Vol V:	0 0
Cell Highest Temp °C:	0 0
Cell Lowest Temp °C:	0 0
Charger Socket Pos. Temp	0
Charger Socket Neg. Temp	0
Charge Total Hrs Ah:	0
DisCharge Total Hrs Ah:	0

3.If cell lowest voltage < 1.8V, Replace the battery pack

4.if Battery Cells Voltage Difference less than 1.5V, Update BMS maintenance program,if Start Charging, Minimum cell voltage is charged to 3V or maximum cell voltage to 3.6V cutoff. Update normal program,

5.if Battery Cells Voltage Difference less than 600mv, charge normally to 100%SOC. Battery Cells Voltage

Difference greater than 600mv, Replace the battery pack

6.if Battery Cells Voltage Difference greater than 1.5V, Replace the battery pack

7. $2V < \text{cell lowest voltage} < 2.5V$, Shielded battery voltage difference fault ,normal charge

8. $2.5V < \text{cell lowest voltage}$, Battery Cells Voltage Difference greater than 600mv, Replace the battery pack

BMS 23 Insulation Resistance Severe Insufficient

Tool: insulation resistance meter



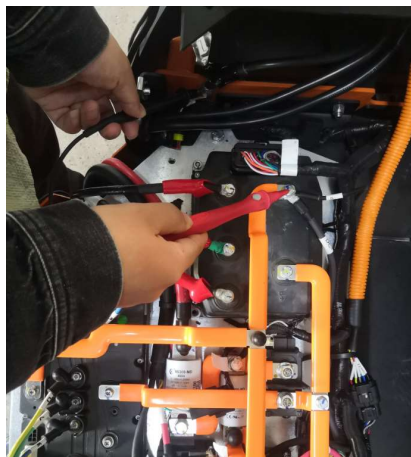
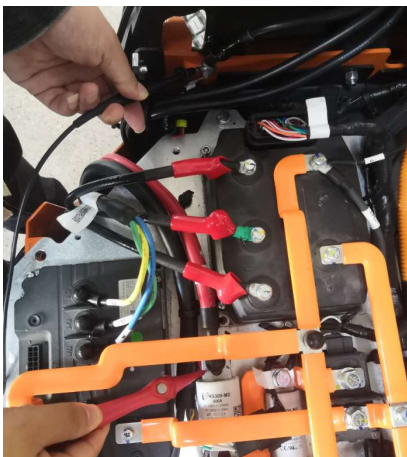
Check:

1.Display pop up BMS 23 fault code.

2.Power off

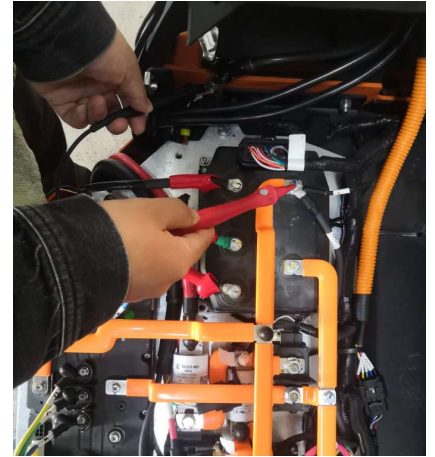
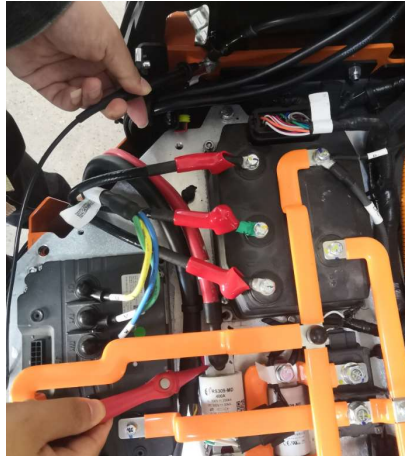
3.Disconnect cover, Test the insulation resistance between B+ and the shell resistance with an insulation resistance meter Test the insulation resistance between B- and the shell resistance with an insulation resistance meter

If the resistance less than 10K ohms, Vehicle damage, Check Vehicle.

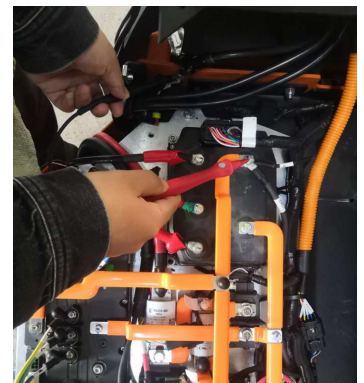
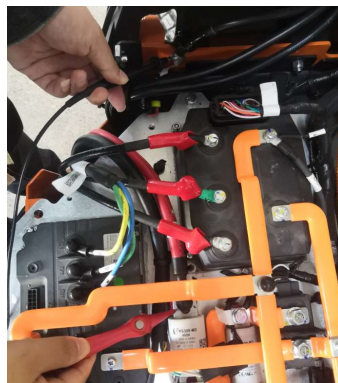
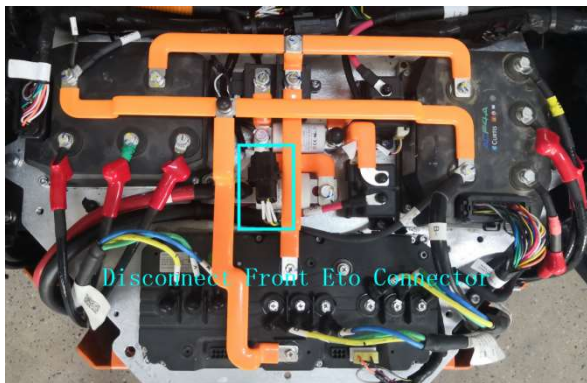


4.If the resistance greater than 10KΩ,Disconnect seat switch, Test the controller B+ and B- loop resistance if

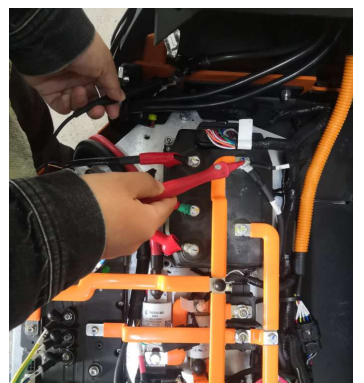
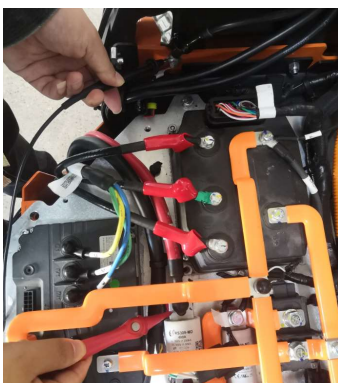
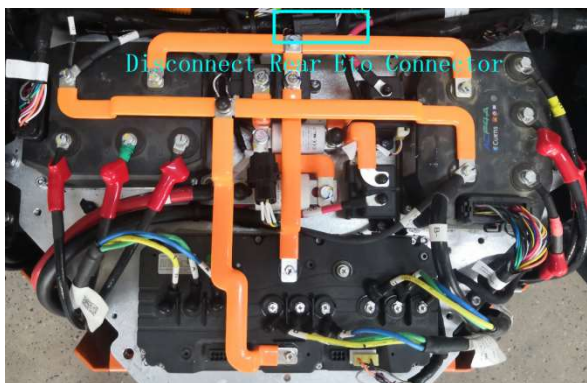
resistance is greater than 10K ohms, seat switch damage



5.if resistance is less than 10K ohms, disconnect front ETO connector Test the controller B+ and B- loop resistance.If resistance is greater than 10K ohms, front ETO damage

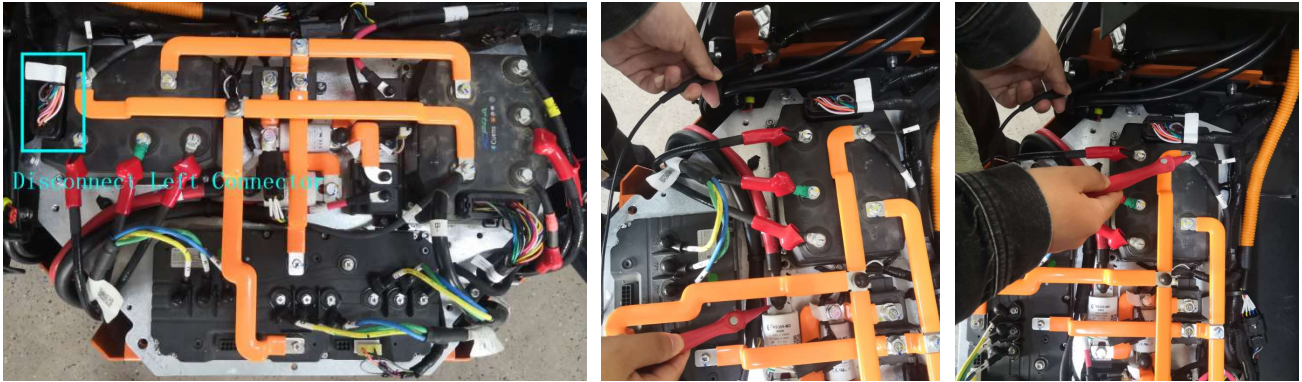


6.If resistance is less than 10K ohms, Disconnect rear ETO connector, Test the controller B+ and B- loop resistance if resistance is greater than 10K ohms, rear ETO damage

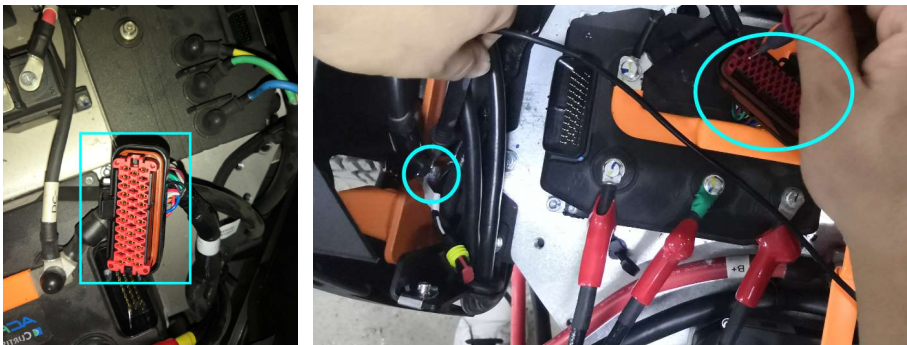


7.if resistance is less than 10K ohms.Disconnect left controller connector, Test the controller B+ and B- loop

resistance, if resistance is greater than 10K ohms, left controller or left controller cable damage,

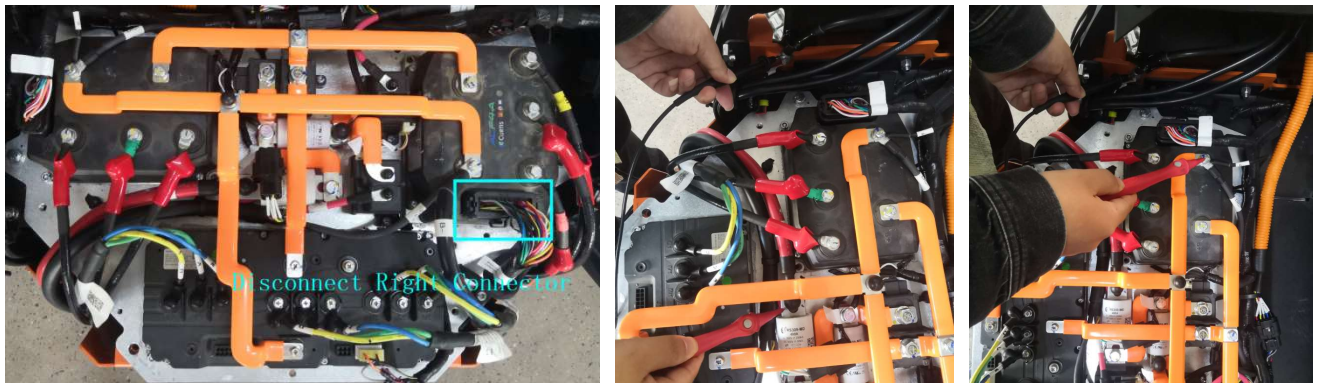


8. Test the resistance of each connector pin to the housing.

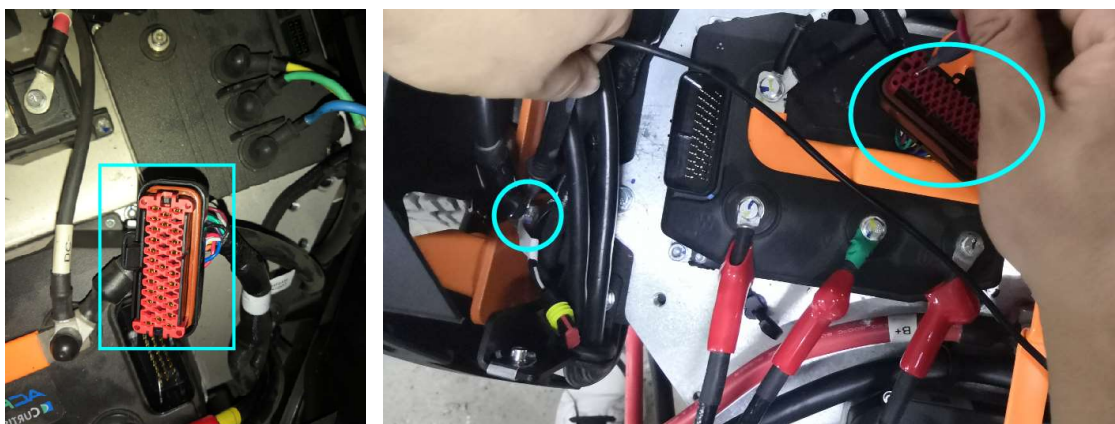


9. if resistance is less than 10K ohms, disconnect right controller connector Test the controller B+ and B- loop

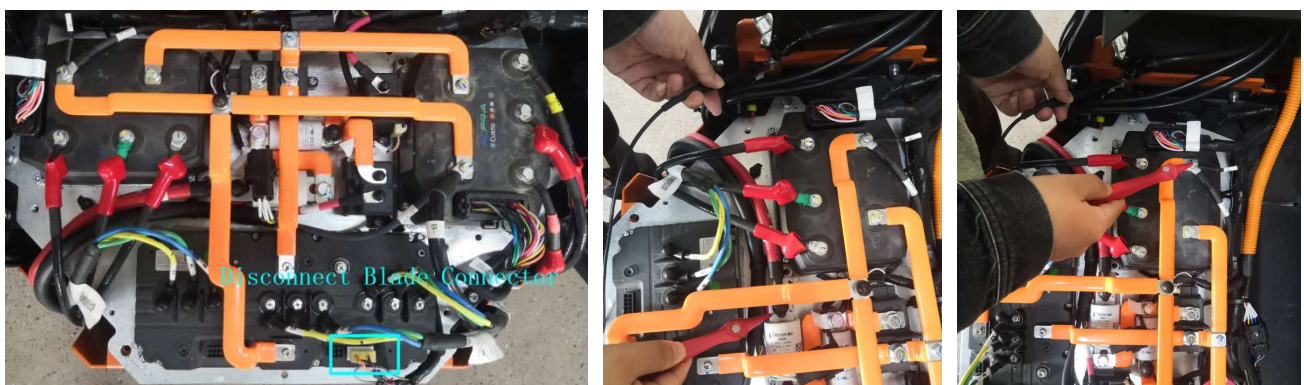
resistance if resistance is greater than 10K ohms, right controller or right controller cable damage,



10. Test the resistance of each connector pin to the housing

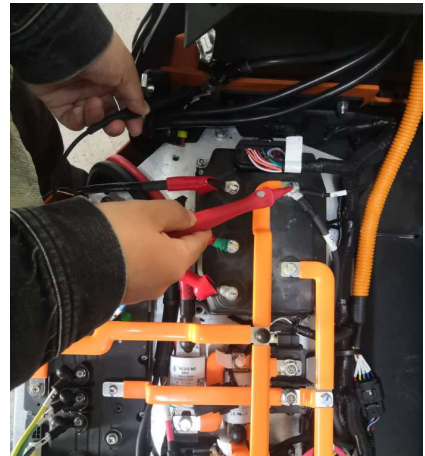
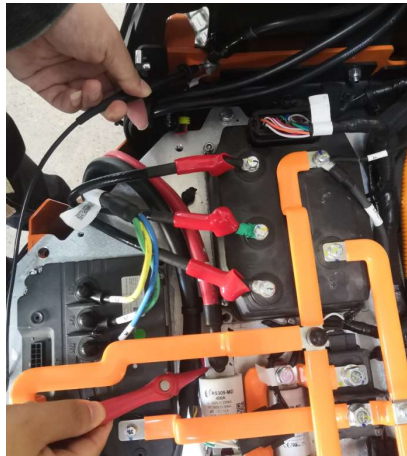


11. If resistance is less than 10K ohms, Disconnect 3-in-1 blade controller connector, Test the controller B+ and B- loop resistance resistance is greater than 10K ohms, 3-in-1 blade controller damage

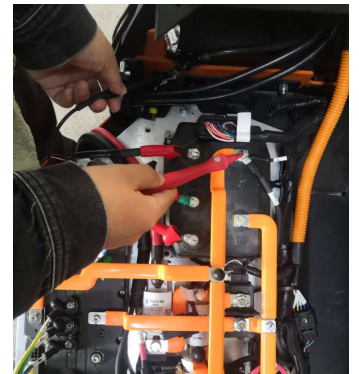
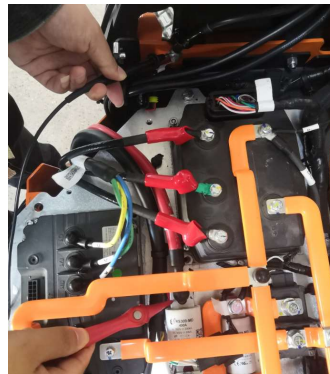


12. If resistance is less than 10K ohms, Disconnect left park switch, Test the controller B+ and B- loop resistance

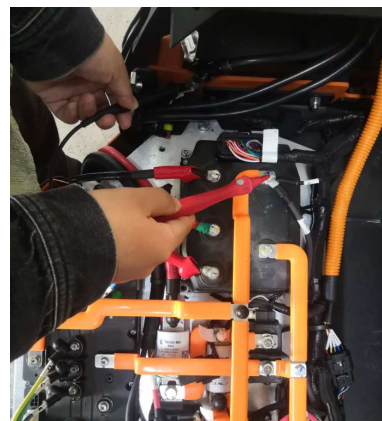
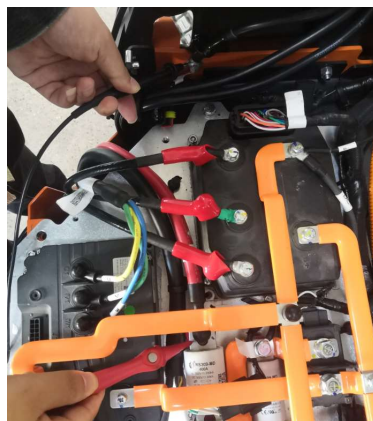
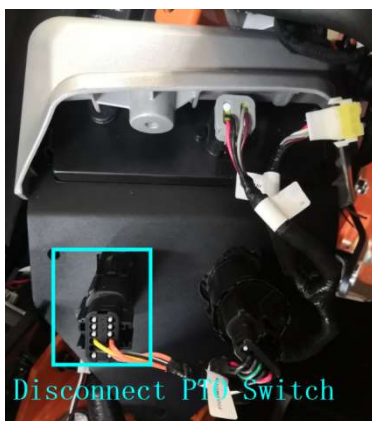
resistance is greater than 10K ohms, left park switch damage



13.if resistance is less than 10K ohms, Disconnect right park switch, Test the controller B+ and B- loop resistance
if resistance is greater than 10K ohms, right park switch damage

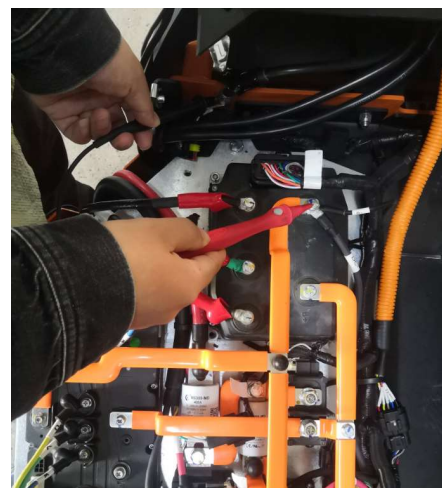
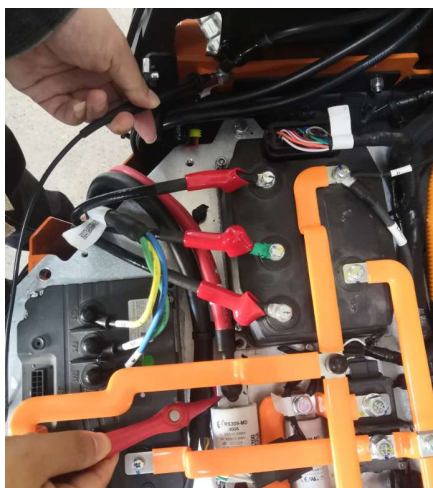


14.if resistance is less than 10K ohms, Disconnect PTO switch, Test the controller B+ and B- loop resistance
if resistance is greater than 10K ohms,PTO switch damage

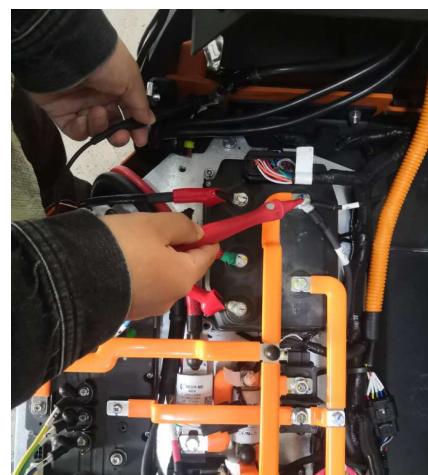
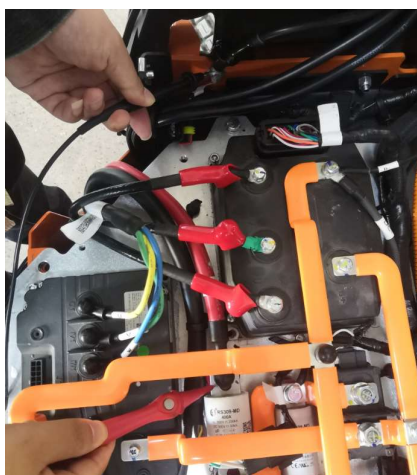
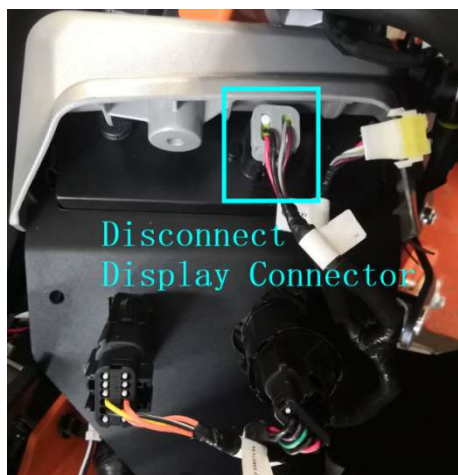


15.if resistance is less than 10K ohms, Disconnect key switch, Test the controller B+ and B- loop resistance if

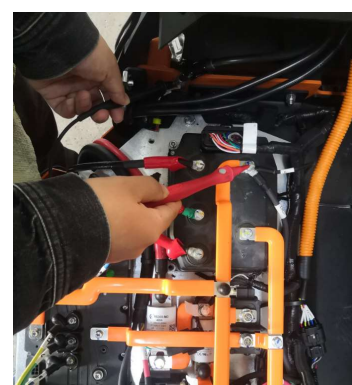
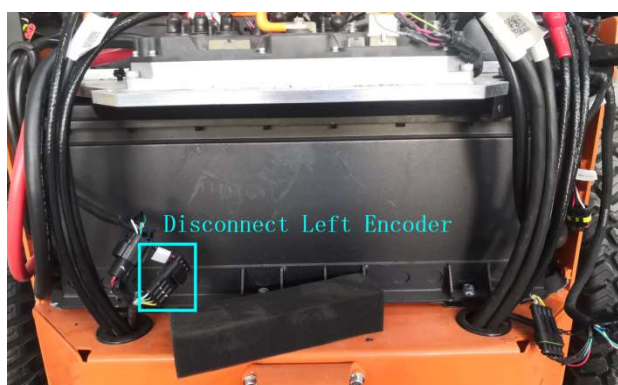
resistance is greater than 10K ohms, key switch damage



16.if resistance is less than 10K ohms, Disconnect display connector, Test the controller B+ and B- loop resistance
if resistance is greater than 10K ohms, display damage

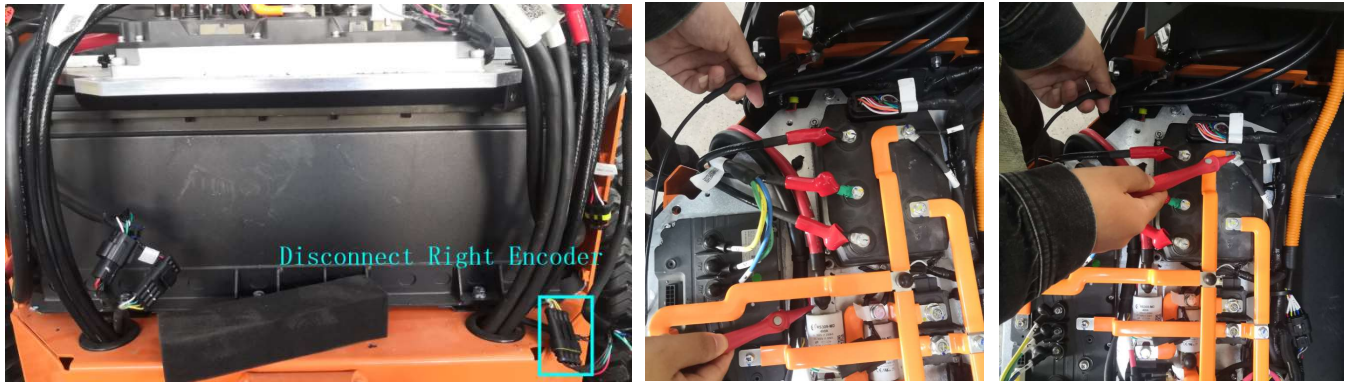


17.if resistance is less than 10K ohms, Disconnect left position encoder, Test the controller B+ and B- loop
resistance if resistance is greater than 10K ohms, Left position encoder damage

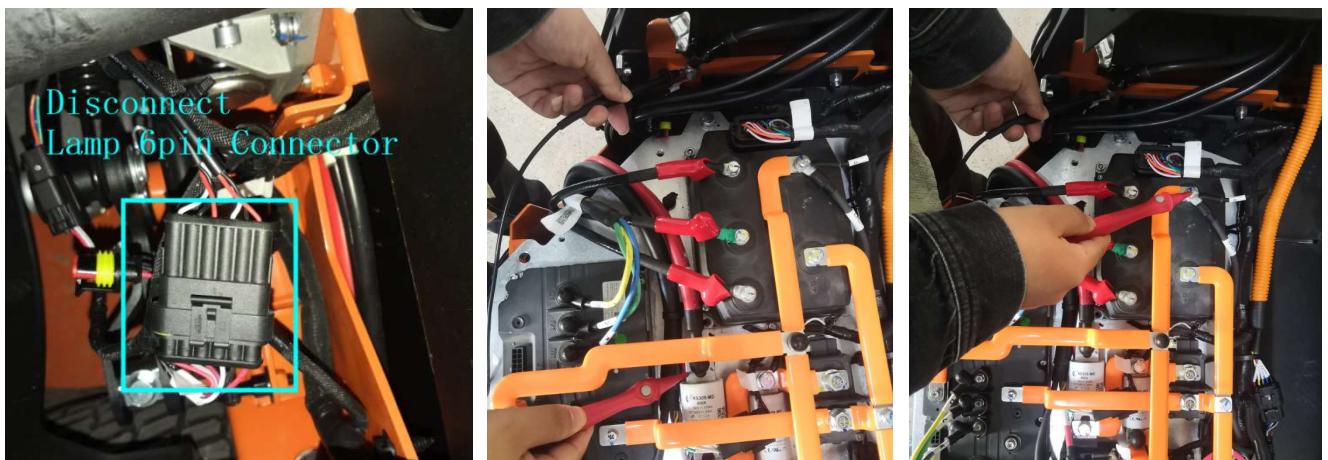


18.if resistance is less than 10K ohms, Disconnect right position encoder, Test the controller B+ and B- loop

resistance if resistance is greater than 10K ohms, Right position encoder damage



19.if resistance is less than 10K ohms, Disconnect lamp 6pin connector, Test the controller B+ and B- loop resistance if resistance is greater than 10K ohms, lamp damage



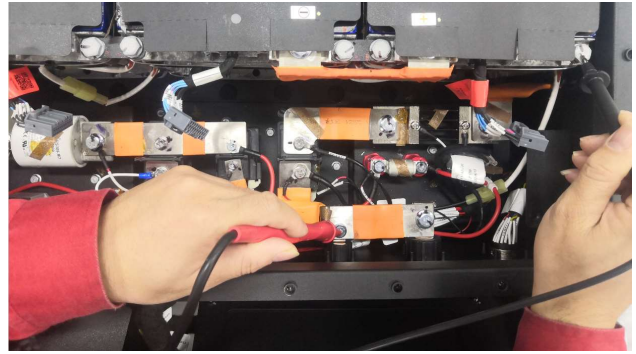
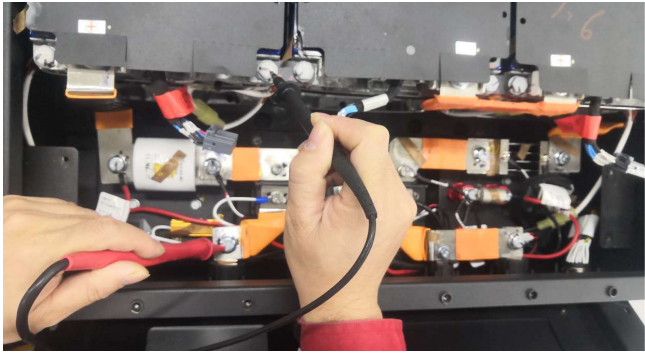
20.Check battery ,Disconnect the P+、P-、C+、C-、CAN ports from the battery, Remove the battery box cover,

Check inside the battery,

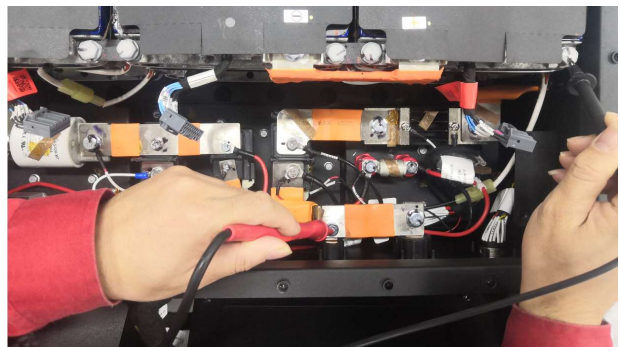
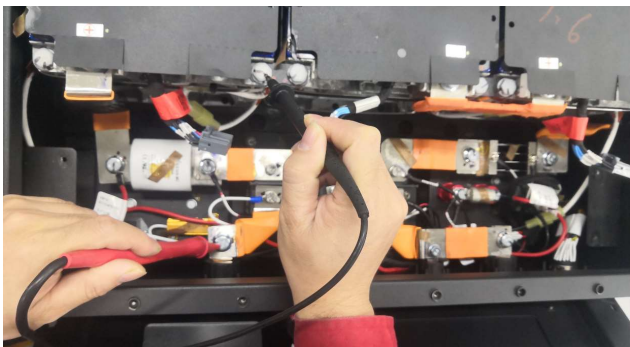


21. Disconnect the BMS, Test the battery relay B+ and B- loop resistance with an insulation resistance meter, if

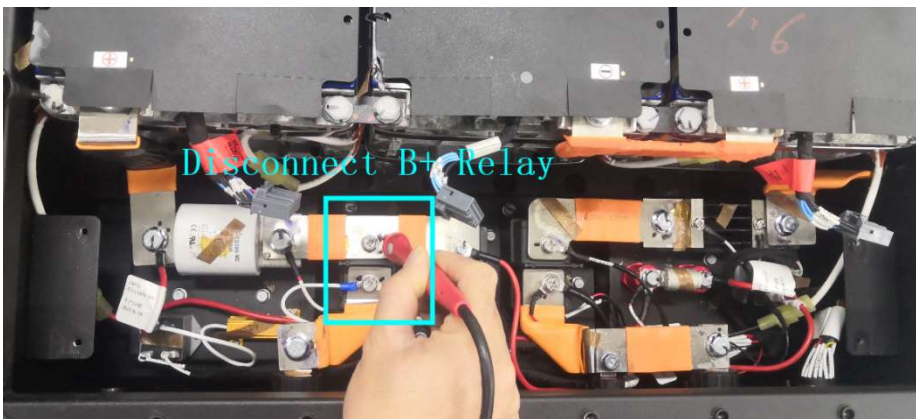
resistance is greater than 10K ohms, BMS damage



22.if resistance is less than 10K ohms, Disconnect DC power, Test the battery B+ and B- loop resistance with an insulation resistance meter if resistance is greater than 10K ohms, DC power damage

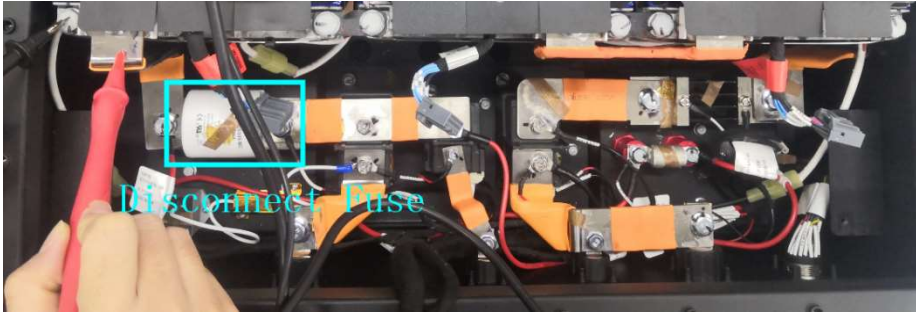


23.if resistance is less than 10K ohms, Disconnect B+ relay, Test the battery B+ loop resistance, if resistance is greater than 10K ohms, B+ relay damage

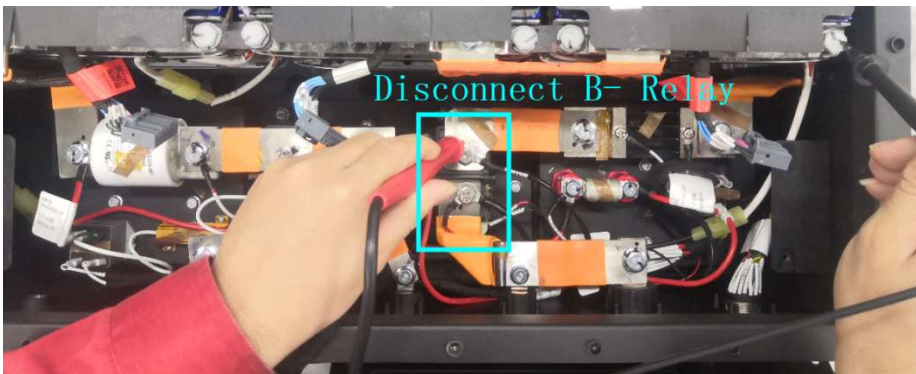


24.If resistance is less than 10K ohms, Disconnect fuse, Test the battery B+ loop resistance, if resistance is greater

than 10K ohms, fuse damage

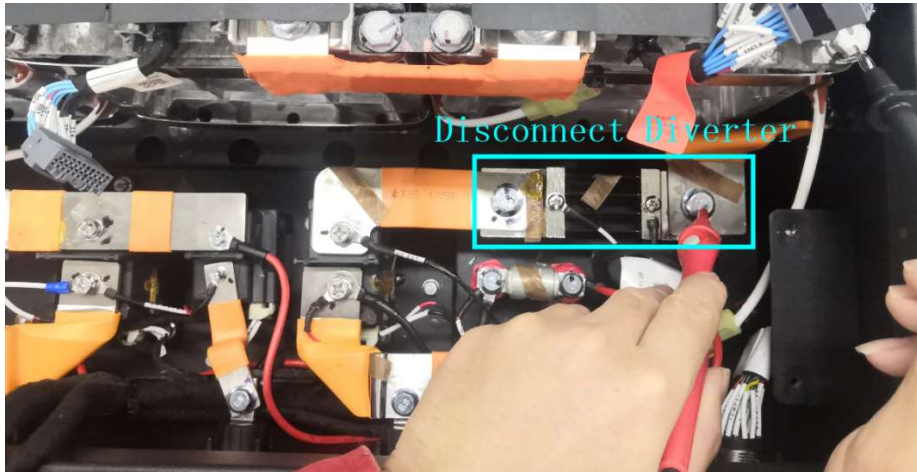


25.if resistance is less than 10K ohms,Disconnect B- relay, Test the battery B- loop resistance,if resistance is greater than 10K ohms, relay damage



26.if resistance is less than 10K ohms,Disconnect diverter, Test the battery B- loop resistance,if resistance is

greater than 10K ohms, diverter damage if resistance is less than 10K ohms, cell damage



BMS 24 Battery Discharge Slight Overcurrent

Check:

1. Display pop up BMS 24 fault code.
2. Reduce load

BMS 25 Battery Discharge Severe Overcurrent

Check:

1. Display pop up BMS 25 fault code.
2. Reduce load
3. Restart the vehicle

BMS 26 Battery Cells Voltage Slight Difference

Check:

1. Display pop up BMS 26 fault code.
2. Reduce load
3. Charging

BMS 27 Battery Cells Voltage Severe Difference

Tool: Computer、PCAN、Debugging wire、Charger



Check:

1.Display pop up BMS 27 fault code.

2.Connect to host computer View computer interface

ToolsForCAN-PlatformChecker - Made by Mason Xu Release at 20231127

Connect Hardware Select: Record TRC File REC ☐ CT: 0 Temperature Mode ☐ Centigrade-°C ☒ Fahrenheit-°F

Baud Rate: 250 kBit/sec

Residential Gen2 CRZ Residential Gen2 CRT Commercial Optimus ZTR Residential Gen1 CRZ/CRT UTV Residential PMU Troubleshooting

☒ Normal Mode ☐ Small StandOn Mode ☐ PZ60 Mode

Traction System	
L Ctrl Throt Value %:	0
L Ctrl SW Ver:	R Ctrl SW Ver:
L Ctrl HW Ver:	R Ctrl HW Ver:
L Mtr Speed rpm:	0
L Mtr Phase Curr A:	0
L Mtr Temp :	0
L Ctrl Temp:	0
Error Code:	N/A

Blade System	
L Ctrl SW Ver:	R Ctrl SW Ver:
L Ctrl HW Ver:	R Ctrl HW Ver:
L Mtr Speed rpm:	0
L Mtr Phase Curr A:	0
L Ctrl Temp:	0
Error Code:	N/A

Communication Status

Communication Status		
L Traction Ctrl :	L Blade Ctrl :	BMS:
		Instrument/Display:
R Traction Ctrl :	R Blade Ctrl :	GPS:

Read Info Write Info Throttle Range Monitor Traction Motor Self-learning

Select Model: CZ32S8X

Serial Number:

Model:

GPSSW Ver:

Vehicle work hours h:

Blade work hours h:

BMS System Overview | Cell And Discharge/Charge |

SOH%:	0
Cell Highest Vol V:	0
Cell Lowest Vol V:	0
Cell Highest Temp °C:	0
Cell Lowest Temp °C:	0
Charger Socket Pos. Temp	0
Charger Socket Neg. Temp	0
Charge Total Hrs Ah:	0
DisCharge Total Hrs Ah:	0

3.If cell lowest voltage < 1.8V, Replace the battery pack

4.if Battery Cells Voltage Difference less than 1.5V, Update BMS maintenance program, Start Charging, Minimum cell voltage is charged to 3V cutoff. Update normal program, Battery Cells Voltage Difference less than 600mv, charge

normally to 100%SOC. Battery Cells Voltage Difference greater than 600mv, Replace the battery pack

5.if Battery Cells Voltage Difference greater than 1.5V, Replace the battery pack

6.if $2V < \text{cell lowest voltage} < 2.5V$, Shielded battery voltage difference fault, normal charge

7.if $2.5V < \text{cell lowest voltage}$, Battery Cells Voltage Difference greater than 600mv, Replace the battery pack

BMS 28 Battery Slight Overvoltage

Check:

1.Display pop up BMS 28 fault code.

2.Stop charging

3.Normal driving

BMS 29 Battery Severe Overvoltage

Check:

1.Display pop up BMS 29 fault code.

2.Stop charging

3.Restart the vehicle

4.Normal driving

BMS 31 Battery Slight Undervoltage

Check:

1.Display pop up BMS 31 fault code.

2.Reduce load

3.Charging

BMS 32 Battery Severe Undervoltage

Tool: Computer、PCAN、Debugging wire、Charger



Check:

1.Display pop up BMS 32 fault code.

2.Connect to host computer View computer interface

ToolsForCAN-PlatformChecker - Made by Mason Xu Release at 20231127

Connect
Hardware Select: Refresh
Baud Rate: 250 kBit/sec
Record TRC File
REC ☐ CT: 0
Temperature Mode
☐ Centigrade-°C
☒ Fahrenheit-°F

Residential Gen2 CRZ Residential Gen2 CRT Commercial Optimus ZTR Residential Gen1 CRZ/CRT UTV Residential PMU Troubleshooting

☒ Normal Mode ☐ Small StandOn Mode ☐ PZ60 Mode

Traction System	
L Ctrl Throt Value %:	0
L Ctrl SW Ver:	R Ctrl SW Ver:
L Ctrl HW Ver:	R Ctrl HW Ver:
L Mtr Speed rpm:	0
L Mtr Phase Curr A:	0
L Mtr Temp :	0
L Ctrl Temp:	0
Error Code:	N/A

Blade System	
L Ctrl SW Ver:	R Ctrl SW Ver:
L Ctrl HW Ver:	R Ctrl HW Ver:
L Mtr Speed rpm:	0
L Mtr Phase Curr A:	0
L Ctrl Temp:	0
Error Code:	N/A

Communication Status

Communication Status		
L Traction Ctrl :	L Blade Ctrl :	BMS:
		Instrument/Display:
R Traction Ctrl :	R Blade Ctrl :	GPS:

Read Info Write Info Throttle Range Monitor Traction Motor Self-learning

Select Model: CZ32S8X
Serial Number:
Model:
GPSSW Ver:
Vehicle work hours h:
Blade work hours h:

BMS System
Overview | Cell And Discharge/Charge |
SOH%: 0
Cell Highest Vol V: 0 0
Cell Lowest Vol V: 0 0
Cell Highest Temp °C: 0 0
Cell Lowest Temp °C: 0 0
Charger Socket Pos. Temp 0
Charger Socket Neg. Temp 0
Charge Total Hrs Ah: 0
DisCharge Total Hrs Ah: 0

3.If cell lowest voltage < 1.8V, Replace the battery pack

4.if Battery Cells Voltage Difference less than 1.5V, Update BMS maintenance program, Start Charging, Minimum cell voltage is charged to 3V cutoff. Update normal program, Battery Cells Voltage Difference less than 600mv, charge

normally to 100%SOC. Battery Cells Voltage Difference greater than 600mv, Replace the battery pack

5.if Battery Cells Voltage Difference greater than 1.5V, Replace the battery pack

6.if $2V < \text{cell lowest voltage} < 2.5V$, Shielded battery voltage difference fault, normal charge

7.if $2.5V < \text{cell lowest voltage}$, Battery Cells Voltage Difference greater than 600mv, Replace the battery pack

BMS 33 Battery Charging Slight Overtemp

Check:

- 1.Display pop up BMS 33 fault code.
- 2.Request half the current limit
- 3.The charger output current drops by half
- 4.Temperature is within standard range fault will automatically clear and normal charging operation will resume

BMS 34 Battery Charging Severe Overtemp

Check:

- 1.Display pop up BMS 34 fault code.
- 2.Stop Charging
- 3.Waiting the unit to cool down
- 4.Recharge

BMS 35 Charging Port Slight Overtemp

Check:

- 1.Display pop up BMS 35 fault code.
- 2.Request half the current limit
- 3.The charger output current drops by half
- 4.Temperature is within standard range fault will automatically clear and normal charging operation will resume

BMS 36 Charging Port Severe Overtemp

Check:

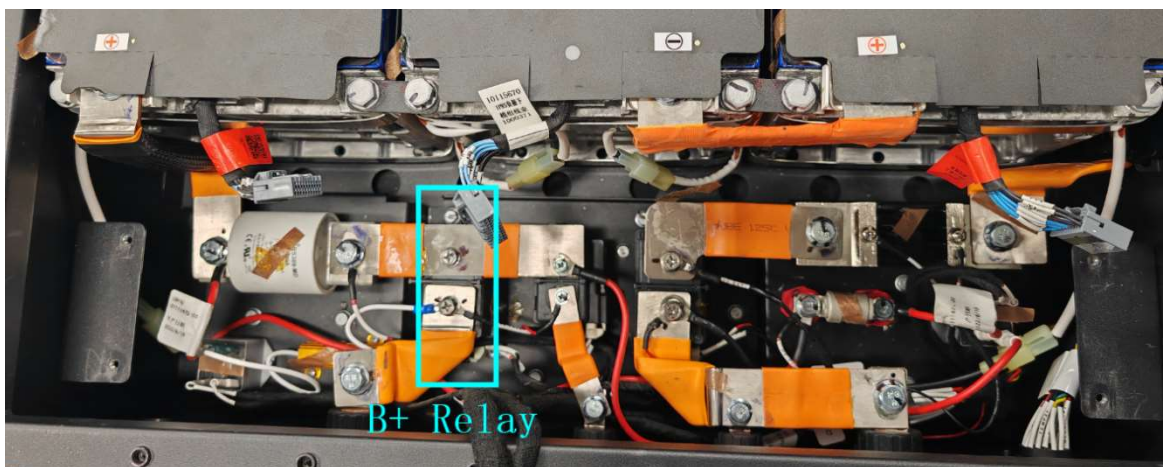
- 1.Display pop up BMS 36 fault code.

- 2.Stop Charging
- 3.Waiting the unit to cool down
- 4.Recharge

BMS 37 Discharging B+ Contactor Welded

Check:

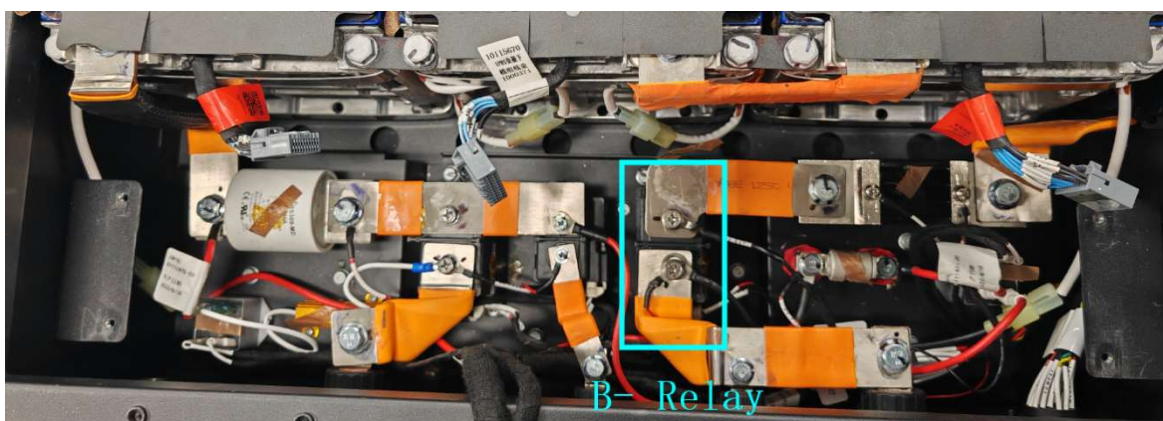
- 1.Display pop up BMS 37 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Replace the battery pack discharge B+ relay



BMS 38 Discharging B- Contactor Welded

Check:

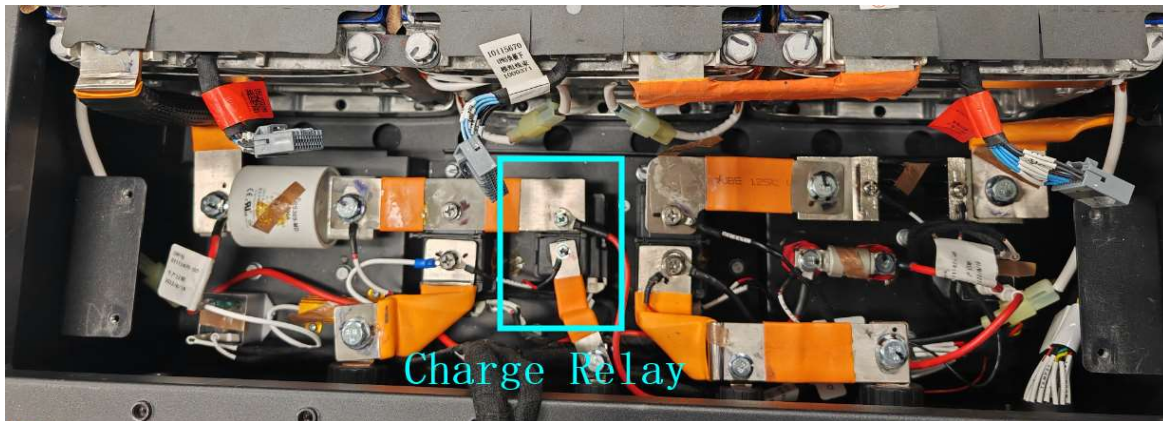
- 1.Display pop up BMS 38 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Replace the battery pack discharge B- relay



BMS 39 Battery Charging Contactor Welded

Check:

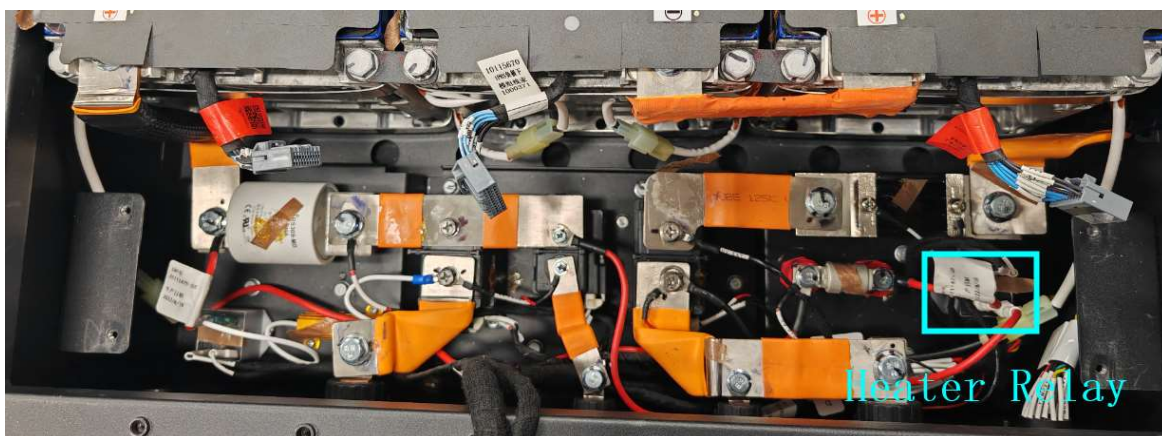
1. Display pop up BMS 39 fault code.
2. Discontinue Use if restart doesn't fix
3. Remove the battery box cover Replace the battery pack charge relay



BMS 41 Battery Heater Contactor Welded

Check:

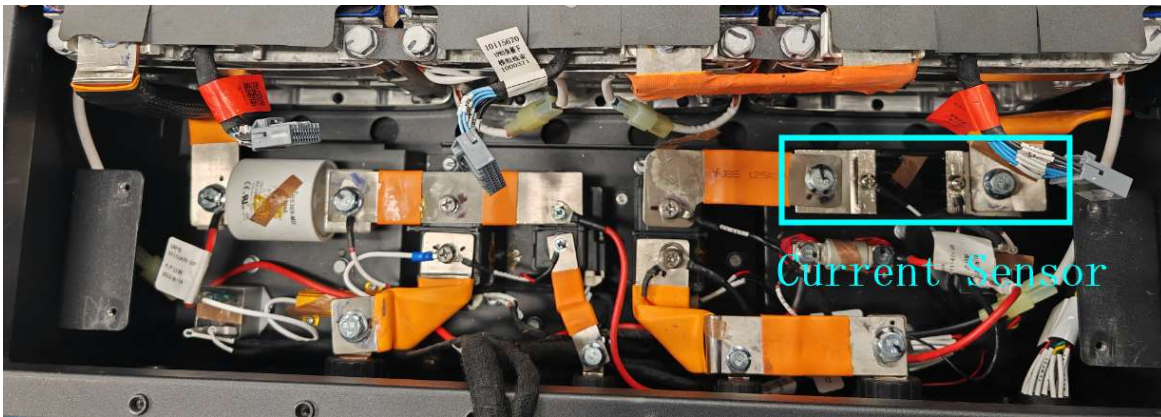
1. Display pop up BMS 41 fault code.
2. Discontinue Use if restart doesn't fix
3. Remove the battery box cover Replace the battery pack heater relay



BMS 42 Battery Current Sensor Abnormal

Check:

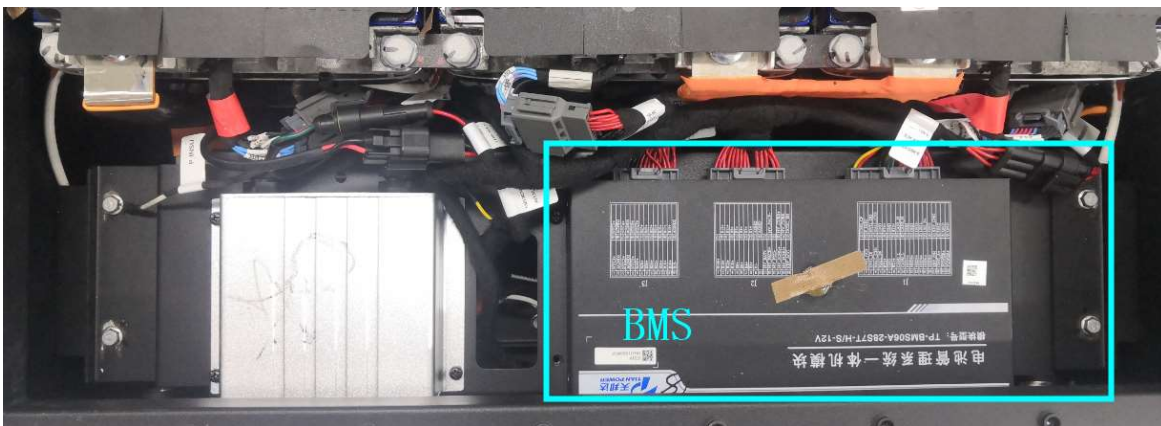
- 1.Display pop up BMS 42 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Replace the battery pack Current Sensor



BMS 43 Battery Sampling Chip Abnormal

Check:

- 1.Display pop up BMS 43 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Replace the battery pack BMS



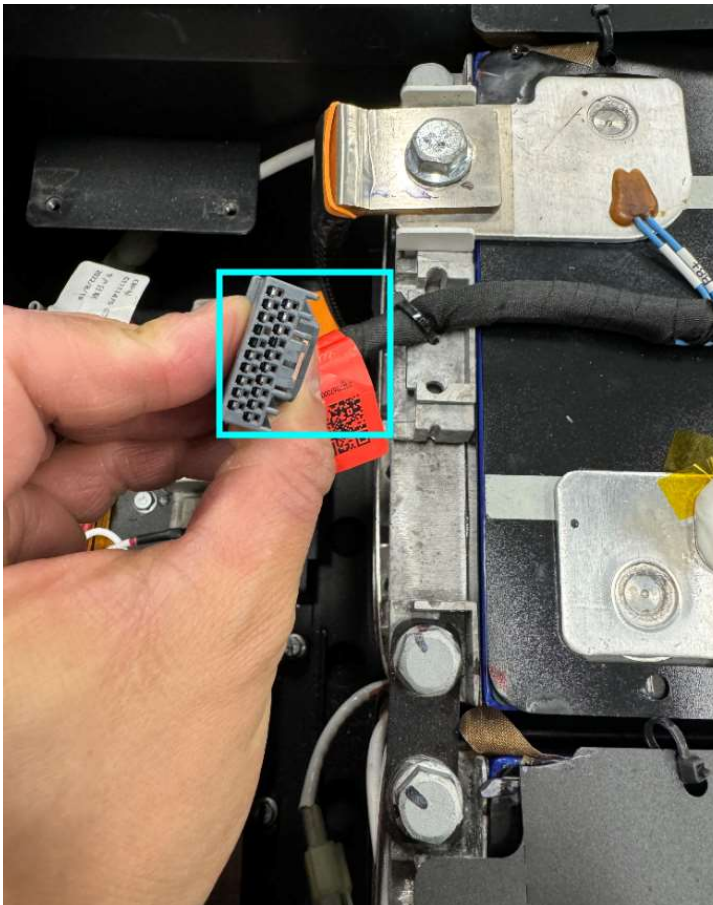
BMS 44 Battery Cell Open Circuit

Check:

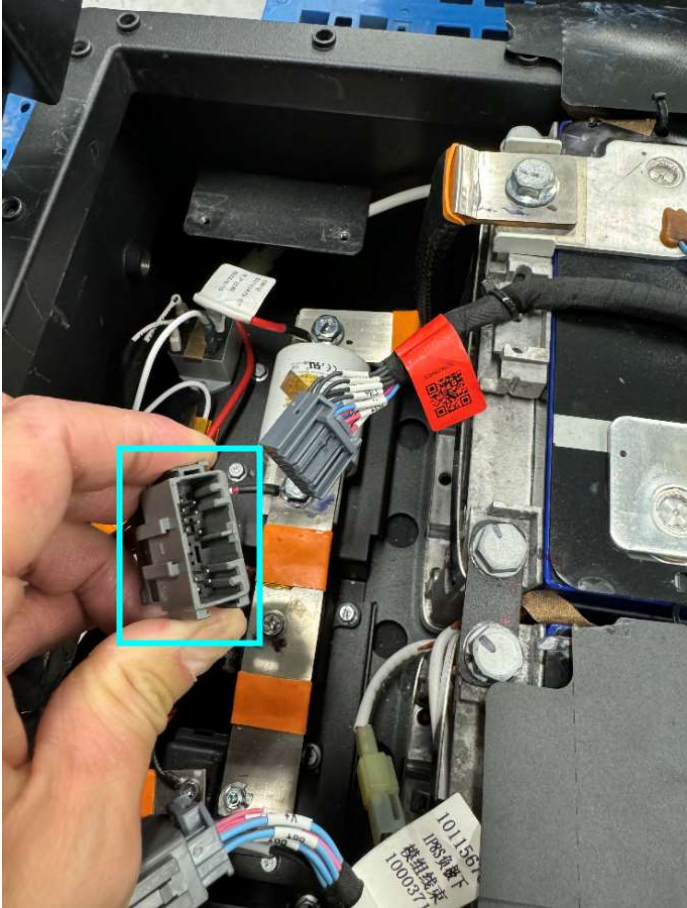
- 1.Display pop up BMS 44 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Check whether the cable on the module is disconnected, no disconnected,



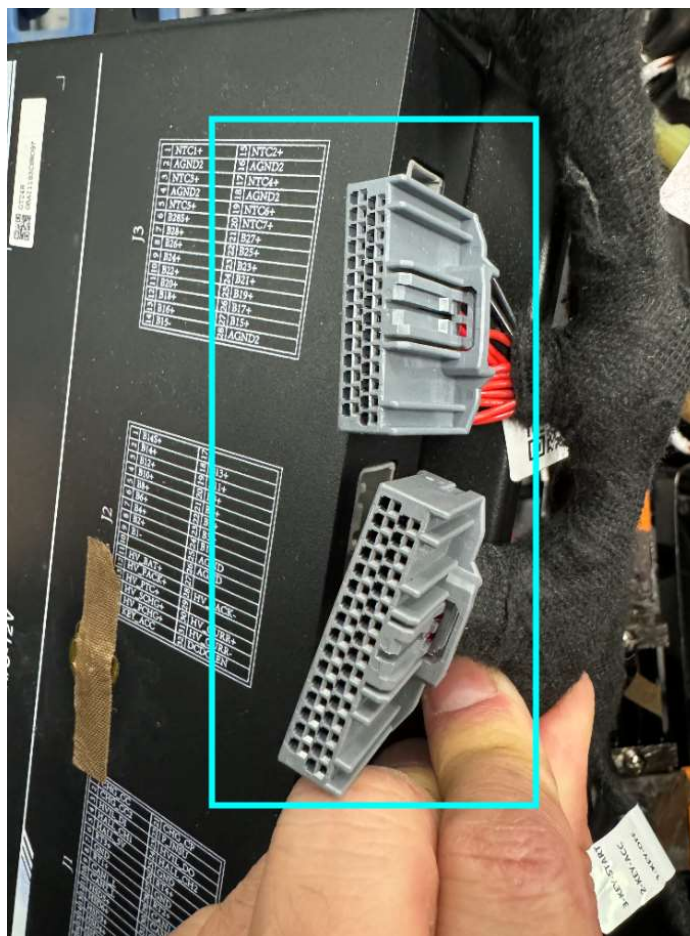
4. Check whether the module connector is burned black,



5. Check whether the BMS module connector is burned black,

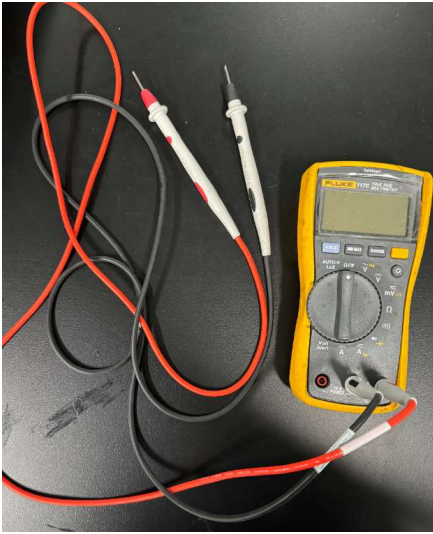


6. Check whether the BMS connector is burned black, if no burned black, BMS damage, Replace the battery pack



BMS 47 Charger CAN Communication Timeout

Tool: multimeter



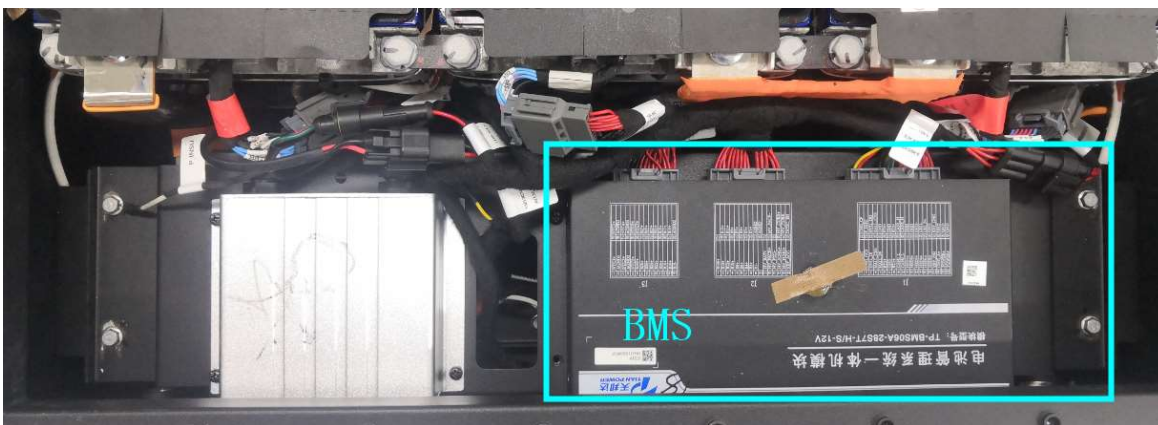
Check:

1. Display pop up BMS 47 fault code.
2. Set the multimeter to resistance, Check whether the CAN between the battery communication port and the

charging base is properly connected.if Displays 0Ω, Replace the charge.



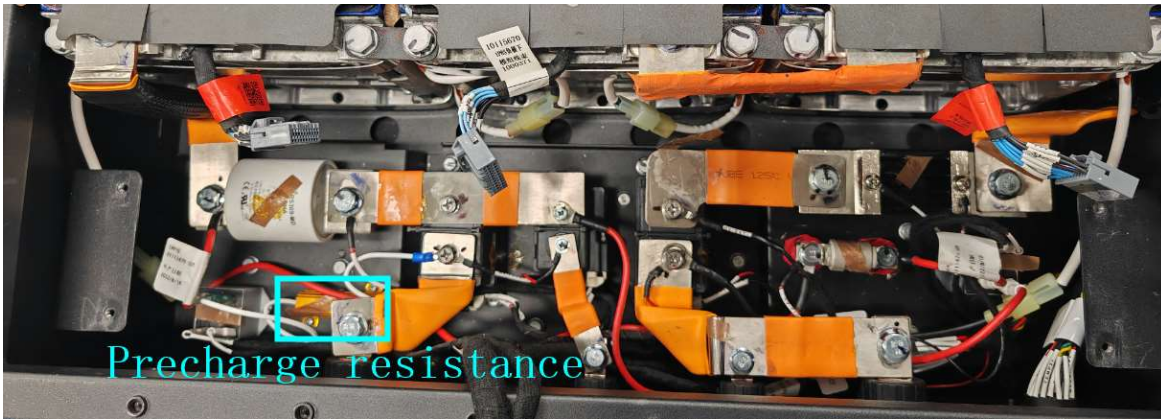
3.if Battery cannot be charged, Remove the battery box cover, Replace the BMS



BMS 48 Battery Precharge Failed

Check:

- 1.Display pop up BMS 48 fault code.
- 2.Discontinue Use if restart doesn't fix
- 3.Remove the battery box cover Replace the battery pack precharge resistance



BMS 49 Battery 12V Abnormal

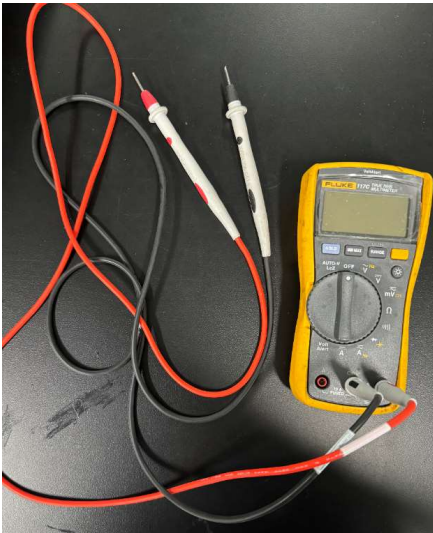
Check:

1. Display pop up BMS 49 fault code.
2. Discontinue Use if restart doesn't fix
3. Remove the battery box cover Replace the battery pack DC/DC power



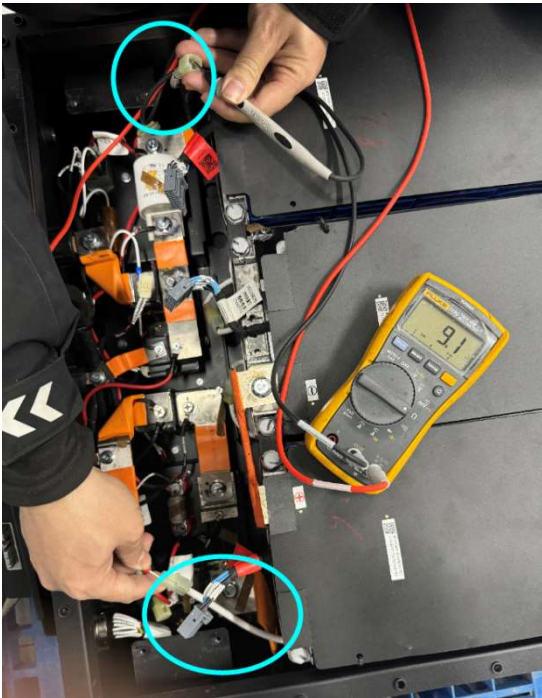
BMS 51 Battery Heater Abnormal

Tool: multimeter

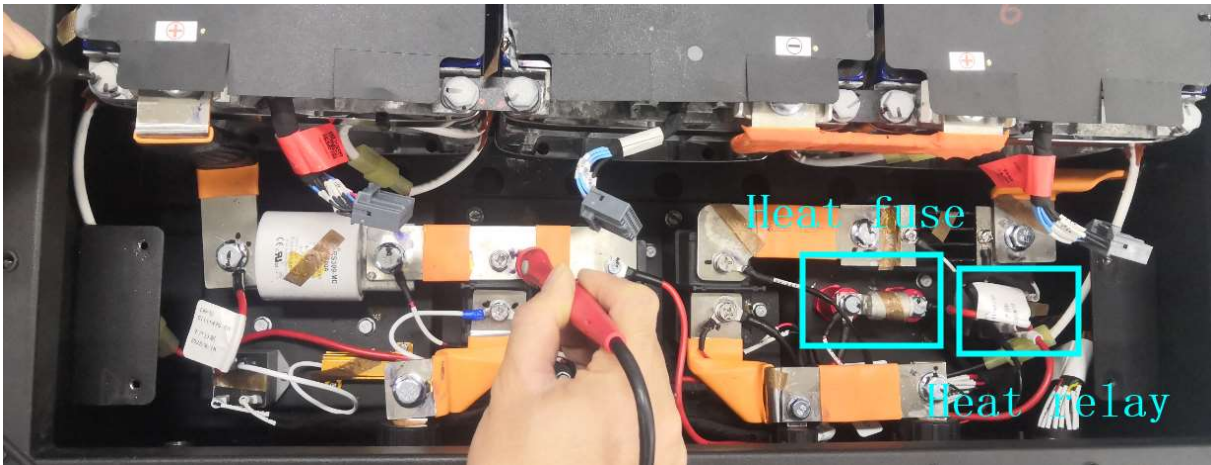


Check:

1. Display pop up BMS 51 fault code.
2. Check the resistance of the battery heating film, The normal range is 9 to 11, if no, heating film damage

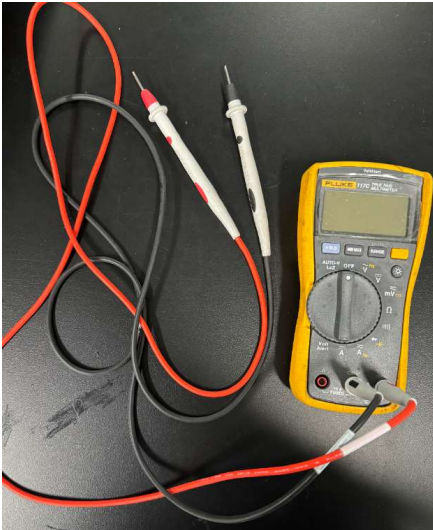


3. Check heat relay and heat fuse, Heating relay and heating fuse intact, Replace the battery pack BMS



BMS 52 Battery CC2 Detection Abnormal

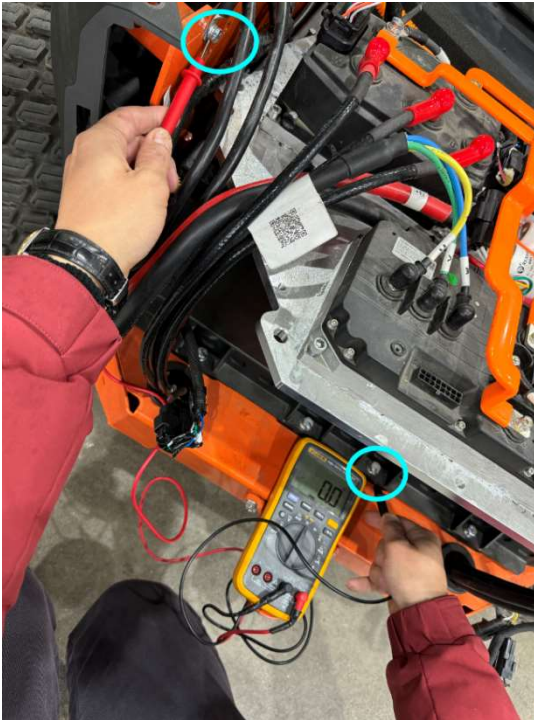
Tool: multimeter



Check:

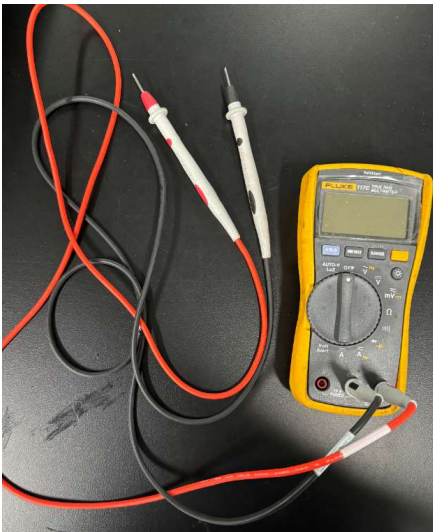
1. Display pop up BMS 52 fault code.
2. Check that the grounding cable on the charging base is properly connected to the frame. If no 0 is displayed,

reconnect the ground cable



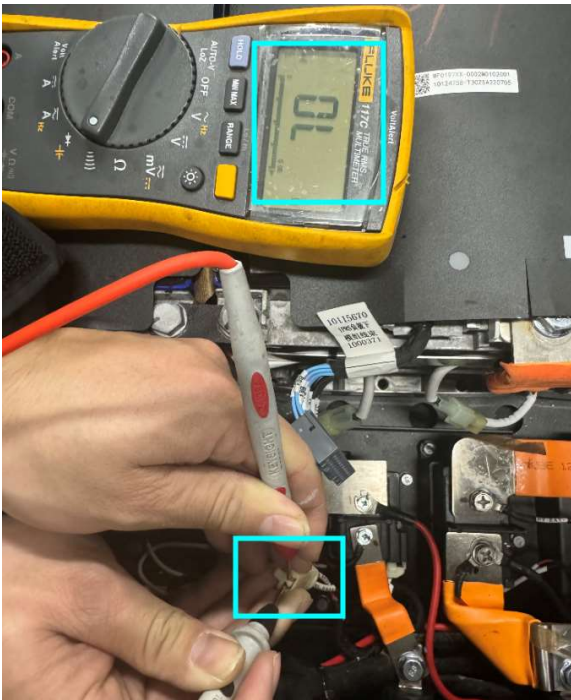
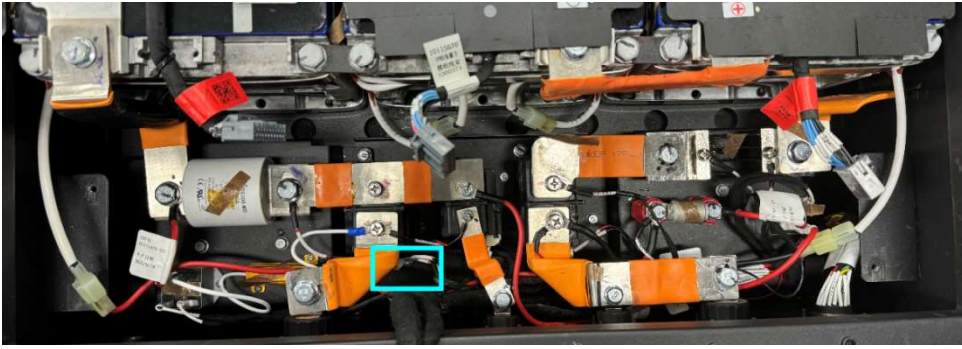
BMS 53 Battery B+ Contactor Coil Short

Tool: multimeter



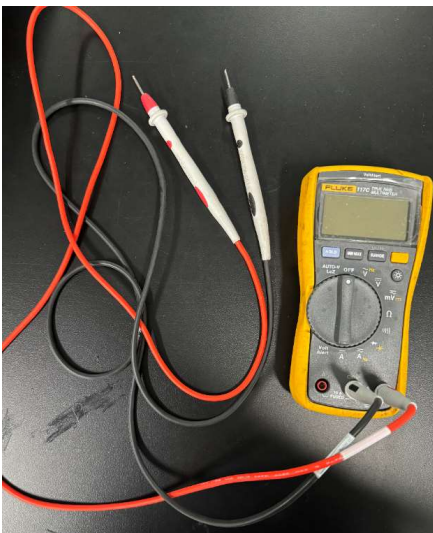
Check:

1. Display pop up BMS 53 fault code.
2. Check the coil of the B+ relay, OL not shown, relay damage, Replace B+ relay



BMS 54 Battery B- Contactor Coil Short

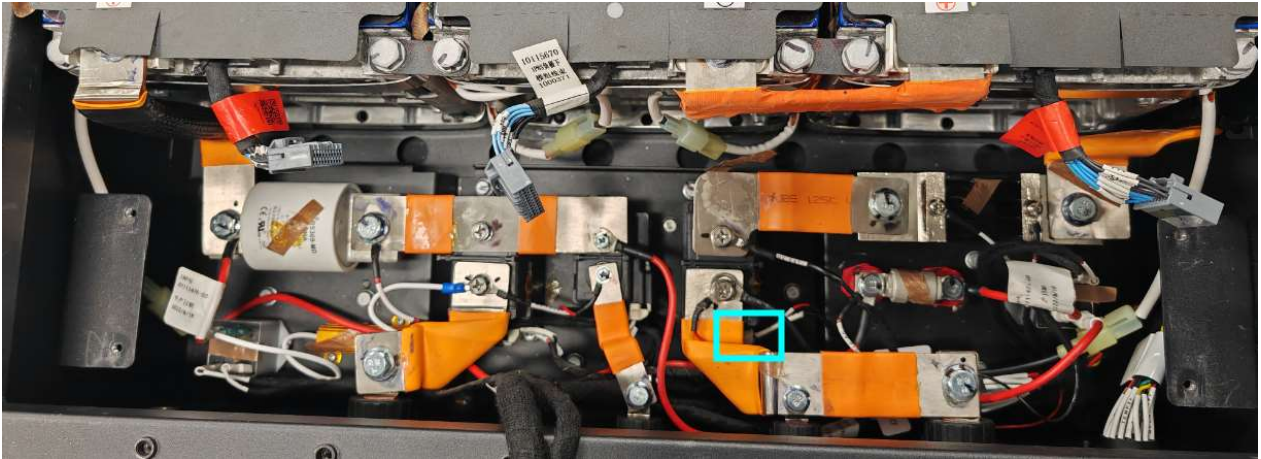
Tool: multimeter



Check:

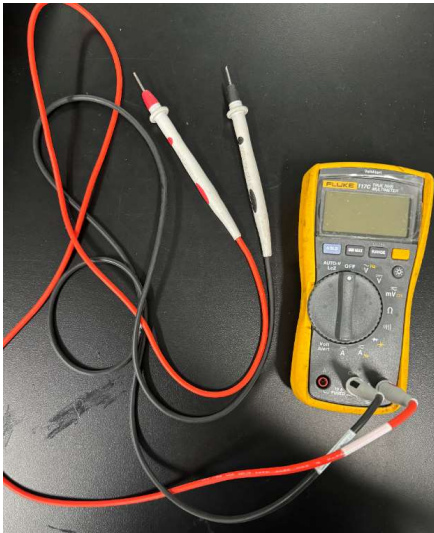
1.Display pop up BMS 54 fault code.

2.Check the coil of the B- relay, OL not shown,relay damage, Replace B- relay



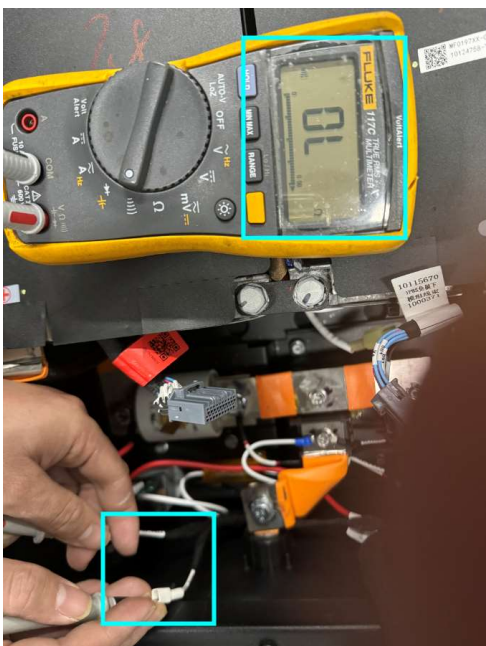
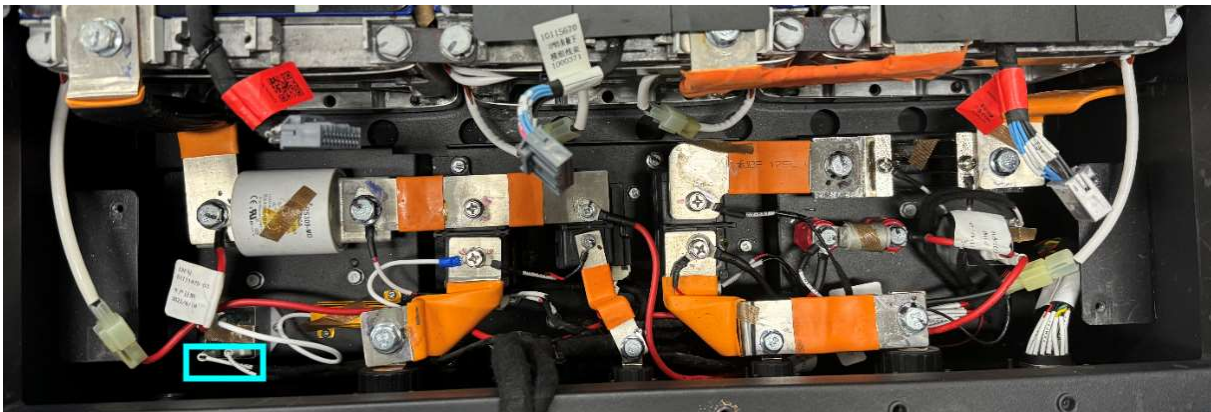
BMS 55 Battery Precharge Contactor Coil Short

Tool: multimeter



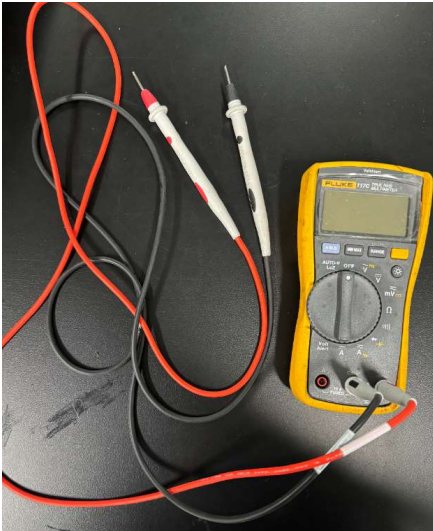
Check:

1. Display pop up BMS 55 fault code.
2. Check the coil of the Precharge relay, OL not shown, relay damage, Replace Precharge relay



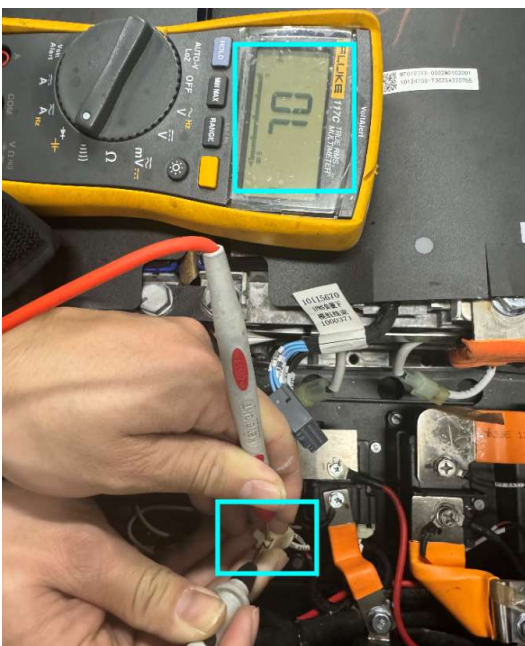
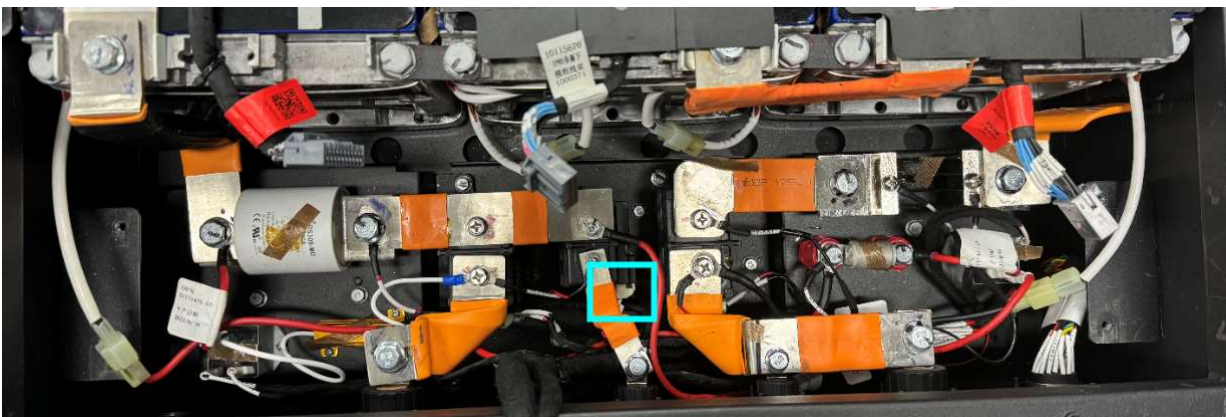
BMS 56 Battery Charge Contactor Coil Short

Tool: multimeter



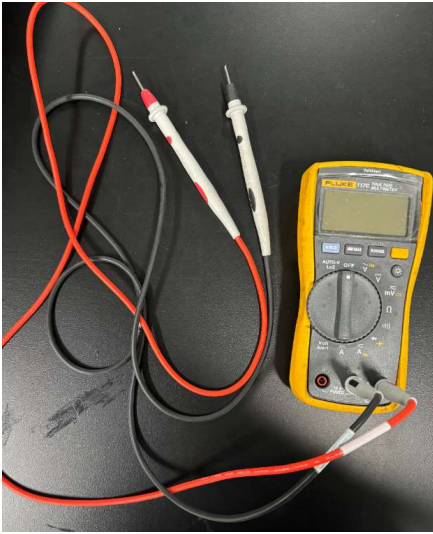
Check:

1. Display pop up BMS 56 fault code.
2. Check the coil of the charge relay, OL not shown, relay damage, Replace charge relay



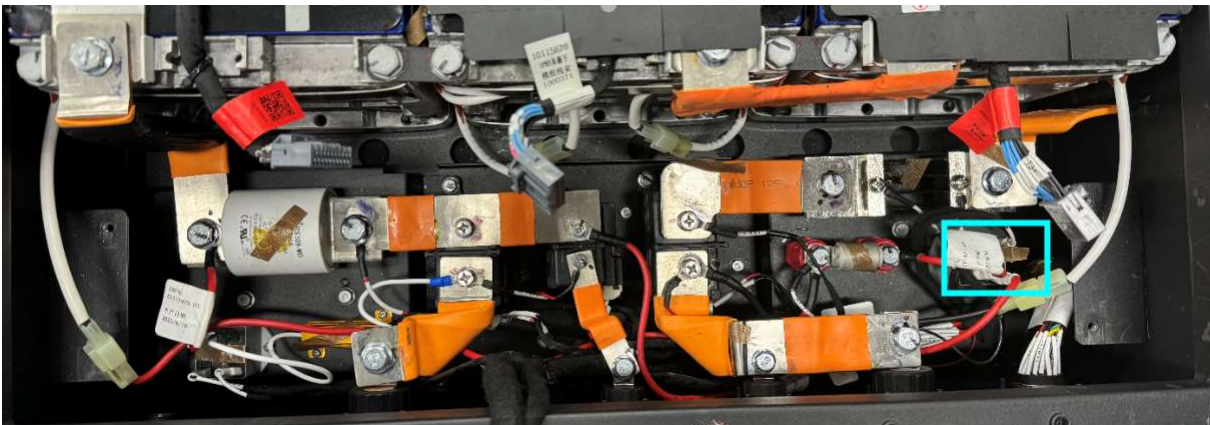
BMS 57 Battery Heating Contactor Coil Short

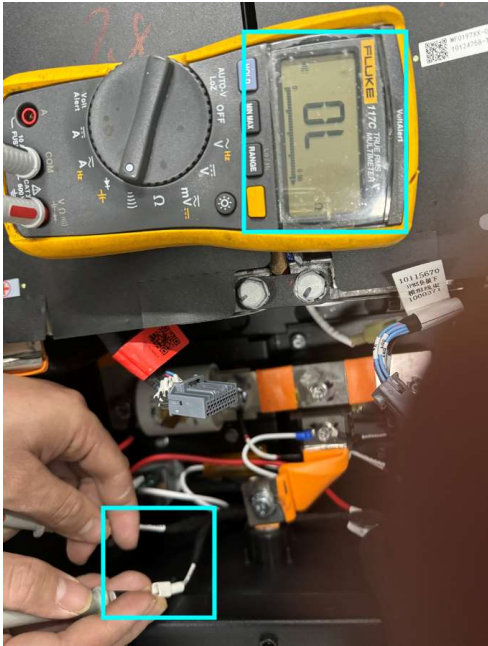
Tool: multimeter



Check:

1. Display pop up BMS 57 fault code.
2. Check the coil of the heating relay, OL not shown, relay damage, Replace heating relay





BMS 58 Battery Continuous Discharge Overcurrent

Check:

1. Display pop up BMS 58 fault code.
2. Reduce load